



Jagannath University
Faculty of Science
Department of Computer Science & Engineering

Curriculum

Program: B.Sc. in Computer Science & Engineering

(8 Semesters)

Session: 2025-2026

(17th Batch)

Jagannath University
Department of Computer Science and Engineering
Curriculum for B.Sc. in Computer Science and Engineering
Session: 2025-2026

Curriculum for the B.Sc. Program for Sessions: 2025-2026

1. **Title of the Program:** Bachelor of Science in Computer Science and Engineering abbreviated as B.Sc. in CSE
2. **Name of the University:** Jagannath University
3. **Vision of the University:** To ensure better academic achievement in world-class higher education, research, and innovation.
4. **Mission of the University:** To achieve equivalency and consistency at the national level in compliance with the fast-growing changes of the advanced world in different sectors of higher education and develop skilled human resources by creating and expanding opportunities for modern learning practices.

M1: To ensure equivalence and consistency in the quality of education in alignment with national standards.

M2: To adapt academic programs and practices in response to rapidly evolving global developments and international standards.

M3: To align higher education with diverse sectors and develop a skilled and competent human workforce.

M4: To expand opportunities for modern, innovative, and technology-enhanced learning processes.

5. **Name of the Degree:** B.Sc. in Computer Science and Engineering
6. **Name of the Faculty Offering the Program:** Faculty of Science
7. **Name of the Department offering the Program:** Computer Science and Engineering
8. **Introduction of the Department:** The Department of Computer Science and Engineering (CSE) in Jagannath University has been at the front positioning subject in the university. The department has the brightest students, and our goal is pushing them to ensure that they reach their greatest potential by putting them through some of the most challenging and rigorous computing courses. The department is part of the Faculty of Science at Jagannath University and started academic activities from 2009. The department is a dynamic and growing community of scholars whose shared passion for research and education is a hallmark of the Vanderbilt environment. The Department of CSE is committed to outstanding undergraduate and graduate education, distinguished research programs, and strong service to our students, professional and technical societies, and the community into the country as well as internationally.
9. **Vision of the Department:** To enlighten the next generation of engineers and scientists and make them national and international leaders in education and research through effective teaching and disciplinary and ethical learning.

10. Mission of the Department:

M1: To advance the field of Computer Science and Engineering through innovative teaching, research, and scholarly activities.

M2: To enhance students' knowledge and skills through quality instruction, global engagement, and experiential learning opportunities.

M3: To provide high-quality undergraduate and graduate education in a modern, state-of-the-art academic environment.

M4: To prepare graduates for successful careers in industry, government, and academia by developing their technical, analytical, and professional competencies.

M5: To contribute to societal development by promoting technology transfer, fostering industry collaboration, and achieving global recognition through education and research.

M6: To conduct impactful and cutting-edge research that addresses contemporary challenges and advances 21st-century technologies.

11. Objective of the Department:

- i. Prepare graduates with comprehensive engineering knowledge in computer science and engineering, encompassing theory and practical application.
- ii. Cultivate proficient problem-solving skills through rigorous analysis and synthesis of complex engineering challenges.
- iii. Foster innovative design thinking by integrating mathematical foundations and computer science theory in system modeling and solution development.
- iv. Develop hands-on expertise in software and hardware implementation, utilizing industry-standard programming languages and methodologies.
- v. Promote research-based inquiry for continuous learning, experimentation, and the development of cutting-edge solutions.
- vi. Instill ethical awareness, social responsibility, and professionalism in engineering practice, emphasizing the impact on society and the environment.
- vii. Prepare graduates for successful integration into diverse professional settings, including industry, academia, entrepreneurship, and further academic pursuits.

12. Description of the Program:

Bachelor of Science (Honors) in Computer Science and Engineering is a 4 (four) academic year program where each academic year is divided into 2 (two) semesters i.e. the whole program is divided into eight semesters. A semester, conducted for a period of 6 (six) months, is named either 1st semester or 2nd semester in each academic year. Examination will be held at the end of

each semester. A minimum requirement is set up by the university authority for the students to be promoted from one year to another year. A credit system is used in this case where a student is required to clear all 160 credit courses for the program. One credit for the theoretical courses requires 15 lecture hours, and for laboratory courses, one credit requires 30 lecture hours.

13. Program Education Objectives (PEOs):

The Program Educational Objectives (PEOs) of CSE-JnU are as follows:

PEO1: Provide students with a strong theoretical foundation as well as industry-relevant skills in various technologies related to ICT to be able to adapt to the evolving technical challenges and changing career opportunities.

PEO2: Instill analytic and critical thinking skills in order to exhibit analytical decision-making and problem-solving abilities when dealing with complex real-time challenges.

PEO3: Indoctrinate professional and ethical aspects of modern computing technology so that students can apply their skills to design, create, and use computing artifacts that are economically feasible and socially acceptable through their innovations.

PEO4: Providing an education that will prepare graduates to create, sustain, and improve ICT infrastructure in organizations, as well as manage ICT resources to support a variety of organizational functions in order to confront contemporary issues in a broader social context.

14. Program Learning Outcomes (PLOs):

A graduate student of CSE-JnU will have an ability to

PLO1: Engineering knowledge - Ability to apply knowledge of computing; mathematics, including discrete mathematics as well as probability and science; and engineering to the solution of complex engineering problems.

PLO2: Professional Skills - The ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, multimedia, web design, big data analytics, and networking for efficient design of computer-based systems of varying complexity

PLO3: Problem Analysis - Identify, formulate, review research literature, and analyze complex engineering problems using first principles of mathematics, statistics, economics, and engineering sciences.

PLO4: Design and Development of solutions - An ability to apply mathematical foundations, algorithmic principles, and computer science theory in the design, implementation, and evaluation of a software or hardware or any problem using programming language concepts, i.e, C/C++, Java, OOP, etc.

PLO5: Conduct investigations of complex problems - Use research-based knowledge and research methods, including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PLO6: Modern tool usage—Ability to use the techniques, skills, and modern engineering tools necessary for practice as a CSE professional, including prediction and modeling of complex engineering activities with an understanding of the limitations.

PLO7: Environment and sustainability - Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PLO8: Ethics - Understanding of professional, ethical, legal, security, and social issues and responsibilities and norms of the engineering practice.

PLO9: Communication - Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PLO10: Individual and team work—Perform effectively as an individual and as a member or a leader in multidisciplinary settings to accomplish a goal

PLO11: Lifelong learning - Recognition of the need for, and an ability to engage in lifelong learning and continuing professional developments with technological changes.

PLO12: Successful Career and Entrepreneurship - The ability to employ modern computer languages, environments, and platforms in creating innovative career paths to be an entrepreneur and a zest for higher studies.

15. Generic Skills/Graduate Profile:

A computer science and engineering graduate, beyond his/her deep understanding of the universe's fundamental laws, develops a strong set of generic skills that are highly valuable in a wide range of careers. The computer science and engineering graduates will have the following generic skills.

A. Analytical and Problem-Solving Skills:

a1) Breaking down complex problems: Graduates in computer science and engineering has the ability to analyze complex problems by breaking them down into smaller, more manageable components and recognizing important connections. They can utilize this expertise to diagnose problems, enhance efficiency, and make informed judgments in any domain.

a2) Quantitative reasoning: With their advanced skills in mathematics and scientific theory, they are able to examine data, construct models, and carry out intricate computations to reach rational solutions.

a3) Critical thinking: They possess the ability to assess information in an unbiased manner, challenge preconceptions, and recognize fundamental patterns and relationships. Proficiency in this talent is essential for making accurate assessments in every situation.

B. Technical Skills:

b1) Programming: The majority of graduates in Computer Science and Engineering acquire programming abilities specifically for the purpose of data analysis, simulations, and scientific computing. The capacity to utilize and modify coding languages can be applied to diverse technical domains.

b2) Laboratory and research skills: They acquire expertise in the process of formulating and implementing experiments, gathering empirical evidence, and analyzing findings. These talents are highly transferable to research and development positions in various organizations.

b3) Instrument and software proficiency: Graduates in computer science and engineering acquire the skills to operate a range of scientific instruments and apply specialized software, developing technological proficiency that is highly valued across multiple industries.

C. Communication Skills:

c1) Scientific writing: They refine their proficiency in effectively conveying intricate scientific knowledge in written reports, proposals, and research publications. Proficiency in this skill is crucial for fostering efficient collaboration and facilitating the exchange of knowledge in any work environment.

c2) Effective presentation: Computer science and engineering graduates has the ability to effectively communicate and explain their research findings to both their peers and the general public. This experience equips them with valuable communication and presentation skills that are highly sought after in a wide range of professional environments.

c3) Technical writing and data visualization: They possess the ability to convert complex technical information into easily comprehensible formats suitable for a wide range of audiences, making it a very valuable skill for engineers, educators, and various other professionals.

D. Other social skills:

d1) Time management and organization: Computer science and engineering studies necessitate the ability to handle challenging coursework, laboratory assignments, and individual research, which in turn develop effective time management and organizational abilities.

d2) Teamwork and collaboration: Collaboration is a common aspect of computer science and engineering projects, which aim to educate graduates the skills of successful teamwork, group dynamics management, and clear communication of ideas.

d3) Adaptability and learning agility: Computer science and engineering are always changing, which means that individuals in these fields must continuously learn and adapt to new concepts and approaches. The capacity to easily adapt and acquire new abilities is highly advantageous in any professional trajectory.

16. Mapping/Alignment University's Mission vs PEO:

PEOs	M1	M2	M3	M4
PEO1	✓	✓		
PEO2			✓	✓
PEO3	✓		✓	✓
PEO4		✓	✓	✓

17. Mapping/Alignment PEOs vs PLOs:

	PEO1	PEO2	PEO3	PEO4
PLO1	H	H	H	H
PLO2	H	H	H	H
PLO3	M	M	H	H
PLO4	H	H	H	H
PLO5	H	H	H	H
PLO6	H	H	H	H
PLO7	M	M	H	H
PLO8	H	H	H	H
PLO9	H	H	H	H
PLO10	H	H	H	H
PLO11	M	M	H	H
PLO12	H	H	H	H

Part B

18. Curriculum Structure

- a. Duration of the program: 4 Years (8 Semesters)
- b. Total minimum credit requirement: 160
- c. Course distribution

Areas of Study	Theor y	Practical	Viva-voce	Projec t	Total
CSE	83	23	4	9	119
Electrical and Electronics	9	3			12
Physics	3				3
Mathematics	14	1			15
Statistics	3				3
Economics	2				2
Accounting	3				3
Humanities	3				3
Total	122	27	4	9	160

Year and Semester-wise Distribution of Credits

Year	Semester	Theory		Practical		Project	Viva-voce	Total
		Major	Related	Major	Related			
First	First	9	9	2				20
	Second	12	5	3			1	21
Second	First	9	9	3				21
	Second	12	5	2	1		1	21
Third	First	15		4		1		20
	Second	12		4		2	1	19
Fourth	First	14		5				19
	Second	9		3		6	1	19
Total		92	28	26	1	9	4	160

The duration of final examination for theoretical and practical courses will be as follows:

Theory		Practical	
Marks (credit)	Exam time in hour	Marks (credit)	Exam time in hour
30 (2.0)	2	30(2.0)	4
60 (3.0)	3	60 (3.0)	6

19. Semester wise Course Outline for the Entire Program:**1st Year 1st Semester**

Course code	Course Title	Major/Non-Major	Marks	Credit
CSE-1101	Introduction to Computer Science and IT	Major	100	3
CSE-1102	Structured Programming Language	Major	100	3
CSEL-1103	Structured Programming Language Lab	Major	50	1
CSER-1104	Math-I (Calculus)	Non-Major	100	3
CSER-1105	Physics	Non-Major	100	3
CSE-1106	Electrical Circuit Analysis	Major	100	3
CSEL-1107	Electrical Circuit Analysis Lab	Major	50	1
CSER-1108	English	Non-Major	100	3
			700	20

1st Year 2nd Semester

Course code	Course Title	Major/Non-Major	Marks	Credit
CSE-1201	Object Oriented Programming-I	Major	100	3
CSEL-1202	Object Oriented Programming-I Lab	Major	50	1
CSE-1203	Data Structures	Major	100	3
CSEL-1204	Data Structures Lab	Major	50	1
CSE-1205	Basic Electronics	Major	100	3
CSEL-1206	Basic Electronics Lab	Major	50	1
CSER-1207	Math- II (Linear Algebra)	Non-Major	100	3
CSE-1208	Discrete Mathematics	Major	100	3
CSER-1209	Economics	Non-Major	50	2
CSEV-1210	Viva-Voce	Major	50	1
			750	21

2nd Year 1st Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSE-2101	Object Oriented Programming-II	Major	100	3
CSEL-2102	Object Oriented Programming-II Lab	Major	50	1
CSE-2103	Digital Logic Design	Major	100	3
CSEL-2104	Digital Logic Design Lab	Major	50	1
CSER-2105	Math- III (Ordinary differential Equation)	Non-Major	100	3
CSE-2106	Probability and Statistics for Computer Science	Non-Major	100	3
CSE-2107	Data Communication	Major	100	3
CSEL-2108	Data Communication Lab	Major	50	1
CSER-2109	Financial and Managerial Accounting	Non-Major	100	3
			750	21

2nd Year 2nd Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSE-2201	Computer Architecture	Major	100	3
CSE-2202	Mathematical Analysis for Computer Science	Major	100	3
CSE-2203	Database Management System	Major	100	3
CSEL-2204	Database Management System Lab	Major	50	1
CSER-2205	Math-IV (Complex Variable, Fourier and Laplace Transform)	Non-Major	100	3
CSER-2206	Numerical Analysis	Non-Major	50	2
CSEL-2207	Numerical Analysis Lab	Non-Major	50	1
CSE-2208	Design and Analysis of Algorithm	Major	100	3
CSEL-2209	Design and Analysis of Algorithm Lab	Major	50	1
CSEV-2210	Viva-Voce	Major	50	1
			750	21

3rd Year 1st Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSE-3101	Theory of Computation	Major	100	3
CSE-3102	Introduction to Data Science	Major	100	3
CSEL-3103	Data Science Lab	Major	50	1
CSE-3104	Operating Systems	Major	100	3
CSEL-3105	Operating Systems Lab	Major	50	1
CSE-3106	Microprocessor and Assembly Language	Major	100	3
CSEL-3107	Microprocessor and Assembly Language Lab	Major	50	1
CSE-3108	Computer Networks	Major	100	3
CSEL-3109	Computer Networks Lab	Major	50	1
CSEP-3110	Internet and Web Programming (Project)	Major	50	1
			750	20

3rd Year 2nd Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSE-3201	Compiler Design and Construction	Major	100	3
CSEL-3202	Compiler Design and Construction Lab	Major	50	1
CSE-3203	Computer Graphics and Animation	Major	100	3
CSEL-3204	Computer Graphics and Animation Lab	Major	50	1
CSE-3205	Software Engineering	Major	100	3
CSEL-3206	Software Engineering Lab	Major	50	1
CSE-3207	IoT and Embedded System	Major	100	3
CSEL-3208	IoT and Embedded System Lab	Major	50	1
CSEP-3209	Application Design and Development (Project)	Major	50	2
CSEV-3210	Viva-Voce		50	1
			700	19

4th Year 1st Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSE-4101	Artificial Intelligence	Major	100	3
CSEL-4102	Artificial Intelligence Lab	Major	50	1
CSE-4103	Software Quality Assurance and Testing	Major	100	3
CSE-4104	Software Quality Assurance and Testing Lab	Major	50	1
CSE-4105	Digital Image Processing	Major	100	3
CSEL-4106	Digital Image Processing Lab	Major	50	1
CSE-4107	Data Mining and Machine Learning	Major	100	3
CSEL-4108	Data Mining and Machine Learning Lab	Major	50	1
CSE-4109	Cryptography and Network Security	Major	100	3
CSEL-4110	Cryptography and Network Security Lab	Major	50	1
			700	19

4th Year 2nd Semester

Course Code	Course Title	Major/Non-Major	Marks	Credit
CSET- 4201	Thesis	Major	200	6
CSEP- 4201	Project	Major	200	6
CSEV-4202	Viva Voce	Major	50	1
CSE-42**	Option-I	Major	100	3
CSEL-42**	Option-I Lab	Major	50	1
CSE-42**	Option-II	Major	100	3
CSEL-42**	Option-II Lab	Major	50	1
CSE-42**	Option-III	Major	100	3
CSEL-42**	Option-III Lab	Major	50	1
			700	19
Total Marks: 5800		Total Credit		160

Part C

20. Mapping between all Courses with PLOs:

Course Code	Course Title	Program Learning Outcomes											
		PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO10	PLO11	PLO12
CSE-1101	Introduction to Computer Science and IT	H	H	H	H						M		
CSE-1102	Structured Programming Language	M	M	M	M		M						
CSEL-1103	Structured Programming Language Lab	M	M	M	M		M						
CSER-1104	Math-I (Calculus)	H	H	H	H		H	H			M	M	
CSER-1105	Physics	H	H				H				M		M
CSE-1106	Electrical Circuit Analysis	H	H	H		H		H	H	H	H		M
CSEL-1107	Electrical Circuit Analysis Lab	H	H	H		H	H	H	H	H			H
CSER-1108	English	H	H	H			H			H	H		M
CSE-1201	Object Oriented Programming-I	H	H	H	M		M	M					
CSEL-1202	Object Oriented Programming-I Lab	H	H	H	M		M	M					
CSE-1203	Data structure	H	H	H	H		H		H		M	H	H
CSEL-1204	Data structure Lab	H	H	H	H		H		H		M	H	H
CSE-1205	Basic Electronics	H	H	H		H	H	H		M	M	M	M
CSEL-1206	Basic Electronics Lab	H	H	H		H			H		H		M
CSER-1207	Math- II (Linear Algebra)		H	H	H			H			M	M	M
CSE-1208	Discrete Mathematics		M	M	M								
CSER-1209	Economics	H	H		H	H	H	H		H		H	H

CSE-2101	Object Oriented Programming-II	H	H	H	M		M	M					
CSEL-2102	Object Oriented Programming-II Lab	H	H	H	M		M	M					
CSE-2103	Digital Logic Design	H	H	H		H	H			H			M
CSEL-2104	Digital Logic Design Lab	H	H	H	H	H	H			H		M	M
CSER-2105	Math- III (Ordinary differential Equation)	H	H	H	H								M
CSE-2106	Probability and Statistics for Computer Science	M	H	H									
CSE-2107	Data Communication	H	H	M	M	H							
CSEL-2108	Data Communication Lab	H	M	H	H	H	H						
CSER-2109	Financial and Managerial Accounting	H		H			H			H			H
CSE-2201	Computer Architecture	H	H	M	H								
CSER-2202	Mathematical Analysis for Computer Science	H	M	M						M			M
CSE-2203	Database Management System	H	H	H	M		M	M					
CSEL-2204	Database Management System Lab	M	H	H	H	H	H			H			M
CSER-2205	Math-IV (Complex Variable, Fourier and Laplace Transform)	H	H	H	H			H			M	M	M
CSER-2206	Numerical Analysis	H	H		H			H					

CSEL-2207	Numerical Analysis Lab	H	H		H			H					
CSE-2208	Design and Analysis of Algorithm	M	H	H	H	H				H			M
CSEL-2209	Design and Analysis of Algorithm Lab	M	H	H	H	H	H			H			M
CSE-3101	Theory of Computation	H	H	H	M		M	M					
CSE-3102	Introduction to Data Science	H	H	M	M	H						H	H
CSEL-3103	Data Science Lab	H	H	M	H	H		M	M				M
CSE-3104	Operating Systems	M	H	H	M	H	H	M		M			M
CSEL-3105	Operating Systems Lab		H	H	H	H	H			H			
CSE-3106	Microprocessor and Assembly Language		H	H	H			H			M	H	M
CSEL-3107	Microprocessor and Assembly Language Lab		H	H	H			H			H	M	M
CSE-3108	Computer Networks	H	H	M		M							
CSEL-3109	Computer Networks Lab	M	M	M		M		H	M	M	M	H	
CSEP-3110	Internet and Web Programming (Project)		H	H		H	H			H	H		
CSE-3201	Compiler Design and Construction	H	H	H	M		M	M					
CSEL-3202	Compiler Design and Construction Lab	M	H	H	H	H				H			M
CSE-3203	Computer Graphics and Animation	H	H	H	M		M	M					
CSEL-3204	Computer Graphics and Animation Lab	M	H	H	H	H				H			M
CSE-3205	Software Engineering	M	H	H	H	H			H	H		H	H

CSEL-3206	Software Engineering Lab	M	H	H	H	H	M		H	H		H	H
CSE-3207	IoT and Embedded System	H		H		M	M	M				H	H
CSEL-3208	IoT and Embedded System Lab	H		H		M	M	M				H	H
CSEP-3209	Application Design and Development (Project)		H	H		H	H			H	H		
CSE-4101	Artificial Intelligence	H	H	H	H	H		H	M		M		
CSEL-4102	Artificial Intelligence Lab	H	H	H	H	H		H	M		M		
CSE-4103	Software Quality Assurance and Testing	H	H	H	H	H	M	H	M		M		M
CSEL-4104	Software Quality Assurance and Testing Lab	H	H	H	H	H	M	H	M		M		M
CSE-4105	Digital Image Processing	H	H	H	H	M	H	M	M	M	M	M	M
CSEL-4106	Digital Image Processing Lab	H	H	H	H	M	H	M	M	M	H	M	M
CSE-4107	Data Mining and Machine Learning	H	H	H	M		M	M					
CSEL-4108	Data Mining and Machine Learning Lab	M	H	H	H	H	H			H			M
CSE-4109	Cryptography and Network Security	H	H	H		M		H	M				M
CSEL-4110	Cryptography and Network Security Lab	M	H	H	H	H	H			M			
CSE-4207	Network Routing and Switching	H	H	H	M		M	M					
CSEL-4208	Network Routing and Switching Lab	H		H		H	H	H				H	H
CSE-4213	Neural Network	M	H	H	H				M	H		M	M

	and Fuzzy System												
CSEL-4214	Neural Network and Fuzzy System Lab		H	H	H	M			M				M
CSE-4221	Natural Language Processing	M	H	H	H	M	M						
CSEL-4222	Natural Language Processing Lab	M	H	H	H	H	H		M	M			
CSE-4223	System Design and Analysis	H	H	H	M		M	M					
CSEL-4224	System Design and Analysis Lab	M	H	H	H	H	H			H			M
CSE-4225	Blockchain Technology and Application	H	H	H	H			M				M	M
CSEL-4226	Blockchain Technology and Application Lab	H	H	H	H		H		H	H	H		
CSE-4227	VLSI Design	H	H	H	H							M	
CSEL-4228	VLSI Design Lab	H	H	H	H								H
CSE-4229	Cloud Computing		H	H	H	M	M	M			M	M	M
CSEL-4230	Cloud Computing Lab	H	H	M	H	M	M	M			H	M	M

21. Description of all Courses/Subjects of the Program:

CSE-1101: Introduction to Computer Science and IT

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)		
				Semester Final Exam.	Continuous Assessment	Total

					Two Mid-Sem ester Exams	Class Attenda nce	Quiz/Case Study/Assign ment/ Class Performance	
CSE-11 01	Introduction to Computer Science and IT	Theory	3	60	20	10	10	100

Course Description: Computer Architecture is a compulsory course for the B.Sc. Program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 48 hours, 42 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

Knowledge in information technology can teach students essential systems and networking concepts, security practices, and application development. Information Technology deals with the development, implementation, support, and management of computers and information systems. This course is designed to introduce the students with fundamental knowledge on computer basics, its types, evolution and components of computers. Information Technology involves working both with hardware (e.g. CPUs, RAM, hard-disks) and software (e.g. operating systems, web browsers, mobile applications). The difference between Computer Science and Information Technology is that computer scientists design and develop the software programs that IT professionals use and maintain. IT specialists take what they've learned about computers and apply that knowledge in businesses across all industries.

Course Objectives

The major objectives of the course are:

- i. to analyze complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. to apply successful troubleshooting strategies for common hardware and software issues.
- ii. to design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline. to understanding transferrable functions between different types of software.
- iii. to recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
- iv. to use systemic approaches to select, develop, apply, integrate, and administer secure computing technologies to accomplish user goals.to apply computer fundamentals knowledge to other technology, including mobile devices.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Explain the history and fundamental concepts of computers, including the structure and functions of the CPU.

CLO2: Describe various input and output devices, analyze their functions, advantages, and limitations, and explain basic memory organization.

CLO3: Demonstrate an understanding of basic computer software concepts and fundamental programming principles.

CLO4: Explain the significance, components, and ethical issues of Information Technology, along with the basic concepts of data communication.

CLO5: Describe the concepts of e-commerce, cybersecurity, and modern computing technologies.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	HISTORY AND HISTORICAL EVALUATION OF COMPUTERS: 1. Counting devices used before the advent of computers 2. Characteristics of early computers 3. Generations of computer 4. Outstanding characteristics of each generation	• Lecture • Rapport building • Demonstration • Group work	3	CLO1
2.	INTRODUCTION TO COMPUTER & COMPUTER SYSTEM: 1. Computer & it's working principles 2. Classification of computer 3. Outstanding characteristics of computers 4. Block diagram of a typical computer 5. Basic components of a computer 6. Hardware vs. Software	• Lecture • Multimedia Projector • Hand Note	3	CLO1
3.	CPU USED IN PERSONAL COMPUTERS:	• Lecture • Group work • Demonstration	3	CLO1, CLO2

	1. Elements of CPU and their functions 2. Types of processors: RISC Vs. CISC 3. Bus system and port in microcomputer 4. Factors affecting the processing speed of a computer			
4.	INPUT AND OUTPUT DEVICES: 1. Definition and classification 2. Functions and applications of popular input and output devices 3. Common I/O devices 4. Speech recognition Vs. Speech synthesis	• Lecture, • Rapport building • Multimedia Projector • Cooperative learning	3	CL02
5.	MEMORY FUNDAMENTALS: 1. Definition and functions 2. Primary Vs. Secondary memory 3. Semiconductor, optical and magnetic memories 4. RAM Vs. ROM 5. CPU-Memory interconnection	• Lecture • Case Study • Group work	3	CLO2
6.	SOFTWARE AND THEIR USAGE: 1. Software and its classification 2. Application Vs. System software 3. Operating system Vs. Utility programs 4. Functions, objectives, and features of an operating system 5. Software development life-cycle.	• Lecture • Rapport building • Demonstration	3	CLO2
7.	BASIC PROGRAMMING CONCEPTS: 1. High-level Vs. Low-level programming language 2. Translator programs 3. Flowchart and algorithm 4. Data types, variables and operators 5. Control statements: Branching and looping 6. Errors while programming 7. Number system and conversions	• Lecture • Group work • Hand Note	3	CLO3
8.	BASIC INFORMATION TECHNOLOGY: 1. Data vs. Information	• Lecture • Case Study • Group work	3	CLO3

	2. Significance of information in daily life 3. Information Technology vs. Information System 4. Components and functions of a computer-based information system 5. Management Information System 6. Ethical issues in information system 7. Overview of information security			
9.	DATA COMMUNICATION, COMPUTER NETWORK AND THE INTERNET: 1. Telecommunication Vs. Data Communication 2. Components and modes of a Data Communication System 3. Types of computer network 4. Merits and demerits 5. Network topologies and protocols 6. History of the Internet 7. Features of Internet 8. Providing security of Internet and network	• Lecture • Case Study • Group work	3	CLO3
10.	CONCEPTS OF E-COMMERCE: 1. E-commerce Vs. E-Business, 2. Merits, demerits and outstanding features of e-commerce 3. Various e-commerce models 4. Payment system in e-commerce 5. Status of e-commerce in Bangladesh	• Lecture • Case Study • Group work	3	CLO3, CLO4
11.	CONCEPTS OF E-GOVERNANCE: 1. Why e-governance 2. Components of e-governance 3. Evolution of e-governance in Bangladesh 4. Factors for implementing Digital Bangladesh 5. Key development sectors in Bangladesh 6. E-government initiatives from government entities	• Lecture • Case Study • Group work	3	CLO5, CLO4

12.	CYBER SECURITY AND CYBER LAW: 1. Cyber world vs. Real world 2. What is cyber security 3. The state of security today 4. Why security matters today 5. Cybercrime vs. Cyber law	• Lecture • Case Study • Group work	3	CLO5
13	Modern Computing: Cloud Computing, Mobile Cloud Computing (MCC), Internet of Things (IoT), Computer Privacy and Security.	• Lecture • Case Study • Group work	3	CLO5
14	Revision and Solutions		3	CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO 5	PLO 6	PLO7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO 1	H											
CLO 2	H	H										
CLO 3	H									M		
CLO 4	H	M										
CLO 5	H	H	H	H								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√	√		
Mid Semester2				√	√
Class Performance	√	√	√	√	√

Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Case Study/Assignment/ Class Performance (10)	Class Attendance (10)	Other forms
Remember	25%				
Understand	50%	50%	25%		
Apply	25%				
Analyze			40%		
Evaluate		25%	10%		
Create		25%	25%		

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows:

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1

2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions, one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Herbert Schildt, "C: The Complete Reference", Osborne / McGraw-Hill; 4th edition, July 1, 2017.
2. Dennis Ritchie Brian W. Kernighan, "The C Programming Language ", Prentice Hall, January 1, 2015.
3. Balagurusamy, "Programming in ANSI C", McGraw-Hill Education, January 1, 2011.
4. K. N. King, "C Programming: A Modern Approach", W. W. Norton & Company, 2nd Edition, April 19, 2008.
5. Behrouz A. Forouzan, Richard F. Gilberg, "Computer Science: A Structured Programming Approach Using C", Course Technology, 3rd Edition, February 6, 2006.
6. Paul J. Deitel, "C How to Program", Prentice Hall, 5th Edition, September 4, 2006.
7. Stephen G. Kochan, "Programming in C", Sams, 3rd Edition, July 18, 2004.
8. Shafazand, Hassan, and A. Min Tjoa, eds. EurAsia-ICT 2002: Information and Communication Technology. Berlin, Heidelberg: Springer Berlin Heidelberg, 2002. <http://dx.doi.org/10.1007/3-540-36087-5>.

CSE-1102: Structured Programming Language

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Attendance	Quiz/Case Study/Assignment/ Class Performance	
CSE-1102	Structured Programming Language	Theory	3	60	20	10	10	100

Course Description: Structured Programming Language is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the C programming language. It aims to develop the students' logical, critical, and problem-solving skills on programming basics and also on the modern context of Structured Programming Language through exercises and hands-on activities. C is fundamental among professional programmers because it can be used to build logical foundation and programming skills of students. This course familiarizes students with basic as well as advanced features of Structured Programming. Data types, Identifiers and Keywords, Variable Declaration, Expression, Statement, Operators, Library functions, Data Input and Output Functions, Control Statements, Function, Recursion, Automatic, External & Static variable, Array, Pointer, Structure and Unions, Data Files, Some Additional Features of C, Advanced Data types, Access Modifiers, Storage Class Specifiers, Type Conversion in Assignments, Function Type Modifiers, Pointer to Function. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- Apply basic programming concepts to solve substantial problems in C.
- Develop, and test substantial programs using a modern IDE.
- Use, implement, and evaluate fundamental data structures.
- Create, implement, debug, and evaluate algorithms for solving substantial problems.
- Deliver and evaluate basic technical documents, presentations, and group interactions using appropriate tools.
- Prepare themselves for national & international programming contests.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding Structured Programming concepts using basic syntaxes of control Structures, strings for developing skills of logic-building activity using C

CLO2: Identify and use data types, identifiers, keywords, variables, operators, expressions, and input/output functions to construct C programs.

CLO3: Implement functions, recursion, arrays, pointers, structures, and unions, including the use of different storage classes in C programming.

CLO4: Analyze and utilize file handling, advanced data types, type conversion, function modifiers, and pointers to functions in solving programming problems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hour	CLO s
1	Introduction to Computer Programming, Problem Solving Techniques, Algorithm Specification and Development, Programming Style, Debugging and Testing	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
2	Documentation, Program Design Methodologies, Structured and Modular Program Designs, Classification, Assembler and Translators Source and Object Programs,	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1

3	Structured Language, Procedural and Non-procedural Programming, Modular Programming, Object Oriented Programming	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
4	Data types, Identifiers and Keywords, Variable Declaration, Expression, Statement,	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
5	Operators, Library functions, Data Input and Output Functions, Control Statements.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
6	Control structures	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
7	Array	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the	3	CLO 3

		following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
8	Pointers and Strings, Arrays of Pointers	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
9	Functions	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
10	Structure, Union and Bit-field	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
11	External files.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
12	Data Files, Some Additional Features of C	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based	3	CLO 4

		lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
13	Advanced Data types, Access Modifiers, Storage Class Specifiers, Type Conversion in Assignments, Function Type Modifiers, Pointer to Function.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓	✓				✓						
CLO2	✓	✓	✓	✓		✓						
CLO3	✓	✓	✓	✓	✓	✓				✓		
CLO4	✓	✓	✓	✓		✓		✓				

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	✓			
Mid Semester2		✓	✓	
Quiz/Performance	✓	✓	✓	✓
Class Attendance	✓	✓	✓	✓
Final	✓	✓	✓	✓

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows:

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO1 or Question from CLO1
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO3
5	Question from CLO4 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions, one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks,

the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Herbert Schildt, "C: The Complete Reference", Osborne / McGraw-Hill, 4th edition, July 1, 2017.
2. Dennis Ritchie Brian W. Kernighan, "The C Programming Language ", Prentice Hall, January 1, 2015.
3. Balagurusamy, "Programming in ANSI C", McGraw-Hill Education, January 1, 2011.

Reference Books:

1. K. N. King, "C Programming: A Modern Approach", W. W. Norton & Company, 2nd Edition, April 19, 2008.
2. Behrouz A. Forouzan, Richard F. Gilberg, "Computer Science: A Structured Programming Approach Using C", Course Technology, 3rd Edition, February 6, 2006.
3. Paul J. Deitel, "C How to Program", Prentice Hall, 5th Edition, September 4, 2006.
4. Stephen G. Kochan, "Programming in C", Sams, 3rd Edition, July 18, 2004.

CSE-1103: Structured Programming Language Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSEL-1103	Structured Programming Language Lab	Lab	1	30	10	5	5	50

Course Description: Structured Programming Language is a compulsory course for the B.Sc. programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the C programming language. It aims to develop the students logical, critical, and

problem-solving skills on programming basics and also on the modern context of Structured Programming Language through exercises and hands-on activities. C is fundamentals among professional programmers because it can be used to build logical foundation and programming skills of students. This course familiarizes students with basic as well as advanced features of Structured Programming. Data types, Identifiers and Keywords, Variable Declaration, Expression, Statement, Operators, Library functions, Data Input and Output Functions, Control Statements, Function, Recursion, Automatic, External & Static variable, Array, Pointer, Structure and Unions, Data Files, Some Additional Features of C, Advanced Data types, Access Modifiers, Storage Class Specifiers, Type Conversion in Assignments, Function Type Modifiers, Pointer to Function. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- i. Apply basic programming concepts to solve substantial problems in C.
- ii. Develop, and test substantial programs using a modern IDE.
- iii. Use, implement, and evaluate fundamental data structures.
- iv. Create, implement, debug, and evaluate algorithms for solving substantial problems.
- v. Deliver and evaluate basic technical documents, presentations, and group interactions using appropriate tools.
- vi. Prepare themselves for national & international programming contests.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding Structured Programming concepts using basic syntaxes of control Structures, strings for developing skills of logic building activity using C

CLO2: Identification of Data types, Identifiers and Keywords, Variable Declaration, Expression, Statement, Operators, Library functions, Data Input and Output Functions, Control Statements.

CLO3: Illustration to achieve Function, Recursion, Automatic, External & Static variable, Array, Pointer, Structure and Unions.

CLO4: Understanding concept Data Files, Some Additional Features of C, Advanced Data types, Access Modifiers, Storage Class Specifiers, Type Conversion in Assignments, Function Type Modifiers, Pointer to Function.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction to Computer Programming, Problem Solving Techniques, Algorithm Specification and Development, Programming Style, Debugging and Testing	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 1
2	Documentation, Program Design Methodologies, Structured and Modular Program Designs, Classification, Assembler and Translators Source and Object Programs,	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 1
3	Structured Language, Procedural and Non-procedural Programming, Modular Programming, Object Oriented Programming	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 1
4	Data types, Identifiers and Keywords, Variable Declaration, Expression, Statement,	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 2

5	Operators, Library functions, Data Input and Output Functions, Control Statements.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 2
6	Control structures	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
7	Array	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
8	Pointers and Strings, Arrays of Pointers	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
9	Functions	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO 3

		d) Problem-based class		
10	Structure, Union and Bit-field	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
11	External files.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
12	Data Files, Some Additional Features of C	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
13	Advanced Data types, Access Modifiers, Storage Class Specifiers, Type Conversion in Assignments, Function Type Modifiers, Pointer to Function.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
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CLO1	M	M	M	M		M						
CLO2	M	M	M	M		M						
CLO3	H	H		M		M						
CLO4			H	M								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	√			
Mid Semester2		√	√	
Quiz/Performance	√	√	√	√
Class Attendance	√	√	√	√
Final	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Semester 1 (5)	Mid Semester 2 (5)	Quiz/Performance (5)	Class Attendance (5)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 4 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO1 or Question from CLO1
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO4
5	Question from CLO3 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

- Herbert Schildt, "C: The Complete Reference", Osborne / McGraw-Hill, 4th edition, July 1, 2017.
- Dennis Ritchie Brian W. Kernighan, "The C Programming Language ", Prentice Hall, January 1, 2015.
- Balagurusamy, "Programming in ANSI C", McGraw-Hill Education, January 1, 2011.

Reference Books:

- K. N. King, "C Programming: A Modern Approach", W. W. Norton & Company, 2nd Edition, April 19, 2008.
- Behrouz A. Forouzan, Richard F. Gilberg, "Computer Science: A Structured Programming Approach Using C", Course Technology, 3rd Edition, February 6, 2006.
- Paul J. Deitel, "C How to Program", Prentice Hall, 5th Edition, September 4, 2006.
- Stephen G. Kochan, "Programming in C", Sams, 3rd Edition, July 18, 2004.

CSER-1104: MATH-I (Differential & Integral Calculus)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)			
				Semester Final Exam.	Continuous Assessment		
					Two Mid-Semester Exams	Class Performance (Quiz,	Class Attendance

						Assignment, etc.)		
CSER-1104	MATH-I (Differential & Integral Calculus)	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is a foundational course designed to provide computer science students with a solid understanding of differential and integral calculus concepts and their applications in computer science and related fields. The course emphasizes the development of mathematical techniques essential for analyzing algorithms, modeling data structures, and understanding the principles of optimization.

Justification of this course in the program entity

The course MATH-I (Differential & Integral Calculus) is essential for providing students with foundational mathematical tools crucial for their understanding of complex algorithms, data structures, and optimization techniques in computer science. By mastering calculus concepts, students can model and solve a wide range of computational problems, enhancing their ability to develop efficient software solutions and comprehend the theoretical underpinnings of computation. This course equips students with analytical skills and mathematical fluency vital for success in both academic and professional pursuits within the rapidly evolving field of computer science.

Course Objectives

The major objectives of the course are to:

- i. Develop proficiency in applying differential and integral calculus techniques to solve problems relevant to computer science.
- ii. Understand the mathematical foundations of algorithms and data structures through calculus concepts.
- iii. Gain insight into optimization methods commonly used in computer science.
- iv. Enhance problem-solving skills through mathematical reasoning and abstraction.
- v. Prepare students for advanced courses in computer science and related fields that require mathematical rigor.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: To identify and describe various elementary functions both algebraically and graphically.

CLO2: Understand the basic concepts of limit, continuity, and differentiability, and be able to identify the end behavior of functions. They will also determine the continuity of various types of functions.

CLO3: Illustrate the practical applications of differentiation in real-world problems. They will develop techniques for finding derivatives and apply them to determine maxima, minima, tangents, and normals.

CLO4: Comprehend the concepts of anti-derivatives and indefinite integrals. They will learn different techniques for evaluating integrals and understand the meaning of the definite integral as both a limit of Riemann sums and as the area under a curve.

CLO5: Understand the relationship between differentiation and anti-differentiation as inverse operations. They will develop skills in applying definite integrals to evaluate areas between curves, volumes of solids of revolution, surface areas, and arc lengths using various methods. Additionally, they will be able to draw diagrams in polar coordinates, find tangent curves, and evaluate areas enclosed by polar curves and arc lengths of the curves.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Functions and their Graphs: Polynomials and rational functions, logarithmic and exponential functions, trigonometric functions and their inverses, hyperbolic functions and their inverses, combinations of such functions	• Lecture • Rapport building	3	CLO1
2	Functions and their Graphs: Polynomials and rational functions, logarithmic and exponential functions, trigonometric functions and their inverses, hyperbolic functions and their inverses, combinations of such functions	• Lecture • Rapport building • Demonst ration • Group work	3	CLO1
3	Functions and their Graphs: Polynomials and rational functions, logarithmic and exponential functions, trigonometric functions and their inverses, hyperbolic functions and their inverses, combinations of such functions	• Lecture • Rapport building • Demonst ration • Group work	3	CLO2

4	Limit and Continuity: Definitions and basic theorems on limit and continuity, Properties of continuous function, Limit at infinity and infinite limits and computation, L'Hôpital's Rules, Intermediate value theorem with applications.	• Lecture • Assignment	3	CLO2
5	Limit and Continuity: Definitions and basic theorems on limit and continuity, Properties of continuous function, Limit at infinity and infinite limits and computation, L'Hôpital's Rules, Intermediate value theorem with applications.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2
6	Differentiation: Tangent lines and rates of change, Definition of derivative. One-sided derivatives. Rules of differentiation (proofs and applications). Chain rule theorem. Successive differentiation. Leibnitz theorem (proof and application). Related rates. Linear approximations and differentials.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
7	Differentiation: Tangent lines and rates of change, Definition of derivative. One-sided derivatives. Rules of differentiation (proofs and applications). Chain rule theorem. Successive differentiation. Leibnitz theorem (proof and application). Related rates. Linear approximations and differentials.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
8	Applications of Derivatives: Rolle's theorem, Mean value theorem, Intermediate value theorem for derivation and its applications, Concavity and points of inflection. Maximum and minimum values of functions, Absolute maximum and minimum values of functions, Optimization problems.	• Lecture • Rapport building • Demonstration • Group work	3	CLO4
9	Applications of Derivatives: Rolle's theorem, Mean value theorem, Intermediate value theorem for derivation and its applications, Concavity and points of inflection. Maximum and minimum values of functions, Absolute maximum and	• Lecture • Rapport building • Demonstration • Group	3	CLO4

	minimum values of functions, Optimization problems.	work		
10	Technique Integration: Anti-derivatives and indefinite-integrals, techniques of integration, definite integration using anti-derivatives, definite integration using Riemann sums, fundamental theorems of calculus, basic properties of integration, integration by reduction.	• Lecture • Rapport building • Demonstration • Group work	3	CLO5
11	Technique Integration: Anti-derivatives and indefinite-integrals, techniques of integration, definite integration using anti-derivatives, definite integration using Riemann sums, fundamental theorems of calculus, basic properties of integration, integration by reduction.	• Lecture • Rapport building • Demonstration • Group work	3	CLO5
12	Application of integration: Plane areas, solids of revolutions, volumes by cylindrical shells, volumes by cross-sections, arc length surface area of revolution. Improper integrals, gamma and beta functions, graphing in polar co-ordinates, successive integrations, area and length in polar coordinates.	• Lecture • Rapport building • Demonstration • Group work	3	CLO5
13	Application of integration: Plane areas, solids of revolutions, volumes by cylindrical shells, volumes by cross-sections, arc length surface area of revolution. Improper integrals, gamma and beta functions, graphing in polar co-ordinates, successive integrations, area and length in polar coordinates.	• Lecture • Rapport building • Demonstration • Group work	3	CLO5
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
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CLO1			H							M		
CLO2		H		H							M	
CLO3			H				H					
CLO4			H									
CLO5	H						H					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2

3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Calculus with Analytic Geometry-Howard Anton, latest edition.
2. Differential Calculus-Abdul Matin & Bidhubuson Chakraborty, latest edition.

Reference Books:

1. Calculus – Thomas Finley, Latest edition.

CSER-1105: Physics

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-105	Physics	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course intends to understand fundamental laws of nature and enhance student knowledge to know how the classical

understanding become crucial ingredient in the development of new technology. It covers variety of contents fundamental concepts of physics.

Justification of this course in the program entity

This course intends to understand fundamental laws of nature and enhance student knowledge to know how the classical understanding become crucial ingredient in the development of new technology.

Course Objectives

The major objectives of the course are to:

- i. Understanding Fundamental Principles
- ii. Applying Physics to Computing
- iii. Problem-Solving Skills
- iv. Understanding Physical Limitations
- v. Interdisciplinary Connections

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Understanding Harmonic Motion and Acoustics

CLO2: Mastery of Thermodynamics Principles

CLO3: Understanding Solid State Physics

CLO4: Proficiency in Band Theory and Dielectric Materials

CLO5: Comprehensive Understanding of Optics

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Simple harmonic motion, differential equation of simple harmonic motion, total energy and average energy calculation, Combination of simple harmonic oscillation, Lissajous figures,	Lecture, Learning Video, Problem Based Learning Reading Materials	3	CLO-1

2	spring-mass system, calculation of time period of torsional pendulum, damped oscillation, forced oscillation, resonance,	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-1
3	Wave: basic definitions, Differential equation of a progressive wave, Power and intensity of wave motion, Stationary wave, Group velocity and phase velocity; Architectural acoustics, Reverberation and Sabine's formula.	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-1
4	Heat and Thermodynamics: Kinetic theory of gases: ideal gas equation and its correction, Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy; First law of thermodynamics and its applications;	• Lecture • Learning Video • Problem Based Learning • Reading Materials	4	CLO-2
5	Reversible and irreversible processes, Second law of thermodynamics, Carnot cycle, Efficiency of heat engines, Carnot's theorem, Entropy and disorder; Thermodynamic functions, Third law of thermodynamics.	• Lecture • Learning Video • Problem Based Learning • Reading Materials	4	CLO-2
6	Properties of Matter: Crystalline and non-crystalline solids, Unit cell, Lattice and Basis, Bravais Lattices, Crystal Planes and Miller indices, Interplanar Spacing, Simple crystal structures: NaCl, CsCl,	• Lecture • Learning Video • Problem Based Learning • Reading Materials	5	CLO-3
7	Bragg's Law, methods of determination of crystal structure; Defects in solids: point defects, line defects; Bonds in solids, inter-atomic distances, calculation of cohesive and bonding energy;	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-3
8	Introduction to band theory: distinction between metal, semiconductor and insulator; Dielectric Properties of matter:	• Lecture • Learning Video • Problem Based Learning	5	CLO-4

	meaning of dielectric; Magnetic properties of matter: different types of magnetic materials; Introduction of superconductivity.	• Reading Materials		
9	Physical Optics: Theories of light, Interference of light, Young's double slit experiment; Displacement of fringes and its uses, Fresnel's Bi-prism, Interference at wedge shaped films	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-5
10	Newton's rings, Interferometers; Diffraction of light, Fresnel and Fraunhofer diffraction;	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-5
11	Polarized light, Brewster's law, Malus law, Polarization by double refraction, Retardation plates, Nicol prism, Optical activity, Polarimeters, Polaroid.	• Lecture • Learning Video • Problem Based Learning • Reading Materials	3	CLO-5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H				H				M		M
CLO2	H	H				H				M		M
CLO3	H	H				H				M		M
CLO4	H	H				H				M		M
CLO5	H	H				H				M		M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√

Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will

be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Fundamentals of Physics (Extended, 10th Edition), David Halliday, Robert Resnick, Jearl Walker
2. University Physics with Modern Physics (14th Edition), Hugh D. Young, Roger A. Freedman
3. Physics for Engineers Part-I, Dr. Gias Uddin Ahmed
4. Waves and Oscillations: Brijlal Subramaniam

CSE-1106: Electrical Circuit Analysis

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-1106	Electrical Circuit Analysis	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The course is intended to teach students about the basic electrical components, structures, electrical circuit structures, principles of operation, electrical characteristics, and circuit models and to be able to use this knowledge to analyze and design basic electrical circuits. Also, to extend the understanding of how electrical circuits and their functions fit into larger electrical systems. It covers the fundamental concept in electrical engineering for the intended CSE graduates.

Justification of this course in the program entity

The course is intended to teach students about the basic electrical components, structures, electrical circuit structures, principles of operation, electrical characteristics, and circuit models and to be able to use this knowledge to analyze and design basic electrical circuits. Also, to extend the understanding of how electrical circuits and their functions fit into larger electrical systems.

Course Objectives

The major objectives of the course are to:

- i. **Understanding Fundamental Circuit Principles:** Students will learn the foundational principles of electrical circuits, including Ohm's Law, Kirchhoff's Laws, and circuit element behavior (such as resistors, capacitors, and inductors).
- ii. **Analysis Techniques:** Students will be equipped with various analysis techniques for solving electrical circuits, including nodal analysis, mesh analysis, superposition theorem, Thevenin's theorem, Norton's theorem, and transient analysis.
- iii. **Application of Circuit Theorems:** The course aims to develop students' ability to apply circuit theorems to simplify complex circuits and solve practical engineering problems efficiently.
- iv. **DC and AC Circuits:** The course will cover both direct current (DC) and alternating current (AC) circuits, providing a comprehensive understanding of circuit behavior under different conditions.
- v. **Practical Circuit Design and Troubleshooting:** Students will gain hands-on experience in designing and troubleshooting electrical circuits, enabling them to apply theoretical knowledge to real-world scenarios and engineering projects.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CO1- Understand the basic properties of electrical circuits.

CO2- Apply knowledge to construct different circuits

CO3- Inspect the operation of complex circuits.

CO4- Implement basic electrical circuit's tools

CO5 - Evaluate the integration of electrical circuits within larger systems and demonstrate proficiency in analyzing, designing, and troubleshooting complex electrical systems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Circuit variables and elements: Voltage, current, power, energy, independent and dependent sources, resistance Basic laws: Ohm's law, Kirchhoff's current and voltage laws. Simple resistive circuits: Series and parallel circuits, voltage and current division, Techniques of circuit	● Lecture ● Rapport building	3	CO1

	analysis: Nodal and mesh analysis including supernode and supermesh, KVL and KCL			
2	Network theorems: Source transformation, Thevenin's, with applications in circuits having independent and dependent sources	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO1, CO2
3	Network theorems: Norton's and Superposition theorems with applications in circuits having independent and dependent sources	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO3, CO4, CO5
4	Network theorems: Maximum power transfer condition and reciprocity theorem	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO3, CO4, CO5
5	Energy storage elements: Inductors and capacitors, series parallel combination of inductors and capacitors. Charging and discharging of a capacitor	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO2
6	Magnetic quantities and variables: Flux, permeability and reluctance, magnetic field strength, magnetic potential, flux density, magnetization curve	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO2
7	Laws in magnetic circuits: Ohm's law and Ampere's circuital law. Magnetic circuits: series, parallel and series-parallel circuits.	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO3, CO4
8	Electro-magnetism Induction: Relation between Magnetism and electricity, induced E.M.F and current, Faraday's law, Lenz's	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO3, CO5
9	Sinusoidal functions: Instantaneous current, voltage, power, effective current and voltage, average power, phasors and complex quantities, impedance, real and reactive power, power factor.	● Lecture ● Rapport building ● Demonstration ● Group work	3	CO3, CO4

10	A.C. Fundamentals: Generation and equation of alternating current and voltage, Phase, RMS value, A.C. through Resistance, capacitance and inductance ,complex numbers	• Lecture • Rapport building • Demonstration • Group work	3	CO3
11	Analysis of single phase AC circuits: Series and parallel LR, RC and LRC circuits, nodal and mesh analysis, application of network theorems in AC circuits, circuits with non-sinusoidal excitations	• Lecture • Rapport building • Demonstration • Group work	3	CO3, CO5
12	Resonance in AC circuits: Resonance in RLC circuit, resonance frequency ,graphical representation of resonance, Parallel AC circuits	• Lecture • Rapport building • Demonstration • Group work	3	CO3
13	A.C. Filters: Different types of filters, Low-pass,High-pass, band-pass ,bandstop filtes,	• Lecture • Rapport building • Demonstration • Group work	3	CO3, CO4
14	Revision and Solution	;	2	

Mapping between CLOs and PLOs:

	PL O1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PL O 10	PL O 11	PL O 12
CLO1	H											
CLO2		H	H									
CLO3									H			M
CLO4	H				H				H			M
CLO5							H	H	H	H		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			

Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. B. L. Theraja, “A Text of Electrical Thechology”, Volume-1, S.Chand Publications

Reference Books:

1. Robert L. Boylestad, “Introductory Circuit Analysis”, Prentice Hall, 13th Edition, March 23, 2015.
2. Charles Alexander, Matthew Sadiku, “Fundamentals of Electric Circuits”, McGraw-Hill, 4th Edition, August 27, 2012.
3. William Hayt, Jack Kemmerly, Steven Durbin, “Engineering Circuit Analysis”, McGraw-Hill, 7th Edition, February 20, 2007.

CSEL-1107: Electrical Circuit Analysis Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/ Lab Report	Class Performance	Class Attendance	
CSEL-1107	Electrical Circuit Analysis Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. Electrical Circuit Analysis Lab provides hands-on experience in applying theoretical concepts to real circuits. Students learn to design, build, and analyze electrical circuits using various components and instrumentation. The lab reinforces understanding of circuit principles and techniques through practical experimentation and troubleshooting.

Justification of this course in the program entity

The course is intended to exercise Electrical Circuits laboratory- specially obtaining the characteristics of DC circuits including series and parallel circuits, characteristics of ohms' law, the basic circuit theorems and their applications to design and maintain such circuits.

Course Objectives

The major objectives of the course are:

- i. Practical Application: Enable students to apply theoretical concepts learned in the classroom to real-world electrical circuits.
- ii. Hands-on Experience: Provide students with hands-on experience in designing, building, and testing electrical circuits using lab equipment.
- iii. Troubleshooting Skills: Develop students' ability to troubleshoot and diagnose issues in electrical circuits, enhancing problem-solving skills.
- iv. Instrumentation Proficiency: Familiarize students with the use of instrumentation such as multimeters, oscilloscopes, and function generators for circuit analysis and measurement.
- v. Reinforcement of Theory: Reinforce theoretical concepts through practical experimentation, deepening understanding of circuit behavior and analysis techniques.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1- Understand the basic properties of electrical circuits.

CLO2- Apply knowledge to construct different circuits

CLO3- Inspect the operation of complex circuits.

CLO4- Implement basic electrical circuits tools

CLO5 - Analyze and interpret experimental data obtained from electrical circuits.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Name of the Experiment	Teaching Strategy	Lecture Hours	CLOs
1	Familiarization of electrical equipment, component and accessories	• Demonstration • Discussion • Hand note	2	CO1
2	Verification of Ohm's Law and its application	• MMP • Discussion • Group Work • Demonstration	2	CO1, CO2
3	Verification of KVL for dc circuits.	• Demonstration • Text Book • Group Work	2	CO3, CO4

4	Verification of KCL for dc circuits	• Case Study • Demonstration • Discussion • Hand Note	2	CO3, CO4, CO5
5	Verification of Thevenin's theorem.	• MMP • Discussion • Case Study • Group Work	2	CO2
6	Verification of Norton's theorem	• Demonstration • Discussion • Group Work • WB	2	CO2
7	Verification of Superposition theorem.	• Hand Note • Group work • Demonstration • MMP	2	CO3, CO4, CO5
8	Verification of Maximum power transfer theorem.	• Demonstration • Discussion • Group work • Text Book	2	CO3
9	Familiarization with a Cathode Ray Oscilloscope (CRO) and its application	• Demonstration • Discussion • Hand note	2	CO3, CO4
10	Analysis of RL and RC series circuits.	• MMP • Discussion • Group Work • Demonstration	2	CO3, CO5
11	Verification of RC circuit as a low pass filter.	• Demonstration • Text Book • Group Work	2	CO3
12	Verification of RC circuit as a high pass filter.	• Case Study • Demonstration • Discussion • Hand Note	2	CO3, CO5
13	Analysis of RLC series circuit	• MMP • Discussion • Case Study • Group Work	2	CO3, CO4, CO5
14	Revision and Solution		2	

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H											

CLO2		H	H									
CLO3									H			H
CLO4	H				H				H			H
CLO5						H	H	H	H			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class Attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3

4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. B. L. Theraja, "A Text of Electrical Thechology", Volume-1, S.Chand Publications

Reference Books:

1. Robert L. Boylestad, "Introductory Circuit Analysis", Prentice Hall, 13th Edition, March 23, 2015.
2. Charles Alexander, Matthew Sadiku, "Fundamentals of Electric Circuits", McGraw-Hill, 4th Edition, August 27, 2012.
3. William Hayt, Jack Kemmerly, Steven Durbin, "Engineering Circuit Analysis", McGraw-Hill, 7th Edition, February 20, 2007.

CSER-1108: English

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-108	English	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course is designed to develop accuracy in using English language as well as to use appropriate grammar in proper context. In addition, the course will also discuss various aspects for developing the listening, speaking, reading and writing skills.

Justification of this course in the program entity

The course is intended to teach the basics related to developing the knowledge of the usage of the different aspects of English Language. This course is designed to develop accuracy in using English language as well as to use appropriate grammar in proper context. In addition, the course will also discuss various aspects for developing the listening, speaking, reading and writing skills.

Course Objectives

The major objectives of the courses: are

- i. Technical Communication Skills: Develop proficiency in technical writing and communication tailored to the computer science field, including documentation, reports, and presentations.
- ii. Understanding Professional Jargon: Acquire vocabulary and comprehension of technical terms and jargon commonly used in computer science to effectively communicate with peers and industry professionals.
- iii. Critical Reading and Analysis: Enhance critical reading skills to evaluate technical documents, research papers, and software documentation critically, extracting relevant information and identifying key concepts.
- iv. Cross-Cultural Communication: Develop cross-cultural communication skills to collaborate effectively in multicultural teams, understand diverse perspectives, and communicate with international stakeholders in the global tech industry.
- v. Ethical and Legal Considerations: Explore ethical and legal considerations relevant to computer science, including issues related to intellectual property, privacy, and responsible use of technology, through analysis of case studies and ethical dilemmas.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Develop proficiency in various aspects of the English language usage.

CLO2: Demonstrate proper application of English grammar in language usage.

CLO3: Employ strategies for scriptwriting through conversations and dialogues.

CLO4: Equip students to effectively navigate professional and academic contexts within the engineering domain.

CLO5: Develop comprehensive skills in reading, writing, listening, and speaking in English, fostering proficiency across all language domains crucial for effective communication in academic and professional settings within the field of computer science.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	General discussion: Introduction, various approaches to learning English.	• Lecture • Rapport building	3	CLO-1
2	Grammatical Problems: Construction of sentences, grammatical errors.	• Lecture • Rapport building • Demonstration • Group work	3	CLO1, CLO-2
3	Grammatical Problems: Sentence variety and style, conditionals, vocabulary and diction.	• Lecture • Rapport building • Demonstration • Group work	3	CLO-2
4	Reading Skill: Developing readability, scan and skin reading, generating ideas through purposive reading.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO-3
5	Writing Skill: Principles of effective writing; Organization, Planning and development of writing.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO-3
6	Writing Skill: Composition, precis writing, amplification.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO-3
7	General strategies for the writing process: Generating ideas, identifying audiences and purposes.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO-3
8	General strategies for the writing process: Construction arguments, stating problems, drafting and finalizing.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO-3

9	Approaches to Communication: Business communication, and different types of business communication.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO-4
10	Listening Skill: Strategies for developing listening skills.	• Lecture • Rapport building • Demonstration • Group work	3	CLO-3
11	Speaking Skill: Practicing dialogue; Story telling; Effective oral presentation.	• Lecture • Rapport building • Demonstration • Group work	3	CLO-3
12	Report Writing: Defining a report, classification of reports.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO-4
13	Report Writing: Structure of a report, and writing of reports.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO-4
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1		H										
CLO2						H						
CLO3						H				H		M
CLO4									H	H		M
CLO5	H	H	H						H			H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and

the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Michael Swan, *Practical English Usage*.
2. Wren & Martin, *Practical English Grammar*.
3. Imhoof & Hudson, *From Paragraph to Essay*.

Reference Book:

1. Thomas E. Berry, *Common Mistake in English*
2. Raymond & Murphy, *Intermediate English Grammar*.
3. Mosback & Mosback, *Practical Faster Reading*.
4. Chowdhury & Haq ed. *A Prose of our Time*.

CSE-1201: Object Oriented Programming - I

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment		Total	
					Two Mid-Semester Exams	Class Performance		Class Attendance
CSE-1201	Object Oriented Programming - I	Theory	3	60	20	10	10	100

Course Description: Object Oriented Programming - I is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

Object Oriented Programming –I (C++) stands as a foundational programming language renowned for its efficiency, versatility, and widespread applicability across various domains including software development, systems programming, game development, and scientific computing. By including a C++ course in the curriculum, students not only gain proficiency in a language widely used in industry but also develop critical problem-solving skills, logical thinking abilities, and a deeper understanding of computer science principles. Moreover, as C++ serves as a precursor to other object-oriented languages, mastering it lays a solid foundation for students to adapt to new programming paradigms and languages throughout their academic and professional careers.

Course Objectives

The major objectives of the course are:

- i. To adept students at writing clean, efficient, and maintainable code in C++, mastering its syntax, data types, control structures, and functions.
- ii. To instill a strong grasp of object-oriented programming concepts such as classes, objects, inheritance, polymorphism, and encapsulation. This understanding forms the basis for designing robust and scalable software solutions.
- iii. To develop problem-solving abilities by applying C++ to solve real-world problems. This involves analyzing problems, designing algorithms, and implementing solutions using C++ language constructs.
- iv. To learn about memory management techniques in C++, including dynamic memory allocation and deallocation. They also explore strategies for optimizing code performance, such as using pointers, references, and efficient data structures.
- v. To develop applications using C++, covering topics such as graphical user interface (GUI) programming, file handling, networking, and multithreading. This enables students to create a wide range of software applications, from desktop applications to system-level programs.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: demonstrate a solid understanding of the fundamental concepts of the C++ programming language, including syntax, data types, operators, and control structures.

CLO2: apply object-oriented programming principles, such as encapsulation, inheritance, polymorphism, and abstraction, to design and implement classes and objects in C++.

CLO3: utilize the Standard Template Library (STL) and other standard C++ libraries to implement common data structures (e.g., vectors, lists, maps) and algorithms (e.g., sorting, searching) efficiently.

CLO4: develop the ability to create software applications using C++, including console applications, graphical user interfaces (GUIs) with frameworks like Qt, and applications with file handling, networking, and multithreading capabilities.

CLO5: enhance their problem-solving skills by applying C++ programming techniques to solve algorithmic problems and real-world challenges, demonstrating the ability to analyze problems, design algorithms, and implement solutions efficiently.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introducing C++: Definition of OOP, C++ Console I/O, Introduction to Classes, Basic concept of Object-Oriented Programming, Difference between Structured Programming and Object-Oriented Programming, Difference between C/C++, Introducing Function Overloading, Benefits of OOP, Characteristics of Procedure Oriented Programming, Characteristics of Object-Oriented Programming and Application of Object-Oriented Programming.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
2	Introducing Classes, Arrays, Pointers and References: Constructor and Destructor functions, Constructors with parameters, Introducing Inheritance, Object Pointers, Relation between Classes, Structures and unions.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2

3	In-line functions, Automatic in-line functions, assigning objects, passing objects to functions, returning objects from function.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
4	Defining member functions, Friend functions, Static member functions. Array of objects, Pointer to objects, the pointer, using new and delete, passing references, returning references independent references.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
5	Inheritance: Defining derived classes, Single inheritance, multiple inheritance, multilevel inheritance, Hierarchical inheritance, Virtual base classes.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
6	Constructor and destructor in Inheritance, Constructors in derived classes, nesting of classes	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
7	Polymorphism: Function Overloading and Operator Overloading, Constructor Overloading, Default arguments, Copy	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students,	3	CLO 3

	Constructors.	including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
8	Operator overloading: Introduction, overloading of unary operators, Relational and logical operator overloading.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
9	Operator overloading: Overloading of binary operators, Operator overloading using friend functions, Limitations of operator overloading.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
10	Virtual Functions: Pointers to derived classes, Applying Polymorphism using virtual functions, Polymorphic class, Pure Virtual functions, Abstract classes, early binding, and late binding.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
11	C++ I/O System: Streams, Stream classes, Unformatted I/O, Binary I/O, formatted I/O, I/O manipulators, Inserters, Extractors, File I/O streams, Opening and closing files, Random access files, I/O status checking, Customized	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5

	I/O and files.			
12	Template Exception Handling and Standard Template Library: Generic functions, Generic classes, Components of STL, Container, Algorithms.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
13	Exception handling, throwing mechanism, catching mechanism, Rethrowing mechanism, Specifying exceptions Templates,	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. C++: The Complete Reference, 4th Edition by H. Schildt.
2. C++: How to program. 10th Edition by Deitel H M and Deital P J.

Reference Book

2. The C++ Programming Language, 4th Edition by Bjarne Stroustrup.
3. Object-Oriented Programming with C++, 4th Edition by Robert Lafore.
4. Object Oriented Programming with C++ by E Balagurusamy
5. Programming with C++ by Schaums Outline Series

CSE-1202: Object Oriented Programming – I Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Performance	Lab Assignments	Class Attendance	
CSEL-1202	Object Oriented Programming – I lab	Practical	1	30	10	5	5	50

Course Description: Object Oriented Programming – I Lab is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The

course carries 1 credits and 30 lecture hours. Among the 30 hours, 24 are for lab experiments and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The key rationale of this course is to provide understanding of object-oriented programming using C++ so that they can write programs and develops software accordingly. Its main objective is to teach the basic concepts and techniques which form the object-oriented programming paradigm.

Course Objectives

The major objectives of the course are:

- i. Enable students to apply theoretical concepts learned in lectures to real-world programming scenarios through hands-on lab exercises.
- ii. Familiarize students with the syntax and semantics of the C++ programming language by implementing various programming constructs such as variables, operators, control structures, and functions in practical coding exercises.
- iii. Develop students' proficiency in object-oriented programming principles by implementing classes, objects, inheritance, polymorphism, and encapsulation in C++ programs.
- iv. Introduce students to fundamental data structures such as arrays, linked lists, stacks, queues, trees, graphs, and common algorithms (e.g., sorting, searching, recursion) and guide them in implementing these data structures in C++.
- v. Cultivate students' problem-solving abilities by presenting them with challenging programming tasks and guiding them in devising effective solutions using C++ programming techniques.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: demonstrate proficiency in basic programming concepts such as variables, data types, control structures, and functions through practical exercises and programming assignments.

CLO2: develop a solid understanding of C++ syntax, including language constructs such as loops, conditional statements, arrays, and pointers, through hands-on coding exercises and experimentation.

CLO3: gain practical experience in implementing object-oriented programming principles in C++, including creating classes, defining member functions, implementing inheritance, and utilizing polymorphism.

CLO4: implement fundamental data structures (e.g., arrays, linked lists, stacks, queues) and algorithms (e.g., sorting, searching) in C++, gaining hands-on experience in data structure manipulation and algorithm design.

CLO5: develop the ability to create small-scale software applications using C++, including console-based applications, simple graphical user interfaces (GUIs), and applications with file I/O operations.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction with compiler and installation of the software in Computer. Running a simple program and understanding its basic structure.	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 1
2	Demonstration of Classes and Objects [Sample Program: Write the definition of a class called Grade that stores the information of a student (Name, Roll, Marks, Grade). The class calculates the grade of a student according to the obtained marks (general grading system) using the Grade_Calculation member function. Test the functionality of the class by creating an object of the class.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 2
3	Demonstration of array of objects and static members [Sample Program: Write a program which creates and uses array of objects of a class. Write a static member function named showcount() which has the property of displaying the number of objects created of the class.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 2
4	Demonstration of pointers [Write a program using class to process Shopping list for a departmental store. The list includes details such as the code no, and price of each item and perform the operations like adding,	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 2

	deleting items to the list and printing the Total value of an order.]			
5	Demonstration of friend function [Write a program to find the maximum out of two numbers using friend function. Here one number is a member of one class and the other number is member of some other class.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 3
6	Demonstration of returning objects from functions [Write a program to design a class complex to represent complex numbers. The complex class should use a friend function to add two complex numbers. The function should return an object of type complex representing the sum of two complex numbers.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 3
7	Demonstration of Inheritance [Write a program to design a student class representing student roll no. and a test class (derived class of student) representing the scores of the student in various subjects and sports class representing the score in sports. The sports and test class should be inherited by a result class having the functionality to add the scores and display the final result for a student.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 3
8	Demonstration of this pointer [Write a program to maintain the records of person with details (Name, age) and find the eldest among them. The program must use this pointer to return the result.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 3
9	Demonstration of operator overloading [Write a program to design a class representing complex numbers and having the functionality of performing addition and multiplication of two complex numbers using operator	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 4

	overloading.]			
10	Demonstration of copy constructor [Write a program to highlight the difference between overloaded assignment operator and copy constructor.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 4
11	Demonstration of String IO [Write a program to implement I/O operation on characters. An I/O operation includes inputting a string, calculating length of the string, storing the string in a file, fetching the stored characters from it, etc.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 5
12	Demonstration of Files [Write a program to maintain an elementary database of employees using files.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 5
13	Demonstration of Exception Handling [Write a program to implement exception handling with multiple catch statements.]	a) Problem Based Learning (PBL) b) Group Assignment	3	CLO 5
14	Demonstration of Templates [Write a program to implement Stack and its operations using Template class.]	a) Problem Based Learning (PBL) b) Group Assignment	3	

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report	√	√	√	√	√
Viva/Quiz	√	√	√	√	√
Lab Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains weekly problem based lab tasks carrying 20% marks, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to solve one/two problems. The complete solution of the problem with demonstration contains 45% marks whereas viva/question answering related to the problem contains 25% marks. The marks of final examination and continuous assessment are added, and converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. C++: The Complete Reference, 4th Edition by H. Schildt.
2. C++: How to program. 10th Edition by Deitel H M and Deital P J.

Reference Book

1. The C++ Programming Language, 4th Edition by Bjarne Stroustrup.
2. Object-Oriented Programming with C++, 4th Edition by Robert Lafore.
3. Object Oriented Programming with C++ by E Balagurusamy
4. Programming with C++ by Schaums Outline Series

CSE-1203: Data Structures

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-1203	Data Structures	Theory	3	60	20	10	10	100

Course Description: The Data Structures course is designed to provide students with a comprehensive understanding of fundamental data structures and their applications in computer science. The course covers various abstract data types and the algorithms associated with them. Students will learn to analyze the efficiency and effectiveness of different data structures in solving computational problems.

The course begins with an introduction to basic concepts such as arrays, linked lists, stacks, and queues. Students will understand the principles of abstraction and encapsulation in designing data structures. Through a series of lectures, practical exercises, and programming assignments, students will gain hands-on experience implementing these data structures in a programming language such as Java or C++.

As the course progresses, more advanced topics will be covered, including trees, binary search trees, heaps, graphs, and hash tables. Emphasis will be placed on understanding the underlying principles behind these data structures, their operations, and their applications in solving real-world problems.

Students will also explore algorithms for searching, sorting, and traversing various data structures, as well as techniques for analyzing their time and space complexity. The course will incorporate practical examples and case studies to demonstrate the practical relevance of data structures in software development.

By the end of the course, students will have developed a strong foundation in data structures and algorithms, enabling them to design efficient and scalable solutions to a wide range of computational problems. They will also be equipped with the analytical skills necessary to evaluate and choose the appropriate data structure for a given problem domain.

Prerequisites:

Proficiency in programming fundamentals

Understanding of basic computer science concepts such as variables, control structures, and functions

Justification of this course in the program entity

The "Data Structures" course serves as a crucial component in the program entity, providing students with essential knowledge and skills vital for success in computer science and software development fields. Here's a detailed justification for including this course:

- i. **Foundation of Problem Solving:** Data structures form the backbone of efficient problem-solving techniques in computer science. Understanding and mastering data structures is fundamental for developing algorithms and designing software solutions. By offering this course, students develop a strong foundation in problem-solving skills, enabling them to tackle complex computational problems effectively.
- ii. **Optimized Resource Management:** Efficient data structures are essential for optimizing the use of computational resources such as time and memory. Through this course, students learn various data structures such as arrays, linked lists, trees, and graphs, along with algorithms for their manipulation. This knowledge equips them to choose and implement the most suitable data structure for specific problem domains, leading to optimized resource management in software development.
- iii. **Algorithmic Thinking:** Data structures are closely intertwined with algorithms. By studying data structures, students naturally delve into algorithm design and analysis. They learn various algorithms for searching, sorting, and traversing data structures, along with techniques for evaluating their efficiency. This fosters algorithmic thinking, a critical skill for devising efficient solutions to computational problems.
- iv. **Enhanced Problem-Solving Skills:** Mastery of data structures enhances students' problem-solving abilities. They learn to decompose complex problems into simpler components, identify appropriate data structures and algorithms, and implement efficient solutions. This iterative problem-solving approach nurtures logical thinking, critical

reasoning, and creativity, essential attributes for success in the rapidly evolving field of computer science.

- v. Preparation for Advanced Courses and Real-World Applications: Proficiency in data structures is essential for advanced courses in computer science, such as algorithms, databases, and systems programming. Additionally, data structures are extensively used in real-world software development scenarios, including web development, mobile app development, and systems programming. By completing this course, students are well-prepared to tackle advanced coursework and excel in practical software development projects.
- vi. Industry Relevance: Data structures are fundamental to software engineering and are highly valued by industry employers. Proficiency in data structures is often a prerequisite for job roles in software development, data analysis, and system design. Therefore, offering a comprehensive data structures course aligns with industry demands and enhances students' employability prospects upon graduation.

In summary, the inclusion of the "Data Structures" course in the program entity is justified as it equips students with essential problem-solving skills, algorithmic thinking abilities, and industry-relevant knowledge necessary for success in computer science and software development careers.

Course Objectives

The major objectives of the course are:

- i. Understand the fundamental principles and characteristics of various data structures such as arrays, linked lists, stacks, queues, trees, graphs, and hash tables.
- ii. Implement and manipulate common data structures using a programming language such as C, Python, Java or C++, including insertion, deletion, searching, and sorting operations.
- iii. Analyze the time and space complexity of algorithms associated with different data structures to evaluate their efficiency and suitability for solving specific computational problems.
- iv. Design and develop efficient algorithms and techniques for solving advanced programming challenges by leveraging the power of data structures.
- v. Apply data structures and algorithms to real-world problems in software development, demonstrating the ability to select and implement appropriate data structures based on problem requirements and constraints.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Apply advanced programming concepts of logical structures of data as well as discuss various types of sorting and searching techniques.

CLO-2: Analyze the time complexity and examine various algorithms from this perspective.

CLO-3: Design and Develop algorithms for manipulating stacks, queues, linked lists, hash tables, trees, and graphs.

CLO-4: Create and evaluate real-time software that specifies the required algorithm and technologies.

CLO5: Demonstrate proficiency in implementing and utilizing recursive algorithms for solving complex computational problems related to data structures.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Overview of data structure, Execution Time Cases	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1
2	Preliminaries	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO2
3	Data structure array	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
4	Algorithm	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based	3	CLO1, CLO2

		lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
5	Data structure linked list	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
6	Header List, Two-Way list, List operations	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
7	Data Structure STACK, Queue	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
8	Data structure searching, Hasing	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1, CLO2, CLO3

		b) Problem-based class		
9	Data structure sorting	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO2, CLO3
10	Graph Data Structure	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO2, CLO3
11	Tree Structure	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
12	Special trees: Binary SearchTree, Heap, Huffman Tree	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO3
13	Recursion Basics, Tower of Hanoi, Fibonacci Series	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing	3	CLO1, CLO3, CLO5

		for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
14	Revision and Solution		3	CLO1, CLO2, CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H	H	H	H	M	H	M	H	M	M	H	H
CLO2	H	H	H	H	M	H	M	H	M	M	H	H
CLO3	H	H	H	H	M	H	M	H	M	M	H	H
CLO4	H	H	H	H	M	H	M	H	M	M	H	H
CLO5	H	M	M	M	H	M	M	M	M	M	H	H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Seymour Lipschutz, "Data Structure Schaum Series", Seymour Lipschutz Manual, 5th addition, 2019.

Reference Books:

1. Jon Kleinberg, Eva Tardos, "Algorithm Design", Kindle Edition, 1st Edition, 2019.
2. Douglas B West., "Introduction to Graph Theory", Prentice Hall, 2nd Edition, 2018.

3. Narasimha Karumanchi, "Data Structures and Algorithms Made Easy", Paperback, 3rd Edition, 2018.

CSE-1204: Data Structures Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSEL-1204	Data Structures Lab	Lab	1.5	30	10	5	5	50

Course Description: The Data Structures Lab is designed to complement the theoretical knowledge gained in the Data Structures course by providing hands-on experience in implementing and analyzing various data structures and algorithms. Students will work on programming assignments and practical exercises to reinforce concepts learned in lectures. The lab will cover the implementation of common data structures such as arrays, linked lists, stacks, queues, trees, and graphs, as well as algorithms for their manipulation and analysis. Through lab sessions, students will develop practical programming skills and gain proficiency in using data structures to solve computational problems.

Justification of this course in the program entity

The Data Structures Lab is an essential component of the program entity as it offers students practical experience in applying theoretical concepts learned in the Data Structures course. By engaging in hands-on programming tasks, students deepen their understanding of data structures and algorithms, enhance their problem-solving abilities, and strengthen their programming skills. This lab provides a bridge between theory and practice, preparing students for real-world software development scenarios and advanced coursework in computer science.

Course Objectives

The major objectives of the lab are:

- i. Implement and manipulate common data structures such as arrays, linked lists, stacks, queues, trees, and graphs using a programming language such as C, Python, Java, or C++.
- ii. Analyze the time and space complexity of algorithms associated with different data structures and evaluate their efficiency.
- iii. Develop practical programming skills by working on programming assignments and practical exercises related to data structures and algorithms.
- iv. Demonstrate proficiency in using data structures to solve computational problems and validate solutions through experimentation and analysis.

- v. Prepare comprehensive lab reports documenting the implementation, analysis, and evaluation of data structures and algorithms.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Implement common data structures and algorithms using a programming language.

CLO-2: Analyze the time and space complexity of algorithms and evaluate their efficiency.

CLO-3: Develop practical programming skills by working on lab assignments and exercises.

CLO-4: Apply data structures to solve computational problems and validate solutions through experimentation and analysis.

CLO-5: Prepare comprehensive lab reports documenting the implementation, analysis, and evaluation of data structures and algorithms.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction to Lab Environment	a) Demonstration, Hands-on	3	-
2	Implementation of Arrays	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3
3	Implementation of Linked Lists	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3
4	Implementation of Stacks and Queues	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3
5	Implementation of Trees	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3
6	Implementation of Graphs	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3
7	Sorting Algorithms	a) Hands-on, Lab Assignment	3	CLO-1, CLO-2, CLO-3
8	Searching Algorithms	a) Hands-on, Lab Assignment	3	CLO-1, CLO-2, CLO-3
9	Algorithm Analysis and Complexity Analysis	a) Hands-on, Lab Assignment	3	CLO-2, CLO-4

10	Project Work	a) Hands-on, Lab Assignment	3	CLO-1, CLO-3, CLO-4
11	Viva Voce	a) Viva Voce Assessment	3	CLO-1, CLO-2, CLO-3
12	Revision and Solution		3	CLO-1, CLO-2, CLO-3
13	Lab Report Preparation	a) Presentation, Evaluation	3	CLO-5
14	Presentation of Lab Reports	a) Presentation, Evaluation	3	CLO-5

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H	H	H	H	M	H	M	H	M	M	H	H
CLO2	H	H	H	H	M	H	M	H	M	M	H	H
CLO3	H	H	H	H	M	H	M	H	M	M	H	H
CLO4	H	H	H	H	M	H	M	H	M	M	H	H
CLO5	H	M	M	M	H	M	M	M	M	M	H	H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40% Marks):

Bloom's Category	Mid Semester 1 (10%)	Mid Semester 2 (10%)	Quiz/Performance (10%)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		

Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60% Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

2. JAVA How to Program, 10th Edition by Deitel and Deitel, Publisher: Prentice/Hall International.
3. Introduction to Java Programming, 10th Edition by Y. Daniel Liang, Publisher: Pearson.

Reference Books:

1. Java The Complete Reference, 9th Edition by Herbert Schildt.
2. The Java Programming Language, 4th Edition by Ken Arnold, James Gosling, David Holmes.
3. Java, Java, Java™: Object-Oriented Problem Solving, 3rd Edition by Ralph Morelli, Ralph Walde, Publisher: Prentice Hall.
4. Programming with JAVA, 5th Edition by E. Balagurusamy, Publisher: McGraw-Hill INDIA.

CSE-1205: Basic Electronics

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit	Evaluation (Marks Distribution)
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			Point	Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semest er Exams	Class Performance	Class Attendance	
CSE-12 05	Basic Electronics	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course delves into the understanding of electronic components such as resistors, capacitors, and transistors, elucidating their roles within circuits. Through theoretical concepts and practical experiments, students gain insight into circuit analysis techniques, paving the way for comprehending the inner workings of electronic devices encountered in everyday life.

Justification of this course in the program entity

The course is intended to know electronic devices- Specially details about semiconductor materials, different types of semiconductor devices with their applications, switching devices with their characteristics and different amplifiers in order to design efficient switching devices for power control, to design various converters for various applications and some efficient amplifiers for signal processing units.

Course Objectives

The major objectives of the course are to:

- Understanding Electronic Components: Familiarize students with the properties and functions of basic electronic components such as resistors, capacitors, diodes, and transistors.
- Circuit Analysis Techniques: Introduce students to fundamental circuit analysis methods, including Ohm's Law, Kirchhoff's Laws, and nodal/mesh analysis.
- Application Knowledge: Enable students to apply electronic principles to analyze and design simple electronic circuits, fostering problem-solving skills in circuit construction and troubleshooting.
- Hands-on Experience: Provide practical laboratory sessions to reinforce theoretical concepts, allowing students to build, test, and modify electronic circuits using laboratory equipment.
- Real-world Applications: Illustrate the relevance of basic electronics in various fields, such as consumer electronics, telecommunications, and automation, preparing students for further studies or entry-level positions in related industries.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CO1: Identify features of different materials and their applications

CO2: Illustrate the structure and characteristics of semiconductor and power devices such as PN-Junction diode, FET, MOSFET, SCR, TRIAC, DIAC, UJT, Op-Amp, amplifiers.

CO3: Inspect the operation of complex circuits.

CO4: Implement basic electronics circuits tools

CO5: Analyze the performance and behavior of electronic circuits under varying conditions and design solutions for optimization and improvement

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Semiconductors Physics: Discuss Bohr's model, Atomic Structure, identify problems in electronics world, Explain and classify semiconductor material P-N Junction diode: Define p type and n type material, Describe doping and its effect on semiconductor materials, Discuss the biasing conditions and characteristics of semiconductor diode, Define p type and n type material P-N Junction diode Describe doping and its effect on semiconductor materials, Discuss the biasing conditions and characteristics of semiconductor diode	• Lecture • Rapport building	3	CLO1
2	Practical Usages of Diodes: Design clipper, clamper, rectifier and voltage multipliers for various applications, identify Zener diode and its specialty in voltage regulator circuit, Discuss tunnel diode, Schottky diode and photo diode	• Lecture • Rapport building • Demonstration • Group work	3	CLO1, CLO2
3	Regulated Power Supply: Rectifiers – operation and efficiency, half-wave and full-wave rectifiers, Ripple factor	• Lecture • Rapport building • Demonstration	3	CLO3, CLO4, CLO5
4	Bipolar Junction Transistor (BJT): Explain the operation and characteristics of BJT, Configure and bias BJT in different configurations, Draw DC and AC equivalent circuit of transistors, Design switching circuits using BJT	• Lecture • Rapport building • Demonstration	3	CLO3, CLO4

5	Transistor Amplifier Circuits: Discuss the small signal model of BJT, Determine the h-parameter of transistor, Calculate input resistance, output resistance, voltage gain and current gain of transistor	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO5
6	Field Effect Transistor (FET):Describe the operation and characteristics of JFET, Analyze the small signal model of JFET, Design the switching circuit using JFET,MOSFET	• Lecture • Rapport building • Demonstration • Group work	3	CLO2
7	Feedback Amplifier: negative and positive feedback, characteristics and gain with negative voltage and current feedback	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4
8	Op-amps Basic Op-amps- characteristics, inverting, non-inverting, integrators, differentiators, summing amplifiers.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
9	Single-stage Transistor Amplifier: Different types amplifier, operations of Class A, Class B, Class C amplifier, Power gain	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4
10	Multistage and Feedback Transistor: Amplifier Coupling, Two-stage amplifier, RC coupling, frequency response, direct-coupled two-stage amplifier, Darlington pair, emitter follower	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO5
11	Power Electronics: Operations, characteristics and applications of industrial electronics devices: SCR (Silicon Controlled Rectifier), TRIAC, DIAC, UJT (Unijunction Transistor)	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
12	Oscillators: Theory of oscillation, Tuned collector oscillator, Wien Bridge oscillator, Colpitt's oscillator, Hartley oscillator, Phase shift oscillator, Quartz crystal oscillator.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3

13	Electronic Instruments: analog and digital instruments , DC voltmeter, DC ammeter, Ac voltmeter, AC ammeter, CRO,CRT,GPIB instruments	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4, CLO5
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H											
CLO2		H	H									
CLO3									M			M
CLO4	H				H				M			M
CLO5						H	H			M	M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. B. L. Theraja, "A Text of Electrical Thechology", Volume-IV, S.Chand Publications

Reference Books :

1. Robert L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuit Theory", Prentice Hall, 10th Edition, July 31, 2008.
2. David A. Bell, "Electronic Devices and Circuits", Oxford University Press, 5th Edition, April 30, 2008.
3. V.K. Mehta, Rohit Mehta, "Principle of Electronics", S. Chand, 11th Edition, 2007.

CSEL-1206: Basic Electronics Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)

				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessme nt/Lab Report	Class Performance	Class Attendance	
CSEL-1 206	Basic Electronics Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It offers hands-on experience in applying theoretical concepts to real electronic circuits. Students engage in practical experiments to reinforce understanding of components, circuits, and troubleshooting techniques.

Justification of this course in the program entity

The course is intended to exercise Electronics laboratory- specially obtaining the characteristics of semiconductor diode, BJT, MOSFET, Op-amp along with their applications to design and maintain such circuits and in turn maintain equipment having such circuits.

Course Objectives

The major objectives of the course are:

- Practical Application: Enable students to apply theoretical knowledge of basic electronics principles to construct and analyze circuits in a laboratory setting.
- Component Familiarization: Familiarize students with electronic components such as resistors, capacitors, diodes, and transistors through hands-on experimentation.
- Circuit Construction Skills: Develop students' skills in circuit construction, soldering, and breadboarding, enhancing their ability to build functioning electronic circuits.
- Troubleshooting Proficiency: Cultivate students' ability to troubleshoot and diagnose issues within electronic circuits, honing their problem-solving skills.
- Measurement Techniques: Introduce students to measurement techniques using laboratory equipment such as multimeters and oscilloscopes, enabling accurate analysis and characterization of electronic circuits.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Recognize various semiconductor devices used in the lab.

CLO2: Examine the operation of different circuits such as clipping, clamping, rectifier circuits, BJT, MOSFET, and amplifiers.

CLO3: Apply knowledge to design amplifier circuits in the lab.

CLO4: Implement basic electronic circuit tools.

CLO5: Evaluate the performance of electronic circuits in laboratory settings, analyze data collected from experiments, and propose modifications to improve circuit functionality and efficiency.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Name of the Experiment	Teaching Strategy	Lecture Hours	CLOs
1	Analysis the forward and reverse characteristics of a P-N diode	• Demonstration • Discussion • Hand note	2	CLO1
2	Verification of Zener diode as voltage regulator with variable voltage or variable resistance	• Demonstration • Discussion • Hand note	2	CLO2
3	Verification of half-wave rectifier with a single diode	• MMP • Discussion • Group Work • Demonstration	2	CLO1, CLO2
4	Verification of full-wave rectifier.	• Demonstration • Text Book • Group Work	2	CO3, CO4
5	Verification of Common Emitter transistor amplifier	• Case Study • Demonstration • Discussion • Hand Note	2	CLO3, CLO4
6	Design of clipper and clamper networks	• Demonstration • Discussion • Hand Note	2	CO3, CO4
7	Verification of Op-Amp inverting amplifier using IC 741	• MMP • Discussion • Case Study • Group Work	2	CLO2, CLO5
8	Verification of Op-Amp summing amplifier using IC 741	• Demonstration • Discussion • Group Work • WB	2	CLO2
9	Verification of Op-Amp subtractor amplifier using IC 741	• Hand Note • Group work • Demonstration • MMP	2	CLO3, CLO4

10	Verification of Op-Amp differentiator using IC 741	• Demonstration • Discussion • Group work • Text Book	2	CLO3,CLO 5
11	Verification of Op-Amp integrator using IC 741.	• Demonstration • Discussion • Hand note	2	CLO3, CLO4
12	Characterizing of NMOS or PMOS	• MMP • Discussion • Group Work • Demonstration	2	CLO3,CLO 5
13	Analyze transfer characteristics of enhancement type MOSFET	• Demonstration • Text Book • Group Work	2	CLO3
14	Revision and Solution	Interactive, Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PL O 10	PL O 11	PL O 12
CLO 1	H											
CLO 2		H	H									
CLO 3												M
CLO 4	H				H							M
CLO 5								H		H		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class Attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. B. L. Theraja, "A Text of Electrical Thechology", Volume-IV, S.Chand Publications

Reference Books :

1. Robert L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuit Theory", Prentice Hall, 10th Edition, July 31, 2008.

2. David A. Bell, "Electronic Devices and Circuits", Oxford University Press, 5th Edition, April 30, 2008.
3. V.K. Mehta, Rohit Mehta, "Principle of Electronics", S. Chand, 11th Edition, 2007.

CSER-1207: MATH-II (Linear Algebra)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-1207	MATH-II (Linear Algebra)	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is the study of linear systems of equations, vector spaces, and linear transformations. Solving systems of linear equations is a basic tool of many mathematical procedures used for solving problems in science and engineering. This course will give an overview of linear algebra including the topics of vectors in R_n and C_n , system of linear equations, vector spaces, basis and dimension and the linear transformation. Some applications of linear algebra will be discussed. Also, real-world problems will be analyzed and solved.

Justification of this course in the program entity

It plays a crucial role in computer science, serving as a fundamental tool for modeling and solving a wide range of computational problems. In computer graphics, linear algebra techniques are used to manipulate images and generate visual effects. In machine learning and data analysis, linear algebra is essential for understanding algorithms, representing data, and performing dimensionality reduction. Additionally, linear algebra underpins the theory and implementation of many algorithms in areas such as computer vision, cryptography, and network analysis. Thus, a solid understanding of linear algebra is indispensable for computer science students, enabling them to develop efficient algorithms, analyze data structures, and excel in various computational fields.

Course Objectives

The major objectives of the course are to:

- i. Understand the fundamental concepts and structures of linear algebra.
- ii. Develop proficiency in performing operations with vectors and matrices.

- iii. Learn techniques for solving systems of linear equations and computing determinants.
- iv. Explore applications of linear algebra in various fields, including computer science, engineering, and physics.
- v. Gain insight into the geometric interpretations of linear transformations and eigenvalues.
- vi. Prepare students for advanced courses in mathematics and related disciplines that require linear algebra knowledge.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Define vector spaces and the vectors in R^n and C^n with basic properties and determine the distance between two points in R^n and C^n ;

CLO2: Convert a linear system of equations in matrix form and identify homogeneous and non-homogeneous systems. Determine the row rank, column rank, nullity and rank of matrices;

CLO3: Determine the linear combinations of vectors, linear span, linear independence and dependence. Define and identify the basis and dimension of vector spaces and explain the row space, column space and null spaces;

CLO4: Describe kernel and image of linear transformation and their properties and find the rank and nullity of linear transformation. Define Eigen values and Eigen vectors, Caley- Hamilton theorem;

CLO5: Understand and apply the concepts of orthogonalization, orthogonal complements, and orthogonal projections in vector spaces. Describe the Gram-Schmidt process and utilize it to find orthogonal bases. Additionally, demonstrate proficiency in solving least squares problems using orthogonal projections.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Vectors in R^n and C^n: Review of geometric vectors on R^2 and R^3 space. vectors in R^n and C^n .	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials,	3	CLO1

		question-answer sessions, etc. b) Problem-based class		
2	Inner product spaces: Inner product, norm and distance of vectors in $\mathbb{R}^n$ and $\mathbb{C}^n$.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO1, CLO2
3	Matrices and Determinants: Notion of matrix, types of matrices, matrix operations, minors, cofactors, expansion and evaluation of determinants, elementary row and column operations.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3, CLO4
4	Determinants: Determinant function, properties of determinants, minors, cofactors, expansion and evaluation of determinants, elementary row and column operations.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3, CLO4
5	System of Linear Equations: Linear equations, system of linear	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint	3	CLO2

	equations (homogeneous and non-homogeneous).	presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
6	System of Linear Equations: Gaussian elimination method for solving system of linear equations.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO2, CLO5
7	Vector Spaces: Notions of groups and fields, abstract vector space, subspace, sum and direct sum of subspaces.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3, CLO4
8	Vector Spaces: Linear combination, linear dependence, independence of vectors, basis and dimension of a vector space.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials,	3	CLO3

		question-answer sessions, etc. b) Problem-based class		
9	Vector Spaces: Row and column space of a matrix, rank of matrices, solution spaces of systems of linear equations.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3, CLO4
10	Linear Transformations: Linear transformation, kernel and image of a linear transformation and their properties.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3
11	Linear Transformations: Matrix representation of linear transformations, change of bases.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3
12	Eigenvalues and Eigenvectors: Eigenvalues	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint	3	CLO5

	and eigenvectors, diagonalization.	presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
13	Eigenvalues and Eigenvectors: Cayley Hamilton theorem, application of Cayley Hamilton theorem.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO3, CLO4
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H							M		
CLO2		H		H							M	
CLO3			H				H					
CLO4			H									
CLO5		H		H								M

PART C: ASSESSMENT AND EVALUATION**Mapping/Alignment of CLOs vs Assessment Strategy:**

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Linear Algebra with Applications- Howard Anton, latest edition.

Reference Book:

1. Linear Algebra-Gilbert Story, latest edition.

CSE-1208: Discrete Mathematics

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-1208	Discrete Mathematics	Theory	3	60	20	10	10	100

Course Description: Discrete Mathematics is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The

course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course builds up the students' ability to think and express logically and mathematically. The course will address mathematical reasoning, combinatorial analysis, algorithmic thinking, and discrete structures. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE 1201 Objected Oriented Programming, CSE1203 Data Structures, and CSE 2208 Algorithms.

Course Objectives:

The major objectives of the course are:

1. Interpret and apply propositional logic, predicate logic, and theorem proving for mathematical reasoning.
2. Interpret and apply counting, permutations, and combinations for combinatorial analysis.
3. Interpret and apply the growth of functions, complexity analysis of algorithms, and integer algorithms for algorithmic thinking; demonstrate this knowledge and write report for realistic problem solving.
4. Interpret and apply discrete structures such as sets, functions, relations, graphs, and trees for modeling discrete objects; demonstrate this knowledge and write report for realistic problem solving.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding Propositional Logic, Logical Equivalence, Predicates & Quantifiers and Logical Reasoning. Basics, Set Operations, One-To-One, Onto, Inverse, Composition, Graphs

CLO2: Illustration to achieve usability of Greatest Common Divisor, Euclidean Algorithm. Sequences and Summations.

CLO3: Learning of various Mathematical Reasoning and Induction, Proof Strategies, Mathematical Induction, Recursive Definitions, Structural Induction

CLO4: Understanding concept Counting, Basic Rules, Pigeonhole Principle, Permutations and Combinations, Binomial Coefficients and Pascal Triangle, Probability: Discrete Probability. Expected Values and Variance

CLO5: Learning of Graph Theory: Definition, Graph Representation, Theorem, Types of Graphs, Directed, Undirected Graphs, Sub Graphs, Isomorphic graphs, Bridge and Cut Point, Distance Matrix.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
2	Rules of Inference, Introduction to Proofs, Mathematical Induction.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
3	State Function and relation, Compare function and relation	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
4	Describe different types of function such as one to one, on to function.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
5	State Logic, Describe different types of set, Develop operation of sets, Demonstrate Venn Diagram related math	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over	3	CLO1

		lecture materials, question-answer sessions, etc.		
6	Integers: Greatest Common Divisor, Euclidean Algorithm, Sequences and Summations.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO2
7	Mathematical Reasoning and Induction: Proof Strategies	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO3
8	Mathematical Induction, Recursive Definitions, Structural Induction	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO3
9	Counting: Basic Rules, Pigeonhole Principle	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO4
10	Permutations and Combinations, Binomial Coefficients and Pascal Triangle.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over	3	CLO4

		lecture materials, question-answer sessions, etc.		
11	Probability: Discrete Probability. Expected Values and Variance.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO4
12	Graph Theory: Definition, Graph Representation, Theorem	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO5
13	Types of Graphs, Directed, Undirected Graphs, Sub Graphs, Isomorphic graphs, Bridge and Cut Point, Distance Matrix.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

CLOs	Program Outcomes											
	PO-1	PO-	P	PO-4	PO-	PO	PO-	PO-	PO-	PO-	PO-	PO-
CLO1		M										
CLO2		M										
CLO3			M									
CLO4				M								
CLO5				M								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√				
Mid Semester2		√	√		
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- c. Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- d. Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1

2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Susanna S. Epp, “Discrete Mathematics with Applications”, Cengage Learning, 5th Edition, January 01, 2019.
2. Kenneth H. Rosen, “Discrete Mathematics and its Applications”, McGraw-Hill, Inc., 8th Edition, August 10, 2018.

Reference Books:

1. Tremblay J. P. and Manohar R., “Discrete Mathematical Structures with applications to Computer Science”, Tata Mc Graw Hill Publishing Co., 35th Edition, 2008, Re-print: January 22, 2018.
2. Narsing Deo, “Graph Theory with applications to Engineering and Computer science”, Prentice-Hall of India pvt. Ltd., September, 2015.
3. C.L. Liu, “Elements of Discrete Mathematics”, McGraw Higher ED, 2nd Edition, July 16, 2012.

CSER-1209: Economics

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class Attendance	
CSER-1209	Economics	Theory	2	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is the study of how societies allocate scarce resources to satisfy unlimited wants and needs. It examines production, distribution, and consumption of goods and services, as well as the behavior of individuals, firms, and governments in making economic decisions. Key concepts include supply and demand, market structures, inflation, unemployment, fiscal and monetary policies, international trade, and economic growth. Economics helps understand and analyze various issues such as poverty, inequality, environmental sustainability, and the functioning of markets and economies on both micro and macro levels.

Justification of this course in the program entity

The course is intended to teach the basics related to economics. This course is designed to reflect theoretical learning as well as methods and techniques necessary to understand and utilize the economic terms and methods in both real life and in computer science. In addition, the course will also discuss various aspects of developing tools and digital systems related to economic aspects.

Course Objectives

The major objectives of the course are:

- I. Understand the fundamental principles of microeconomics and macroeconomics to analyze economic behavior and decision-making within the context of computer science applications.
- II. Apply economic theories and models to evaluate the impact of technological advancements, such as automation and digitalization, on markets, industries, and society.
- III. Explore the relationship between information technology and economic development, including the role of innovation, entrepreneurship, and intellectual property rights.
- IV. Analyze the economic implications of digital platforms, online markets, and data-driven business models, considering issues like market power, network effects, and privacy concerns.
- V. Develop quantitative skills to interpret economic data, conduct economic analysis using computational tools, and make informed decisions related to technology investment, project management, and resource allocation in the realm of computer science.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understand core concepts of economics.

CLO2: Understand and acquire knowledge of various economic terms and methods.

CLO3: Design, implement, and analyze tools and digital systems related to economic aspects.

CLO4: Analyze the impact of technological advancements on economic systems.

CLO5: Apply economic principles to inform strategic decision-making in computer science.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction: Definition, Microeconomics vs. macroeconomics, scope of economics, meaning of economic theory, some basic concepts- product, commodity, want, need utility, consumption, factors of production.	• Lecture • Rapport building	2	CLO-1
2	Demand: Law of demand, factors determining demand, shifts in demand, demand functions, deriving demand curves, substitution and income effects.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-1 , CLO-2
3	Demand: Deriving aggregate demands, various concepts of demand elasticity and measurements, discussion on the method of estimating demand functions and demand forecasting.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2
4	Supply: Law of supply and supply function, determination of supply, shifts in supply, elasticity of supply, market equilibrium.		2	CLO-2 , CLO-3
5	Economic Theory of Consumer Behavior: reasons for consumption, Principle of diminishing marginal utility, indifference curves, budget constraint, utility maximization and consumer equilibrium.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2
6	Consumer Demand: change in budget constraints, price consumption curve, income consumption curve, consumer demand, market demand, Engel curve.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2 , CLO-3

7	Production: Production functions, total, average and marginal products, law of diminishing marginal products, production isoquants, marginal rate of technical substitution (MRTS).	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2 , CLO-3
8	Production: Optimal combination of inputs, expansion path, returns to scale, estimation of production function and estimation of cost function.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2 , CLO-3
9	Cost: Concepts of cost, fixed cost, variable cost, total cost, marginal cost, average cost, relation between average cost and marginal cost. short-run costs.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2 , CLO-3
10	Cost: Relation between short-run costs and production, long run costs, economies and diseconomies of scale, relation between short run and long run costs, cost function and estimation of cost function.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-4 , CLO-5
11	Markets and Revenue: Meaning of market, different forms of market, concepts of total, average and marginal revenue.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-4 , CLO-5
12	Markets and Revenue: Relation between average revenue and marginal revenue curves, relation between different revenues and elasticity of demand, equilibrium of the firm.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-4 , CLO-5
13	Price and Output: Price and output determination under perfect competition, monopoly, monopolistic competition and oligopoly, profit maximization, price discrimination, plant shut down decision, barriers to entry.	• Lecture • Rapport building • Demonstration • Group work	2	CLO-2 , CLO-3
14	Revision and Problem Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1							H					
CLO2						H	H					
CLO3				H	H						H	H
CLO4		H										
CLO5	H								H			H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class Attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will assess the final questions tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. P. A Samuelson, Economics.
2. John Sloman, Economics.
3. Koutsoyannis, Modern Microeconomics.
4. R.G. Lipsy, An Introduction to Positive Economics.

Reference Books:

1. Stainlake and Grant, Introductory Economics.
2. N. Gregory Mankin, Principles of Economics

CSE-2101: Object Oriented Programming - II

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-2101	Object Oriented Programming - II	Theory	3	60	20	10	10	100

Course Description: Object Oriented Programming - II is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the Java programming language. It aims to develop the students logical, critical, and problem-solving skills on programming basics and also on the modern context of object-oriented programming through exercises and hands-on activities. Java is popular among professional programmers because it can be used to build visually interesting graphical user interface (GUI) and Web-based applications. This course familiarizes students with basic as well as advanced features of Java Programming Emphasis will be given to GUI and event-driven programming, Database Connectivity, Socket Programming, Servlets and JSP Technology, and Distributed Programming. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- vi. understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
- vii. make themselves able to write a computer program to solve specified problems.
- viii. use the Java SDK environment to create, debug and run simple Java programs.
- ix. build themselves as a professional developer in Java.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding Object-Oriented programming concepts using basic syntaxes of control Structures, strings for developing skills of logic building activity using Java

CLO2: Identification of classes, objects, members of a class and the relationships among them needed for a finding the solution to specific problem

CLO3: Illustration to achieve reusability using inheritance, interfaces, and packages and describes faster application development can be achieved with exception handling mechanisms

CLO4: Understanding concept of multithreading for robust faster and efficient application development and applications of collection interfaces in Java

CLO5: Learning of various I/O operations, GUI -programming, connecting Java with databases using JDBC and implementation of networking with Java

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction to Java: History of Java, Java Class Libraries, Introduction to Java Programming, A simple Program, Memory Concepts, Decision Making.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
2	Class and Object, Class Methods, Declaring Methods with multiple parameters.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
3	Methods: Introduction to Program Module in Java, Static Methods, Math Class Methods, Java API Packages.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
4	Strings, Arrays and Methods	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students,	3	CLO 2

		including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
5	Abstraction, Method Overloading, Method Overriding	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
6	Inheritance: Introduction to Superclasses and Subclasses, Protected Members, Using Constructors and Finalizers in Subclasses, Composition vs. Inheritance,	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
7	Polymorphism: Introduction to polymorphism, Dynamic method building, Final Methods and Classes.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
8	Interfaces	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
9	Multithreading	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for	3	CLO 4

		discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
10	Exceptions, Try and Catch blocks	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
11	GUI Components: Overview of Swing Components, JOptionPane	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
12	Introduction to Event Handling with Nested Classes, Common GUI Event Types and Listener Interfaces, Mouse Event Handling,	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
13	Socket Programming	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- a. Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- b. Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. JAVA How to Program, 10th Edition by Deitel and Deitel, Publisher: Prentice/Hall International.
2. Introduction to Java Programming, 10th Edition by Y. Daniel Liang, Publisher: Pearson.

Reference Books:

1. Java The Complete Reference, 9th Edition by Herbert Schildt.
2. The Java Programming Language, 4th Edition by Ken Arnold, James Gosling, David Holmes.
3. Java, Java, Java™: Object-Oriented Problem Solving, 3rd Edition by Ralph Morelli, Ralph Walde, Publisher: Prentice Hall.
4. Programming with JAVA, 5th Edition by E. Balagurusamy, Publisher: McGraw-Hill INDIA.

CSEL-2102: Object Oriented Programming – II Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Continuous Evaluation	Class Participation	Class attendance	

CSEL-210 2	Object Oriented Programming – II Lab	Practical	1.0	30	10	5	5	50
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Course Description: Object Oriented Programming - II Lab is a compulsory course for the B.Sc. (Honours) program in the Department of Computer Science and Engineering, Jagannath University. The course is designed for the students to give practical knowledge about how to apply different features of Java programming language in developing interactive applications. The course carries 1.0 credits and 30 lecture hours. Among the 30 hours, 24 are for computer based practical implementation and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the Java programming language. It aims to develop the students logical, critical, and problem-solving skills on programming basics and also on the modern context of object-oriented programming through exercises and hands-on activities. Java is popular among professional programmers because it can be used to build visually interesting graphical user interface (GUI) and Web-based applications. This course familiarizes students with basic as well as advanced features of Java Programming Emphasis will be given to GUI and event-driven programming, Database Connectivity, Socket Programming, Servlets and JSP Technology, and Distributed Programming. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- i. understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
- ii. make themselves able to write a computer program to solve specified problems.
- iii. use the Java SDK environment to create, debug and run simple Java programs.
- iv. build themselves as a professional developer in Java.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Analyze the various important features of a pure object-oriented programming language (Java).

CLO2: Use a development environment to design, code, test, and debug simple programs, including multi-file in object-oriented programming language.

CLO3: Implement the key features of object oriented programming concepts using Java language.

CLO4: Know in preparing a professional looking package for each business project using java advance programming features such as GUI design, exception handling and multi-threading.

CLO5: Create a complete real world software solution by applying the knowledge of various features of java language

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Demonstration of Compiler and Classes and Objects [Write a Program to define a class, describe its constructor, and instantiate its object.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2
2	Demonstration of Java Math Library and Java String Library [Write a Java program to practice and explore java String and Math library class and its member methods.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
3	Demonstration of Object passing and return [Write a Java Program to develop a class for Rational numbers that implement object passing and returning objects mechanism using member methods.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
4	Demonstration of Array of Objects [Write a Java Program to instantiate objects of a java class using an array of objects.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
5	Demonstration of Polymorphism [Write a Program to implement runtime polymorphism in constructors]	Problem Based Learning (PBL)	2	CLO1, CLO2, CLO3, CLO4

	and member methods in a java class]	Group Assignment		
6	Processing Graphical User Inputs using JOptionPane [Write a simple addition application that uses two input dialogs to obtain integers from the user and a message dialog to display the sum of the integers the user enters. Use JOptionPane Dialog boxes to take your input from the user and display the output.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
7	Demonstration of Inheritance [Write a program that creates a vehicle class hierarchy in Java to demonstrate the concept of Inheritance.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
8	Demonstration of Abstract Class and Interfaces [Write a java program to create an abstract class named shape that contains two integers and an empty method named print Area(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class shape. Each one of the class contains only the method print Area() that print the area of the given shape.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
9	Demonstration of Thread [Write a Java program that implements a multi-threaded program has three threads. First thread generates a random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd the third thread will print the value of cube of	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4

	the number.]			
10	Demonstration of Files [Write a program that loads names and phone numbers from a text file where the data is organized as one line per record and each field in a record are separated by a tab (\t). It takes a name 2or phone number as input and prints the corresponding other value from the file.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
11	Demonstration of Exceptions [Write a program that creates a user interface to perform integer divisions. The user enters two numbers in the textfields, Num1 and Num2. The division of Num1 and Num2 is displayed in the Result field when the Divide button is clicked. If Num1 or Num2 were not an integer, the program would throw NumberFormatException. If Num2 were Zero, the program would throw an ArithmeticException. Display the exception in a message dialog box.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
12	Demonstration of Swing [Write a program that creates a scientific calculator using even-driven programming paradigm of Java.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
13	Demonstration of JDBC [Write a program that creates a simple library management system using event-driven and concurrent programming paradigms and java database connectivity.]	Problem Based Learning (PBL) Group Assignment	2	CLO1, CLO2, CLO3, CLO4
14	Demonstration of Socket Programming [Write a program that creates a Multi-threaded Echo Server	Problem Based	2	CLO1, CLO2, CLO3, CLO4

	and Client in Java.]	Learning (PBL) Group Assignment		
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Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO 10	PLO 11	PLO 12
CLO1	H				H							
CLO2						H						
CLO3			H									
CLO4							H				H	
CLO5											H	H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report	√	√	√	√	√
Viva/Quiz	√	√	√	√	√
Lab Performance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Lab Report (5)	Quiz (5)	Lab Performance (5)	Attendance (5)
Remember				
Understand		25%	25%	
Apply	25%	40%	40%	
Analyze	40%	25%	25%	
Evaluate	25%	10%	10%	
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks
Remember	5

Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains weekly problem based lab tasks carrying 20% marks, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to solve one/two problems. The complete solution of the problem with demonstration contains 45% marks whereas viva/question answering related to the problem contains 25% marks. The marks of final examination and continuous assessment are added, and converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. JAVA How to Program, 10th Edition by Deitel and Deitel, Publisher: Prentice/Hall International.
2. Introduction to Java Programming, 10th Edition by Y. Daniel Liang, Publisher: Pearson.

Reference Books:

1. Java The Complete Reference, 9th Edition by Herbert Schildt.
2. The Java Programming Language, 4th Edition by Ken Arnold, James Gosling, David Holmes.
3. Java, Java, Java™: Object-Oriented Problem Solving, 3rd Edition by Ralph Morelli, Ralph Walde, Publisher: Prentice Hall.
4. Programming with JAVA, 5th Edition by E. Balagurusamy, Publisher: McGraw-Hill INDIA.

CSE-2103: Digital Logic Design

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment		Total	
					Two Mid-Semester Exams	Class Performance		Class Attendance
CSE-2103	Digital Logic Design	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It focuses on the study of digital circuits and systems. It covers fundamental concepts such as Boolean algebra, logic gates,

combinational and sequential logic, flip-flops, registers, and basic computer architecture. Students learn to design, analyze, and implement digital circuits using hardware description languages and simulation tools. The course provides the essential groundwork for understanding the inner workings of modern digital systems, including processors, memory units, and peripherals.

Justification of this course in the program entity

Digital Logic Design plays a crucial role in computer science by providing the fundamental building blocks for designing and understanding the inner workings of digital systems. In computer science, almost every aspect of computing, from basic operations to complex algorithms, relies on digital circuits. Understanding digital logic allows computer scientists to design efficient and reliable hardware systems, such as processors, memory units, and input/output devices. Moreover, digital logic concepts form the basis for understanding computer architecture, which is essential for optimizing system performance and developing innovative computing technologies. Additionally, proficiency in digital logic design enables computer scientists to develop and troubleshoot software more effectively by providing insights into how hardware components interact at the lowest level.

Course Objectives

The major objectives of the course are to:

- i. Grasp basic digital concepts.
- ii. Design combinational logic circuits.
- iii. Analyze sequential logic circuits.
- iv. Utilize Hardware Description Languages (HDLs).
- v. Apply digital logic to computer architecture.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Understand core concepts of Boolean algebra and Minimization.

CLO2: Understand the Logic Gates, Combinational Circuits, and Flip-Flops.

CLO3: Design and implement MSI Logic Circuit and Counter and Register.

CLO4: Understand the theoretical knowledge of Integrated-Circuit Logic Families and Memory Devices.

CLO5: Analyze and troubleshoot digital circuits to identify and rectify errors effectively.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hours	CLOs

1	Number system & Codes: Introductory concepts of Binary, Octal and Hexadecimal number systems & their arithmetic operations, conversion of different number systems	• Lecture • Rapport building	2	CLO1
2	Codes: BCD, ASCII and EBCDIC codes, Data representation.	• Lecture • Rapport building	3	CLO1
3	Boolean Algebra and Logic Gates: AND, OR, NOT, NAND, NOR, EXOR, EX-NOR. Basic theorem and properties of Boolean Algebra	• Lecture • Rapport building • Demonstration • Group work	3	CLO1, CLO2
4	Logic Circuit simplification: Boolean algebra, De Morgan's theorem, truth-table	• L Lecture	3	CLO1, CLO2
5	Product of sums and Sum of product simplification	• Lecture	3	CLO1, CLO2
6	Simplification of Boolean Functions: K-map techniques, K-map with don't care condition	• Lecture • Group work • Exercise	3	CLO1, CLO2
7	Combinational Logic: The half adder and full adder, The half subtractor and full subtractor, Parallel adders, IC parallel adders	• Lecture	3	CLO2, CLO3
8	MSI Logic Circuits: Different types of Encoder, Different types of decoder (BCD to decimal, BCD-to-7-segment decoder/drivers), BCD to Decimal Decoders	• Lecture	3	CLO3, CLO4
9	MSI Logic Circuits: Multiplexer and their application, De-Multiplexer and their application. Flip-Flops: SR, JK, D and T flip-flops. The D latch.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO5
10	Flip-Flops: Master-slave FF. Flip-flop applications. FF synchronization. Data storage and transfer. Frequency division and	• Lecture • Rapport building • Demonstration	3	CLO3, CLO5

	counting. Register: Different types of Register, Shift-register.	• Group work		
11	Counters Introduction, synchronous and asynchronous ripple - up and down counters Parallel - up, down and up/down counters, Ring counter, Johnson Counter, Counter applications: frequency counter, digital clock.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO5
12	Integrated-Circuit Logic Families: Digital IC terminologies, TTL logic family, TTL series characteristics, open-collector TTL, tristate TTL, ECL family, MOS digital ICs, MOSFET, CMOS characteristics, CMOS tristate logic	• Lecture • Rapport building • Demonstration • Group work	3	CLO4
13	Memory Devices: Memory technology, general memory operation, semiconductor memory technologies, different types of ROMs, semiconductor RAMs, static and Dynamic RAMs.	• Lecture • Rapport building • Demonstration • Group work	3	CLO4
14	Revision and Solution	Interactive Session	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H						H			
CLO2		H	H						H			
CLO3			H			H			H			
CLO4	H				H				H			M
CLO5		H										H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
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Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination,

students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. T.J. Tocci, Principle of Digital Electronics, latest edition.

Reference Books:

1. Digital Logic Design by Moris Manno, latest edition.
2. Digital Fundamentals by Floyd, latest edition.
3. R.P. Jain, Modern Digital Electronics, latest edition.

CSEL-2104: Digital Logic Design Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class Attendance	
CSEL-2104	Digital Logic Design Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It offers students a hands-on exploration of fundamental principles in digital circuitry. Through a series of structured exercises, students delve into the construction, analysis, and optimization of digital logic circuits. The course covers topics ranging from Boolean algebra and logic gates to sequential and combinational circuits. Utilizing industry-standard software tools and physical hardware components, students gain proficiency in designing and simulating complex digital systems. This practical approach enhances comprehension of abstract concepts and fosters critical thinking skills essential for navigating the intricacies of modern computing technology.

Justification of this course in the program entity

The Digital Logic Design Lab course in computer science provides students with hands-on experience in understanding and implementing fundamental concepts of digital logic circuits.

Through practical exercises, students learn how to design, simulate, and analyze digital circuits using industry-standard tools and hardware components. This course is essential as it forms the foundation for understanding computer architecture, microprocessor design, and hardware-software interfacing.

Course Objectives

The major objectives of the course are:

- i. Understanding basic digital logic concepts.
- ii. Designing combinational circuits.
- iii. Analyzing sequential circuits.
- iv. Utilizing design tools and simulation software.
- v. Applying digital logic to real-world problems.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Capture basic concept on the components of Digital Logic Design with the design and analyze combinational logic circuits, design and analyzes sequential logic circuits.

CLO2: Design and implementation of Digital Logic Circuits using ICs of basic logic gates and some simple digital logic circuits.

CLO3: Design and implement MSI Logic Circuit and Counter and Register

CLO4: Understand and implement Integrated-Circuit Logic Families and Memory Device

CLO5: Consolidate and apply their understanding of basic components, logic gates, MSI circuits, and memory devices to solve intricate problems in digital logic design.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Study of Basic Logic Gates: AND, OR, NOT, NOR, NAND, X-OR, X-NOR	• Lecture • Lab performance test	2	CLO2
1	Study of Basic Logic Gates X-OR, X-NOR	• Lecture, • Lab performance test	2	CLO2
2	Study of Universal NAND and NOR Gate	• Lecture, • Lab performance test	2	CLO3
2	Verification of De Morgan's theorem and Logic circuits	• Lecture, • Lab performance test	2	CLO1
3	REVIEW and TEST	• Lab performance test	2	

3	Study of X-OR and X-NOR gate using other basic gates(AND, OR , NOT)	- Lecture - Lab performance test	2	CLO1
4	Study of Half Adder & Full Adder Circuit	- Lecture - Lab performance test	2	CLO2
4	Study of Half Subtractor & Full Subtractor Circuit	- Lecture - Lab performance test	2	CLO2
5	SOP and PLOS implementation	- Lecture, - Lab performance test	2	CLO2
5	K-map implementation	- Lecture, - Lab performance test	2	CLO2
6	Design and Implementation of Encoder	- Lecture, - Lab performance test	2	CLO2
6	Design and Implementation of Decoder	- Lecture, - Lab performance test	2	CLO2
7	Test & Verification of S – R , J – K, T, D Flip-Flop circuit	- Lecture, - Lab performance test	2	CLO3
7	Test & Verification of T, D Flip-Flop circuit	- Lecture, - Lab performance test	2	CLO3
8	REVIEW and TEST	Lab performance test	2	CLO1,CLO2,CL O3
9	Design and Implementation of MUX	- Lecture, - Lab performance test	2	CLO5
9	Design and Implementation of DEMUX	- Lecture, - Lab performance test	2	CLO5
10	7 segment display	- Lecture, - Lab performance test	2	CLO5
11	Design and Implementation of Asynchronous counter	- Lecture, - Lab performance test	2	CLO4
12	Design and Implementation of Synchronous Up & Down counter.	- Lecture, - Lab performance test	2	CLO4
13	Design and Implementation of A counter BCD counter.	- Lecture, - Lab performance test	2	CLO4
14	Review	Discussion	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H						H			

CLO2		H	H						H			
CLO3			H			H			H			
CLO4	H				H				H			M
CLO5				H							M	M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2

3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

2. T.J. Tocci, Principle of Digital Electronics, latest edition.

Reference Books:

4. Digital Logic Design by Moris Manno, latest edition.
5. Digital Fundamentals by Floyd, latest edition.
6. R.P. Jain, Modern Digital Electronics, latest edition.

CSER-2105: Math- III (Ordinary differential Equation)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-2105	Math- III (Ordinary differential Equation)	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. A differential equation is a mathematical equation for an unknown function of one or several variables that relates the values of the function itself and its derivatives of various orders. Real-life problems occur in many areas of engineering, mathematical & physical sciences, and other applied sciences that can be

modeled by differential equations. For example oscillation of a simple pendulum representing the equation of motion, bending equation in applied mechanics, chemical reaction equations, moon lander model equation in control and optimization theory, Lotka-Volterra equations in biology, and Stellar structure models in astrophysics, etc.

Justification of this course in the program entity

It is essential for computer science students as it provides a solid mathematical foundation for modeling and analyzing dynamic systems encountered in computational fields. By studying ODEs, students gain valuable skills in solving complex problems, understanding system behavior, and applying mathematical concepts to real-world scenarios. This course equips students with tools to tackle challenges in various domains such as population dynamics, mechanical systems, and electrical circuits, laying the groundwork for advanced studies in computational mathematics and related disciplines.

Course Objectives

The major objectives of the course are to:

- i. Develop a thorough understanding of ordinary differential equations (ODEs) and their role in modeling dynamic systems encountered in computer science applications.
- ii. Equip students with the ability to solve first-order and higher-order ODEs using analytical, numerical, and qualitative methods.
- iii. Foster proficiency in analyzing the behavior of solutions, including stability analysis and phase space analysis, to gain insights into system dynamics.
- iv. Explore applications of ODEs in computer science, including modeling population dynamics, mechanical systems, electrical circuits, and chemical reactions, to develop practical problem-solving skills.
- v. Cultivate computational skills through hands-on exercises and projects, enabling students to implement ODE-solving techniques using software tools commonly used in computer science.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Understand the fundamentals of ordinary differential equations (ODEs) and their relevance in modeling dynamic systems within computational domains.

CLO2: Apply analytical, numerical, and qualitative methods to solve first and higher-order ODEs encountered in computer science scenarios.

CLO3: Analyze the stability of solutions and conduct phase space analysis to interpret the behavior of dynamic systems modeled by ODEs.

CLO4: Implement ODE modeling techniques in practical applications such as population dynamics and electrical circuits, demonstrating the ability to translate mathematical concepts into computational solutions.

CLO5: Utilize theoretical knowledge, problem-solving skills, and computational techniques to address complex problems involving ODEs, enhancing proficiency in mathematical modeling and computational problem-solving in computer science.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Ordinary differential equations and their solutions: Classification of differential equations, Solutions, Implicit solutions, Singular solutions.	Lecture, Examples, Discussion	3	CLO1
2	Ordinary differential equations and their solutions: Initial value problems, Boundary value problems.	Lecture, Examples, Problem-solving	3	CLO1, CLO2
3	Ordinary differential equations and their solutions: Basic existence and uniqueness (statement and illustration only), Direction fields, Phase line.	Lecture, Visual Aids, Discussion	3	CLO1, CLO3
4	Solution of first-order equations: Separable equations and equations reducible to this form.	Lecture, Examples, Problem-solving	3	CLO2
5	Solution of first-order equations: Linear differential equations, exact differential equations.	Lecture, Examples, Problem-solving	3	CLO2
6	Solution of first-order equations: Special integrating factors, Substitutions and transformations.	Lecture, Examples, Problem-solving	3	CLO2
7	Modeling with first-order differential equations: Construction of differential equations as mathematical models (exponential growth and decay).	Lecture, Examples, Discussion	3	CLO4

8	Modeling with first-order differential equations: Beating and cooling, mixture of solutions, series circuit, logistic growth, chemical reaction, falling bodies.	Lecture, Examples, Problem-solving	3	CLO4
9	Modeling with first-order differential equations: Model solutions and interpretation of results. orthogonal trajectories.	Lecture, Examples, Discussion	3	CLO4
10	Solutions of higher-order linear differential equations: Linear differential operators. Basic theory of linear differential equations.	Lecture, Examples, Problem-solving	3	CLO2, CLO5
11	Solutions of higher-order linear differential equations: Solution space of homogeneous systems. Reduction of order. Homogeneous linear equations with constant coefficient.	Lecture, Examples, Problem-solving	3	CLO2, CLO5
12	Solutions of higher-order linear differential equations: Non-homogeneous equation. Method of undetermined coefficient. Variation of parameters. Euler-Cauchy differential equations.	Lecture, Examples, Problem-solving	3	CLO2, CLO5
13	Revision and Solution	Review session, Problem-solving	2	CLO1, CLO2, CLO3, CLO4, CLO5
14	Revision and Solution	Review session, Problem-solving	2	CLO1, CLO2, CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H		H								M
CLO2			H									

CLO3			H	H								
CLO4	H	H	H									
CLO5			H	H								M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1

2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. D G Zik-Ordinary Differential Equations, latest edition.
2. B D Sharma- Ordinary Differential Equations, latest edition.

Reference Book:

1. S. L. Ross-Ordinary Differential Equations, latest edition.

CSER-2106: Probability and Statistics for Computer Science

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSER-2106	Probability and Statistics for Computer Science	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course introduces students to

the fundamental principles of probability theory and statistical methods essential for applications in computer science. Students will develop a solid understanding of probability concepts, including probability distributions, random variables, and basic combinatorial analysis. Emphasis will be placed on the application of probability theory to solve problems in computer science, such as analyzing algorithms, modeling data structures, and understanding randomness in computational processes. Additionally, students will learn various statistical techniques for data analysis, including descriptive statistics, hypothesis testing, and regression analysis. Through hands-on exercises and real-world applications, students will gain practical skills in implementing statistical methods using computational tools and programming languages commonly used in computer science. The course aims to equip students with the knowledge and skills necessary to make informed decisions, analyze complex datasets, and develop robust computational solutions in various domains of computer science.

Justification of this course in the program entity

The course is designed to develop students' knowledge about basic statistical methods. The course introduces descriptive statistics, correlation and regression analysis, basic rules of probability, concept of random variable and measuring its probability, probability models, and their interpretations.

Course Objectives

The major objectives of the course are to:

- i. Understand Probability Fundamentals: Gain a solid understanding of basic probability concepts, including probability distributions, random variables, and combinatorial analysis.
- ii. Apply Probability in Computer Science: Learn how to apply probability theory to analyze algorithms, model data structures, and understand randomness in computational processes.
- iii. Develop Statistical Analysis Skills: Acquire practical skills in descriptive statistics, hypothesis testing, and regression analysis for analyzing data sets relevant to computer science applications.
- iv. Utilize Computational Tools: Learn to use computational tools and programming languages to implement statistical methods and analyze data effectively.
- v. Enhance Decision-Making Abilities: Develop the ability to make informed decisions based on statistical analysis, and apply statistical reasoning to solve problems encountered in various domains of computer science.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Develop an understanding of the concepts of statistical methods and recognize their significance in various contexts.

CLO2: Demonstrate comprehension of concepts related to data collection, processing, presentation, analysis, and interpretation.

CLO3: Gain proficiency in understanding the concepts of correlation and regression analysis to analyze relationships between multiple variables effectively.

CLO4: Acquire knowledge of events, their probabilities, and mathematical descriptions of discrete and continuous random variables and their probability functions to solve statistical probability problems.

CLO5: Enhance understanding of central measures of random variables and probability models for effective statistical modeling and analysis.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Definitions of statistics, characteristics, Limitations, Classification, Functions of statistics, Scope and application of statistics.	• Lecture • Rapport building	2	CLO1
2	Variables and Data: Variables, Attributes, Measurement Scales, Meaning of data, Sources of statistical data, Primary and secondary sources of data, Questionnaire and its construction..	• Lecture • Rapport building • Demonstration • Group work	3	CLO1, CLO2
3	Presentation of Data: Classification, Frequency distribution. Graphical presentation of data, Details of different types of graphs and charts with their relative merits and demerits. Concept of explorative data analysis, Stem-and-leaf plot.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2
4	Descriptive Measures: Measures of Central Tendency: Mean, median, mode and quantiles. Definition, properties, uses, limitations and relationship.	• Lecture • Rapport building	3	CLO2, CLO3
5	Absolute and relative measures; Definition, properties, uses and limitations. Moments, skewness and kurtosis. The five number summary, Box-and-whiskers plot.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO3

6	Relationship between Variables: Concepts of bivariate data, Scatter diagram. Simple correlation, Types of correlation. Correlation Coefficients: Pearson and Spearman. Assumptions, properties, applications and limitations of correlation coefficient. Concepts, properties, uses, applications	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO3
7	Simple linear regression: Concepts, lines of regression, method of least squares (principles of least squares, fitting of linear regression lines, properties of OLS estimators and estimation of parameters). Properties of regression coefficient.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO3
8	Basic Concepts of Probability: Random Experiment, sample space, events, union and intersection of events, different types of events, interpretations of probability.	• Lecture • Rapport building • Demonstration • Group work	3	CLO2, CLO3
9	Basic Concepts of Probability: Conditional probability, independence and dependence, theorem of total probabilities, Bayes' theorem and its uses and importance in statistics	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4
10	Random variables: Discrete and continuous random variables, probability mass function, probability density function, Distribution function and its properties.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
11	Random variables: Bivariate random variable, joint distribution, marginal and conditional distributions, independence of random variables	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
12	Mathematical expectation: expectation of a function, properties of expectations, expectations of sum and products of random variables, Variance, properties of variance, conditional expectation and conditional variance	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4

13	Discrete and continuous distributions: Detailed study of binomial, Poisson, geometric, normal, uniform and exponential distributions.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	H										
CLO2	M	H										
CLO3	M	H										
CLO4	M	H										
CLO5	H		H									

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
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Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Rahman M. S. (2022): Statistiuics and Probability – An Introductory Analysis, Kazi Prokashoni, 38 Banglabazar, Dhaka.

Reference Book

1. Weiss N. A., (2012): Introductory Statistics, 9th edition, Pearson Education, Inc
2. Sheldon M. Ross (2010): Introduction to Probability Models, 10th edition, Elsevier Book aid international, Sabre foundation.
3. Islam, M. N. (2011): An Introduction to Statistics and Probability, Mullick & Brothers, Dhaka.

CSE-2107: Data Communication

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Attendance	Class Performance	
CSE-2107	Data Communication	Theory	3	60	20	10	10	100

Course Description: Data Communication is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 48 hours, 42 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

To make an effective communication between the gadgets it is critical to play out every single essential apparatus and their fundamental design precisely. Therefore, to comprehend the system structure, segments and other applicable devices and procedure is an absolute necessity. This course enables the student to understand the operation of the components in a data communication network and the functional relationships of these components. The student will gain the ability to analyze, design, and improve data communication networks both wired and wireless.

Course Objectives

The major objectives of the course are:

- to interpret the components, tools and techniques of data communication systems.
- to enable the student to understand the operation of the components in a data communication network and the functional relationships of these components.
- to gain the ability to analyze, design, and improve data communication networks.
- to manage the operations of data communication networks.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Identify the various definitions for data communication, how they are used in the communications industry, different types of communications media and networking protocol.

CLO2: Examine the data and signal through transmission medium and line coding scheme.

CLO3: Learn Different Modulation, demodulation, multiplexing, spread spectrum techniques.

CLO4: Learn different LAN topologies and different components of a LAN.

CLO5: Understand in detail the characteristics of wireless and mobile network.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Explain the concept of data communication, describe the components and standards of data communication, Justify the necessities of network models.	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 1
2.	Learn detailed description of layers and TCP/IP protocol.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
3.	Classify the transmission media, Investigate the characteristics of various transmission media, Examine the data and signal through transmission medium, Identify the causes to signal impairments during transmission, determine the capacity and data rate of the channel.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
4.	Understand line configuration, learn different communication modes, learn different direction of flow.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over	3	CLO 2

		lecture materials, question-answer sessions, etc. b) Problem-based class		
5.	Identify the reasons of errors, analyze the type of errors, apply error detection and correction methods used in data transmission, modify existing methods for detection and correction errors.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
6.	Learn flow control approaches, Data link layer protocols for noiseless and noisy channel, distinguish between various encoding schemes and decide which one is better for which situation, working principle of Pulse Code Modulation.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 2
7.	Learn different analog and digital modulation techniques: AM, FM, PM, ASK, FSK, PSK, understand different types of serial and parallel transmission.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
8.	Classify multiplexing techniques, understand different types of multiplexing techniques, differentiate among different multiplexing techniques, learn switching Techniques.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3

9.	Learn basic concept of Spread Spectrum Techniques: CDMA, TDMA, FDMA.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
10.	LANs: identify the hardware/ in a LAN, identify the topology in a LAN, identify access methods in a LAN.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
11.	Identify the NOS used in a LAN. define gateway, bridge, and router in the context of a LAN.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
12.	Wireless links, characteristics, IEEE 802.11 wireless LANs ("Wi-Fi"), Cellular Internet Access: Architecture, Standards.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 5
13	Mobility, Principles: addressing and routing to mobile users, Mobile IP, handling mobility in cellular networks.	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students,	3	CLO 5

		including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class		
14	Revision and Solutions.		3	

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7
CLO1	H	H					
CLO2	H	H		M			
CLO3	H	M	M	H			
CLO4	H		M		H		
CLO5	H		H				

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Class Performance (10)	Class Attendance (10)	Other forms
Remember		25%			
Understand	50%	50%			
Apply	25%	25%			
Analyze	25%				
Evaluate					
Create					

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10

Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows:

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Behrouz A. Forouzan, Data Communications and Networking Computer Architecture.

Reference Books:

1. Haykin, Communication Systems.
2. William Stallings, Data and Computer Communications.
3. Computer Networking: A Top-Down Approach Featuring the Internet, Jim Kurose and Keith Ross, Addison Wesley, 4th Edition

CSEL-2108: Data Communication Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Attendance	Class Performance	

CSEL-2108	Data Communication Lab	Lab	1	30	10	5	5	50
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Course Description: Data Communication Laboratory is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 1 credit and 30 lecture hours. Among the 30 hours, 24 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Course Objectives:

The objectives of the course are as follows:

1. to develop basic concepts and principles about practical tools used in data communication lab.
2. to create, implement, debug, and evaluate algorithms and techniques for solving substantial problems.
3. to deliver and evaluate basic technical documents, presentations, and group interactions using appropriate tools.

Course Learning Outcomes (CLOs):

After completing the course, successful students will be able to:

CLO1: to develop understanding of MATLAB and CISCO packet tracer simulator environment.

CLO2: to develop understanding various signal properties, how to generate the signals, perform basic signal operations and generate network topologies.

CLO3: to digitize analog signal and data link layer framing.

CLO4: to demonstrate various modulation techniques and Line coding schemes.

CLO5: to implement various error detection and correction schemes and multiplexing techniques.

PART B: CONTENTS OF THE COURSE

Course description and alignment of course content with CLOs:

Course Contents	Lecture Hour	CLOs
1. Learn the basics of MATLAB, develop understanding of MATLAB environment, commands and syntax, understand the use of MATLAB for solving Data Communication problems.	24 Hours	CLO1
2. Learn how to use Cisco Packet Tracer Simulator and develop understanding of Packet Tracer Environment.		CLO2
		CLO3
		CLO4
3. Learn the basics of signals and their properties and learn to		CLO5

generate different types of analog and digital signal. 4. Learn some basic signal operations. 5. Learn and implement different network topologies. 6. Learn how to sample and quantize an analog signal. 7. Learn the data link layer framing method: Character stuffing and bit stuffing. 8. Learn to demonstrate basic analog modulation technique. 9. Learn to demonstrate basic digital modulation technique. 10. Learn and implement various Line encoding schemes in data communication. 11. Learn and implement error detection algorithms. 12. Learn and implement error control and correction methods. 13. Learn and implement different multiplexing techniques. 14. Revision and Solutions.		
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Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7
CLO1				H			
CLO2	H	H	H	H	H	H	
CLO3	H	M	H	H	M	H	
CLO4	H		H	H	H	H	
CLO5	H		H	H		H	

Teaching Strategy:

The following methods will be used to teach this course

- a) Lab work with supervision
- b) Assignment, viva and presentation

Mapping/Alignment of Course outline vs Teaching Strategy:

Part of course outline	Teaching Strategy (a)	Teaching Strategy (b)
1.	√	√
2.	√	√

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of Course outline vs Teaching Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√	√	√	√
Mid Semester 2	√	√	√	√	√
Class Attendance	√	√	√	√	√
Class Performance	√	√	√	√	√
Final	√	√	√	√	√

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. Among 60% of the total marks, 40% carries experiments and 30% is allotted for viva-voce. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class attendance carries 10% marks. In the final examination, students have to perform four/five experiments. The marks of final examination and continuous assessment are added, and converted to the letter grade.

PART D: LEARNING RESOURCES**Text Book:**

1. Behrouz A. Forouzan, Data Communications and Networking

Reference Books:

5. Stephen J. Chapman, MATLAB Programming for Engineers.
6. Haykin, Communication Systems.
7. William Stallings, Data and Computer Communications.
8. Computer Networking: A Top-Down Approach Featuring the Internet, Jim Kurose and Keith Ross, Addison Wesley, 4th Edition

CSER-2109: Financial and Managerial Accounting**PART A: INTRODUCTION**

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-2109	Financial and Managerial Accounting	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It provides students with a foundational understanding of accounting principles and practices essential for decision-making

in business contexts. The course covers topics such as financial statement analysis, budgeting, cost accounting, and internal control systems. Students learn how to interpret financial data, assess business performance, and make informed managerial decisions. Through case studies and real-world examples, students develop practical skills in financial reporting, budget planning, and strategic analysis to support effective financial management in organizations.

Justification of this course in the program entity

The course is intended to teach the basics related to finances and accounting management. This course is designed to reflect both theoretical learning as well as methods and techniques necessary to acquire the skills in accounting for computer science graduates. In addition, the course will also discuss various aspects of accounting that may be needed in the development and maintenance of accounting related issues with the help of computer science.

Course Objectives

The major objectives of the course are to:

- i. Understand Accounting Fundamentals: Gain a basic understanding of accounting principles, terminology, and concepts used in financial and managerial accounting.
- ii. Analyze Financial Statements: Learn how to interpret financial statements, including balance sheets, income statements, and cash flow statements, to assess the financial health and performance of an organization.
- iii. Apply Cost Accounting Techniques: Develop skills in cost accounting methods to analyze and allocate costs accurately, aiding in decision-making and budgeting processes.
- iv. Utilize Managerial Accounting Tools: Acquire proficiency in using managerial accounting tools such as budgeting, variance analysis, and performance measurement to support managerial decision-making and strategic planning.
- v. Evaluate Internal Controls: Understand the importance of internal controls and develop the ability to assess and improve internal control systems to safeguard assets and ensure financial integrity within an organization.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Understand core concepts of financial accounting.

CLO2: Understand the principles and practices of managerial accounting.

CLO3: Gain proficiency in cost accounting techniques and methods.

CLO4: Implement various accounting methods to analyze financial data accurately.

CLO5: Apply accounting principles to interpret financial statements, make informed decisions, and support organizational goals.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Financial Accounting: Objectives and importance of accounting; Accounting as an information system.	Lecture, Rapport building	2	CLO1
2	Financial Accounting: Computerized system and applications in accounting; Recording system: double entry mechanism.	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4
3	Financial Accounting: Accounts and their classification; Accounting equation.	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4
4	Financial Accounting: Accounting cycle: journal, ledger, trial balance.	Lecture, Rapport building, Demonstration	3	CLO1, CLO4
5	Financial Accounting: Preparation of financial statements considering adjusting and closing entries;	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4
6	Financial Accounting: Accounting concepts (principles) and conventions. Financial statement analysis and interpretation: ratio analysis.	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4, CLO5
7	Cost and Management Accounting: Cost concepts and classification.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3
8	Cost and Management Accounting: Overhead cost: meaning and classification; Distribution of overhead cost.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3

9	Cost and Management Accounting: Overhead recovery method/rate; Job order costing: preparation of job cost sheet and quotation price.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3, CLO4
10	Cost and Management Accounting: Inventory valuation: absorption costing and marginal/ variable costing technique.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3, CLO4
11	Cost and Management Accounting: Cost-Volume-Profit analysis: meaning, breakeven analysis.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3, CLO4
12	Cost and Management Accounting: Margin approach sensitivity analysis Short-term investment decisions relevant and differential cost analysis.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3, CLO4
13	Cost and Management Accounting: Long-term investment decisions: capital budgeting, various techniques of evolution of capital investment.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO3, CLO4
14	Problem Solution and Revision	Q/A	2	All

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1						H			H			
CLO2						H			H			
CLO3						H			H			
CLO4			H									
CLO5	H											H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	

Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	15
Analyze	15
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, class attendance carries 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of

these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Needles and Anderson: Principle of Accounting.

Reference Book

1. Weggandt and Kieso: Financial Accounting.
2. S.P. Jain and K.L. Narang: Advanced Accounting.
3. Md. Muinuddin Khan: Advanced Accounting.
4. Basu and Das: Advanced Accounting.

2nd Year 2nd Semester

CSE-2201: Computer Architecture

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Attendance	Class Performance	
CSE-2201	Computer Architecture	Theory	3	60	20	10	10	100

Course Description: Computer Architecture is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 48 hours, 42 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

In an age where computers are becoming increasingly prevalent, it is important to understand how they work. Computer architecture is the study of how computers are designed and how they function. It covers the hardware, the software, and the interaction between the two. It also covers the performance and power consumption of the computer. It is important to study computer architecture because it can help you understand how to make computers more efficient and how to troubleshoot problems. The field of computer architecture is constantly evolving, as new technologies are developed and new challenges are faced. By studying computer architecture, students can stay up-to-date on the latest advancements and learn how to apply them to real-world situations. The subject explores how machines are designed, built, and operate. Knowing what's inside and how it works will help the students design, develop, and implement applications better, faster, cheaper, more efficient, and easier to use because you will be able to make informed decisions instead of assuming.

Course Objectives

The major objectives of the course are:

- to develop computer architecture concepts related to design of modern processors, memories and I/Os.
- to analyze and evaluate the performance of commercially available computers using various performance factors.
- to develop computer architecture concepts related to design of modern processors, memories and I/Os.
- to identify the elements of modern instructions sets and their impact on processor design. build themselves as a professional developer.
- to develop logic for assembly language programming.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: develop computer architecture concepts related to design of modern processors, memories and I/Os and Analyze and evaluate the performance of commercially available computers using various performance factors together various addressing modes.

CLO2: develop concept of Instruction set architecture and logic for assembly language programming.

CLO3: understand different data representation as well as getting familiar with different algorithms for arithmetic operations.

CLO4: understand the memory hierarchy including caches and virtual memory.

CLO5: to demonstrate comprehension of a pipelined architectures including data path and hazards.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Getting familiar with basic components of computers and microprocessors, types of components: registers, control section and ALU, RISC & CISC Processors.	• Q/A • Test • Demonstration	3	CLO 1
2.	Calculate the performance of a modern digital computer from parameters such as processor speed, cycles per instruction.	• Q/A • Test • Demonstration	3	CLO 1
3.	Getting familiar with various addressing modes that are defined in a given instruction set architecture that illustrates how the machine language instructions in that architecture identify the operand(s) of each instruction.	• Q/A • Test • Demonstration	3	CLO 1
4.	Understand how logical operations are performed in machine	• Q/A • Test	3	CLO 2

	language,	● <u>Demonstration</u> Assignment		
5.	Understand how procedure is supported by computer hardware, how flow is controlled and gain the concept of recursion.	● Q/A ● Test ● <u>Demonstration</u> ● <u>Assignment</u>	3	CLO 2
6.	Arithmetic for computers Representation of signed and unsigned numbers. Overflow during addition and subtraction.	● Q/A ● Test ● <u>Demonstration</u> Project	3	CLO 3
7.	Different multiplication and division algorithms: Sequential multiplication algorithm.	● Q/A ● Test ● <u>Demonstration</u> ● Assignment	3	CLO 3
8.	Booth's Multiplication algorithm. Restoring Division algorithm.	c) Q/A d) Test e) Demonstration f) Assignment	3	CLO 3
9.	Learn how different types of data are represented: Fixed and floating (IEEE754) number representations. Floating point addition and Multiplication.	a) Q/A b) Test c) Demonstration Assignment	3	CLO 3
10.	Memory Hierarchy, Cost and Performance, Random Access Memory, Principle of locality, data hierarchy. cache, Direct Mapped Cache.	● Q/A ● Test ● <u>Demonstration</u> ● <u>Project</u>	3	CLO 4
11.	Handling Cache Misses, Multilevel Cache, Handling Write, reducing Cache Misses by More Flexible Placement of Blocks, Set associative cache, Fully associative cache.	● Q/A ● Test ● <u>Demonstration</u>	3	CLO 4
12.	Pipelining and hazards, the importance of pipelining, Structural Hazard, resolving Structural Hazard, Bubbling the pipelining, Renaming.	● Q/A ● Test ● <u>Demonstration</u>	3	CLO 5

13	Understanding data hazard. Forwarding or bypassing data hazard. Control hazard, stall, branch prediction.	• Q/A • Test • Demonstration • Project	3	CLO 5
14	Revision and Solutions		3	

Mapping between CLOs and PLOs:

	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO 11	PLO 12
CLO1	H	H										
CLO2	H	H		H								
CLO3	H											
CLO4	H	M	M									
CLO5	H	H	H	H								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√	√		
Mid Semester2				√	√
Class Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (30 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Class Performance	Class Attendance	Other forms
Remember	25%	25%			
Understand	50%	50%	50%		
Apply	25%	25%	25%		
Analyze			25%		
Evaluate					
Create					

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
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Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows:

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. M. Moris Mano, "Computer Systems Architecture", 4th Edition, Pearson/PHI, ISBN:10:0131755633
2. Carl Hamacher, Zvonko Vranesic, Safwat Zaky, "Computer Organization", 5th Edition, McGraw Hill.

Reference Books:

1. John L. Hennessy and David A. Patterson, "Computer Architecture a quantitative approach", 4th Edition Elsevier, ISBN:10:0123704901
2. William Stallings, "Computer Organization and Architecture", 6th Edition, Pearson/PHI, ISBN:10:0-13-609704-9
3. Donald e Givone, "Digital Principles and Design", TMH.

5. A. Anandkumar, “Fundamentals of digital circuits”, 4th edition, PHI.

CSE-2202: Mathematical Analysis for Computer Science

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-2202	Mathematical Analysis for Computer Science	Theory	3	60	20	10	10	100

Course Description: Mathematical Analysis for Computer Science is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course builds up the students’ ability to think and express logically and mathematically. The course will address mathematical reasoning, combinatorial analysis, algorithmic thinking. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE 1203 Data Structures, CSE 1208 Discrete Mathematics and CSE 2208 Algorithms.

Course Objectives:

The major objectives of the course are:

5. Interpret and apply recurrent problems for mathematical reasoning.
6. Interpret and apply Manipulation of sums
7. Interpret and apply Number theory; Special numbers; generating functions for algorithmic thinking; demonstrate this knowledge and write report for realistic problem solving.
8. Interpret and apply Random variables; stochastic process; Markov chains for realistic problem solving.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding Recurrent problems, Manipulation of sums.

CLO2: Illustration to achieve Number theory, Special numbers, generating functions.

CLO3: Learning of Functions, Recursive Functions, Relations and Their Properties.

CLO4: Understanding concept of Random variables; stochastic process Markov chains: discrete parameter, continuous parameter, birth-death process, network.

CLO5: Learning of Application of queuing models queuing models: birth-death model, Markovian model, open and closed queuing.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lectur e Hour	CLOs
1	Recurrent problems;	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
2	Recurrent problems;	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
3	Manipulation of sums	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO1
4	Manipulation of sums	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among	3	CLO1

		students, including tests/quizzes over lecture materials, question-answer sessions, etc.		
5	Number theory; Special numbers; generating functions.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO2
6	Number theory; Special numbers; generating functions.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO2
7	Number theory; Special numbers; generating functions.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO2
8	Functions, Recursive Functions, Relations and Their Properties.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO3
9	Functions, Recursive Functions, Relations and Their Properties.	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among	3	CLO3

		students, including tests/quizzes over lecture materials, question-answer sessions, etc.		
10	Random variables;	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO4
11	Stochastic process	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO4
12	Markov chains: discrete parameter, continuous parameter, birth-death process; network; Application of queuing models	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO4
13	Queuing models: birth-death model, Markovian model, open and closed queuing	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

CLOs	Program Outcomes											
	PO-	P	P	PO-	P	P	PO	PO	PO	PO	PO	PO
CLO1		M										
CLO2	H	M	M									
CLO3	H	H							M			
CLO4									H			M
CLO5		H	H									M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√				
Mid Semester2		√	√		
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10

Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

- Donald E Knuth, Concrete Mathematics
- Roger Cook, Mathematical Analysis for CS

Reference Books:

- Waller Rudin, Principles of Mathematical Analysis

CSE-2203: Database Management System

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)
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				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performanc e	Class attendan ce	
CSE-2203 :	Database Managemen t System	Theory	3	60	20	10	10	100

Course Description: Database Management System is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The aim of this subject is to get broad understanding of the basic concepts of database management system and the students will develop the skills to design database system and develop application programs to manage & retrieve data from different perspective using Structured Query Language (SQL).

Course Objectives

The major objectives of the course are:

- Understand the role of a database management system in an organization.
- Understand basic database concepts, including the structure and operation of the relational data model.
- Construct simple and moderately advanced database queries using Structured Query Language (SQL).
- Understand and successfully apply logical database design principles, including E-R diagrams and database normalization.
- Describe and discuss selected advanced database topics, such as distributed database systems and the data warehouse.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Explain and analyze the features of database management systems, architecture of database systems, and the role of database users.

CLO2: Analyze the database requirements and determine the entities involved in the system and their relationship to each other and create the logical design of the database using data modeling concepts such as entity-relationship diagrams.

CLO3: Formulate, using relational algebra, solutions to a broad range of query problems.

CLO4: Formulate, using SQL, solutions to a broad range of query and data update problems.

CLO5: Apply, execute and evaluate various Normalization techniques.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction: Introduction to Database Management Systems and Relational Model, Database System Applications, Database Languages.	g) Lecture h) Rapport building	3	CLO1
2	Data Model: Schema And Instances, Data Independence and Interface, Data Definition Languages, Data Manipulation Language, Overall Database Structure, Database architecture, 3 tier-architecture.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO1
3	Entity-Relationship Model: E.R. Model Concept, Notation for ER Diagrams, Mapping Constraints, Keys, Concept of Super Key, Candidate Key, Primary Key.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO2
4	Entity-Relationship Model: Generalizations, Aggregation, Reducing ER Diagrams to Tables, Extended ER Model, Relationships of Higher Degree.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO2
5	Relational Model: Relational Data Model Concepts, Integrity	a) Lecture, b) Rapport building	3	CLO2

	Constraints, Domain Constraints, Referential Integrity, Assertions Triggers, Foreign Key.	c) Cooperative learning		
6	Relational Algebra: Relational Algebra (RA), Relational Calculus, Domain and Tuple Calculus, RA Data Definition Queries and Updates in RA.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO3
7	Relational Databases: Writing Basic and DDL and DML Queries using SQL (Structured Query Language), Constraint	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO4
8	Relational Databases: SQL query, JOIN operations	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO4
9	Relational Databases: Intermediate and Advanced SQL Queries such as Join, nested subqueries, views, authorization, string operation.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO4
10	Relational Databases: Advanced SQL Queries such function, procedure, triggers.	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO4
11	Storage and File Structure: Overview of Physical Storage Media, RAID, File Organization	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO5
12	Database Normalization: Database Normalization	a) Lecture,	3	CLO5

	based on Functional Dependency, Boyce-Codd Normal Form	b) Rapport building c) Cooperative learning		
13	Indexing and Hashing: Database Indexing and Hashing Techniques	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO5
14	Transaction Management: Transaction Management and Concurrency Control Protocols	a) Lecture, b) Rapport building c) Cooperative learning	3	CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		

Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Avi Silberschatz, Henry F. Korth and S. Sudarshan, "Database System Concepts", McGraw-Hill, 7th Edition, March 2019.

Reference Books:

1. Carlos Coronel, Steven Morris and Peter Rob, “Database Systems: Design, Implementation, and Management”, Course Technology, 13 Edition, January 2018.
2. Ramez Elmasri and Shamkant Navathe, “Fundamentals of Database Systems”, Pearson, 7th Edition, March 2017.

CSEL-2204: Database Management System Lab**PART A: INTRODUCTION**

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-2204	Database Management System Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The aim of this subject is to get broad understanding of the basic concepts of database management system and the students will develop the skills to design database system and develop application programs to manage & retrieve data from different perspective using Structured Query Language (SQL).

Justification of this course in the program entity

The primary goal of this course is to provides a platform for students to apply theoretical concepts such as normalization, indexing, querying, and transaction management in a hands-on environment. The DBMS lab allows students to develop skills in designing, implementing, and managing databases. These skills are highly relevant in various professional settings where database management is a critical aspect of information systems. Students must evaluate different approaches to database design, query optimization, and performance tuning. They learn to assess trade-offs and make informed decisions based on their analysis.

Course Objectives

The major objectives of the course are:

- i. Provide a sound introduction to the discipline of database management as a subject in its own right, rather than as a compendium of techniques and product-specific tools.

- ii. Familiarize the participant with the nuances of database environments towards an information-oriented data-processing oriented framework.
- iii. Give a good formal foundation on the relational model of data.
- iv. Present SQL and procedural interfaces to SQL comprehensively.
- v. Give an introduction to systematic database design approaches covering conceptual design, logical design and an overview of physical design.
- vi. Present the concepts and techniques relating to query processing by SQL engines.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: To understand the formal foundation of the relational model of data, implementing and managing database systems using popular relational database management systems (RDBMS), including tasks such as installation, configuration, security management, and backup and recovery.

CLO2: To execute SQL and procedural interfaces to SQL comprehensively.

CLO3: To formulate systematic database design approaches covering conceptual design, logical design and an overview of physical design.

CLO4: To present the concepts and techniques relating to query processing by SQL engines.

CLO5: develop skills in developing database-driven applications, including web applications and desktop applications, by integrating databases with programming languages and frameworks, ensuring seamless interaction and data manipulation.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to MYSQL, tools (XAMPP), Oracle installation	Lectures, Coding Exercises	2	CLO1
2	Implementation of DDL, commands of SQL with suitable examples: Create table, Alter table, Drop Table	Theoretical Explanation, Coding Practice	2	CLO1, CLO2
3	Implementation of DML, commands of SQL with suitable examples. Insert table, Update table Delete Table	Lectures, Problem-solving Sessions	2	CLO1, CLO2
4	Basic query pattern, single row function, Simple DML queries	Interactive Lectures,	2	CLO2, CLO3

5	Basic query pattern, different types of constraint.	Practical Examples, Coding Sessions	2	CLO2, CLO3
6	Joining Multiple Tables, Introduction to view & sequence: Inner Join, Outer Join, Natural Join	Theoretical Explanation, Coding Practice	2	CLO2, CLO3, CLO4
7	Joining Multiple Tables, Introduction to view & sequence: Inner Join, Outer Join, Natural Join	Lectures, Problem-solving Sessions	2	CLO2, CLO3, CLO4
8	Arithmetic Operator Logical Operator Comparison Operator Special Operator Set Operator	Coding Workshops, Practical Examples	2	CLO2, CLO3, CLO4
9	Number Function Aggregate Function Character Function Date Function.	Theoretical Explanation, Problem-solving Sessions	2	CLO2, CLO3, CLO4, CLO5
10	Group by & Having Clause Order by Clause Indexing	Interactive Sessions, Coding Exercises	2	CLO3, CLO4, CLO5
11	Advanced SQL Queries such function, procedure, triggers.	Lectures, Algorithm Implementation	2	CLO3, CLO4, CLO5
12	Advanced SQL Queries such function, procedure, triggers.	Practical Examples, Coding Sessions	2	CLO3, CLO4, CLO5
13	Transforming an ER model into a Relational Schema	Problem-solving Sessions, Coding Practice	2	CLO3, CLO4, CLO5
14	Revise and Q/A	Review Sessions, Open Discussion	2	CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H		H	H						M
CLO2	M		H			H						M
CLO3		H	H			H			H			M
CLO4		H	H	H		H						
CLO5			H		H	H						

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class Attendance	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (35 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Avi Silberschatz, Henry F. Korth and S. Sudarshan, “Database System Concepts”, McGraw-Hill, 7th Edition, March 2019.

Reference Books:

2. Carlos Coronel, Steven Morris and Peter Rob, “Database Systems: Design, Implementation, and Management”, Course Technology, 13 Edition, January 2018.
3. Ramez Elmasri and Shamkant Navathe, “Fundamentals of Database Systems”, Pearson, 7th Edition, March 2017.

CSER-2205: Math-IV (Complex Variable, Fourier and Laplace Transform)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSER-205	Math-IV (Complex Variable, Fourier and Laplace Transform)	Theory	3	70	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. Complex functions have numerous applications in areas such as engineering, physics, differential equations, and number theory, to name just a few. The focus of this course is on the study of holomorphic functions and their most

important basic properties. Topics covered are Analytic function, Harmonic functions, Complex integration, Power series of complex terms; Taylor's and Laurent's expansion of complex functions; Taylor's theorem; zeros of holomorphic functions; Residue theorems, Conformal mappings, Physical significance of Conformal mappings, etc.

Justification of this course in the program entity

This course is essential for computer science students as it equips them with advanced mathematical tools essential for tackling complex problems in their field. Understanding complex variables is crucial for various computer science applications such as signal processing, cryptography, and image processing. Fourier transforms are fundamental for analyzing signals and data, which are ubiquitous in computer science, ranging from digital communication to image and audio processing. Laplace transforms, on the other hand, are essential for solving differential equations, a cornerstone in algorithm analysis and system modeling. In essence, Math-IV provides computer science students with the mathematical foundation necessary to excel in areas requiring sophisticated problem-solving and analytical skills.

Course Objectives

The major objectives of the course are to:

- i. Proficiency in Complex Variables: Understand complex numbers and functions, applying them in engineering and computer science contexts.
- ii. Master Fourier Transform: Learn Fourier series and transforms, applying them in signal processing and data analysis.
- iii. Competence in Laplace Transform: Apply Laplace transforms to solve differential equations and analyze linear systems.
- iv. Problem-Solving Skills Enhancement: Apply mathematical concepts to solve real-world engineering and computer science problems.
- v. Mathematical Modeling: Formulate mathematical models for complex systems encountered in computer science.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CO1: Introduce basic notions and methods of a function of a complex variable

CO2: Define and discuss analytic functions and various theorem & problems

CO3: Discuss complex integrations and residue theorem including branch line integrals. Demonstrate Cauchy's and Morera's theorem

CO4: Calculate the value of contour integral and its corollaries.

CO5: Introduce Fourier series, Fourier transforms and Laplace transforms. Use the Laplace and Fourier transform methods student will be solving ordinary and partial differential equations,

initial and boundary value problems of differential equations arising in Mathematics, Physics and Engineering fields.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Complex Analysis: Complex numbers, properties, modulus & amplitudes of complex numbers.	Lecture, Rapport building	3	CO1
2	Complex Analysis: Analytic functions, Cauchy-Riemann equations.	Lecture, Rapport building, Demonstration, Group work	3	CO1, CO2
3	Complex Analysis: Complex integration, Cauchy's Theorem, Cauchy's Residue theorem.	Lecture, Rapport building, Demonstration, Group work	3	CO3, CO4
4	Complex Analysis: Cauchy's integral formula, Higher order derivatives, Liouville's Theorem	Lecture, Rapport building, Demonstration, Group work	3	CO3, CO4
5	Complex Analysis: Taylor's and Laurent's theorems, singularities, Zero and poles of an analytic function.	Lecture, Rapport building, Demonstration, Group work	3	CO2
6	Complex Analysis: Residues, Evaluation & calculation of residues of real definite integral by contour integrations.	Lecture, Rapport building, Demonstration, Group work	3	CO2

7	Complex Analysis: Bilinear mappings, mappings, Conformal mapping.	Lecture, Rapport building, Demonstration, Group work	3	CO3, CO4
8	Vector Analysis: Limit, continuity and differentiability of scalar and vector point functions, Vector integration, line, surface and volume integrals.	Lecture, Rapport building, Demonstration, Group work	3	CO5
9	Vector Analysis: Gradient, Divergence and Curl of point function, Gauss's Theorem, Stocks Theorem and Green's Theorem.	Lecture, Rapport building, Demonstration, Group work	3	CO5
10	Fourier series: Fourier series, Process of determining the co-efficient, Fourier cosine and sine series.	Lecture, Rapport building, Demonstration, Group work	3	CO5
11	Fourier series: Fourier transform, Fourier sine and cosine transform.	Lecture, Rapport building, Demonstration, Group work	3	CO5
12	Laplace transforms: Basic definitions and properties, Existence theorem, Laplace transforms of periodic functions. Transforms of convolutions.	Lecture, Rapport building, Demonstration, Group work	3	CO5
13	Laplace transforms: Inverse Laplace transforms. Use of Laplace transforms in solving initial value problems.	Lecture, Rapport building, Demonstration, Group work	3	CO5
14	Revision and Solution	Q/A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H							M		
CLO2		H		H							M	
CLO3			H				H					
CLO4			H									
CLO5	H									M		M

PART C: ASSESSMENT AND EVALUATION**Mapping/Alignment of CLOs vs Assessment Strategy:**

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. R.V. Churchill, Complex Variable and Applications, latest edition.
2. M. L. Khanna, Complex Variable, latest edition.

Reference Book:

1. A. K. M Shahidullah, latest edition.

CSE-2206: Numerical Analysis

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Perform ance	Class attend ance	
CSE-2206	Numerical Analysis	Theory	2	30	10	5	5	50

Course Description: Numerical Analysis is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The

course carries 3 credits and 30 lecture hours. Among the 30 hours, 25 are for class teaching and 5 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course will emphasize the development of numerical algorithms to provide solutions to common problems formulated in science and engineering. The primary objective of the course is to develop the basic understanding of the construction of numerical algorithms, and perhaps more importantly, the applicability and limits of their appropriate use. Knowledge of this course will be needed as prerequisite knowledge for future course such as CSE366 Artificial Intelligence.

Course Objectives:

The major objectives of the course are:

9. Understand and use error estimation and root finding algorithms for solving scientific and engineering problems.
10. Understand and use direct and iterative methods of systems of linear equations for solving scientific and engineering problems.
11. Understand and use numerical techniques of interpolation, differential and integral equations for solving scientific and engineering problems.
12. Interpret and apply numerical techniques; demonstrate this knowledge and write report for realistic solution of complex scientific and engineering problems.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understand and use error estimation and root finding algorithms for solving scientific and engineering problems.

CLO2: Understand and use direct and iterative methods of systems of linear equations for solving scientific and engineering problems.

CLO3: Understand and use numerical techniques of interpolation, differential and integral equations for solving scientific and engineering problems.

CLO4: Interpret and apply numerical techniques; demonstrate this knowledge and write report for realistic solution of complex scientific and engineering problems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lectur e Hour	CLOs
1	Applications of round of errors and floating points: Identify round-off errors,	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that	2	CO1

	floating point and convergence, Mature humble sequencers of round off errors, Argument the advantages and applications of round of errors and floating points	include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.		
2	Bisection method and false position method: Define and describe bisection method and false position method, Find roots of equations using bisection and false position methods, Solve systems of algebraic equations using false position method, Implement bisection & false position method in MATLAB	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO1
3	Iterative methods for solving linear equations	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
4	Newton Raphson Method	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
5	Gauss Jordan and elimination method: State and explain both Gauss Jordan and elimination method, Differentiate	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among	2	CO2

	between these, Justify which method is better, Implement to practical work	students, including tests/quizzes over lecture materials, question-answer sessions, etc.		
6	Gauss Jordan and elimination method: State and explain both Gauss Jordan and elimination method, Differentiate between these, Justify which method is better, Implement to practical work	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
7	Euler, Taylor and Maclaurin formula: Define Euler, Taylor and Maclaurin formula, Implement and Examples of each method, Analyzes of Truncation / Global Error, Stability Analysis of each methods	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
8	Runge-Kutta Method: Define and describe Runge-Kutta Method, General form of Multistep, Find convergence of Runge Kutta Method	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3
9	Forward and Backward difference: Apply & solve equation using different differences, Recognize each difference, Integrate forward and backward difference	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3

10	Forward and Backward difference Integrate forward and backward difference	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3
11	Trapezoidal method, Simpson 1/3 and 3/8 rules: Describe initial value problems of Trapezoidal method, Solve Boundary value problems of Simpson 1/3 and 3/8 rules	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4
12	Trapezoidal method, Simpson 1/3 and 3/8 rules: Evaluate the numerical method integral using these methods	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4
13	Classify topics, Solve some problems, Propose solutions to some new problems	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4
14	Revision and Solution		2	

Mapping between CLOs and PLOs:

	Program Outcomes
CLOs	

	PO-	P	P	PO-	P	P	PO	PO	PO	PO	PO	PO
CLO1	✓	✓										
CLO2				✓								
CLO3				✓			✓					
CLO4				✓								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	✓			
Mid Semester2		✓	✓	
Quiz/Performance	✓	✓	✓	✓
Class attendance	✓	✓	✓	✓
Final	✓	✓	✓	✓

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10

Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO3
5	Question from CLO4 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

- Richard L. Burden, J. Douglas Faires, “Numerical Analysis”, Brooks Cole, 9th Edition, August 9, 2019.
- Terrence J. Akai, “Applied Numerical Methods for Engineers”, Wiley, 1st Edition, November 3, 2019.

Reference Books:

- J. Stoer, R. Bulirsch, R. Bartels and W. Gautschi, “Introduction to Numerical Analysis”, Springer, 3rd Edition, December 1, 2018.
- S.S. Sastry, “Introductory Methods of Numerical Analysis”, Prentice-Hall of India Pvt. Ltd, 5th Edition, March 30, 2017.

CSEL-2207: Numerical Analysis Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSEL-207	Numerical Analysis Lab	Lab	1.00	30	10	5	5	50

Course Description: Numerical Analysis is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 30 lecture hours. Among the 30 hours, 25 are for class teaching and 5 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course will emphasize the development of numerical algorithms to provide solutions to common problems formulated in science and engineering. The primary objective of the course is to develop the basic understanding of the construction of numerical algorithms, and perhaps more importantly, the applicability and limits of their appropriate use. Knowledge of this course will be needed as prerequisite knowledge for future course such as CSE366 Artificial Intelligence.

Course Objectives:

The major objectives of the course are:

1. Understand and use error estimation and root finding algorithms for solving scientific and engineering problems.
2. Understand and use direct and iterative methods of systems of linear equations for solving scientific and engineering problems.
3. Understand and use numerical techniques of interpolation, differential and integral equations for solving scientific and engineering problems.
4. Interpret and apply numerical techniques; demonstrate this knowledge and write report for realistic solution of complex scientific and engineering problems.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understand and use error estimation and root finding algorithms for solving scientific and engineering problems.

CLO2: Understand and use direct and iterative methods of systems of linear equations for solving scientific and engineering problems.

CLO3: Understand and use numerical techniques of interpolation, differential and integral equations for solving scientific and engineering problems.

CLO4: Interpret and apply numerical techniques; demonstrate this knowledge and write report for realistic solution of complex scientific and engineering problems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Applications of round off errors and floating points: Identify round-off errors, floating point and convergence, Mature humble sequencers of round off errors, Argument the advantages and applications of round off errors and floating points	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO1
2	Bisection method and false position method: Define and describe bisection method and false position method, Find roots of equations using bisection and false position methods, Solve systems of algebraic equations using false position method, Implement bisection & false position method in MATLAB	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO1
3	Iterative methods for solving linear equations	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2

4	Newton Raphson Method	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
5	Gauss Jordan and elimination method: State and explain both Gauss Jordan and elimination method, Differentiate between these, Justify which method is better, Implement to practical work	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
6	Gauss Jordan and elimination method: State and explain both Gauss Jordan and elimination method, Differentiate between these, Justify which method is better, Implement to practical work	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2
7	Euler, Taylor and Maclaurin formula: Define Euler, Taylor and Maclaurin formula, Implement and Examples of each method, Analyzes of Truncation / Global Error, Stability Analysis of each methods	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO2

8	Runge-Kutta Method: Define and describe Runge-Kutta Method, General form of Multistep, Find convergence of Runge Kutta Method	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3
9	Forward and Backward difference: Apply & solve equation using different differences, Recognize each difference, Integrate forward and backward difference	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3
10	Forward and Backward difference Integrate forward and backward difference	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3
11	Trapezoidal method, Simpson 1/3 and 3/8 rules: Describe initial value problems of Trapezoidal method, Solve Boundary value problems of Simpson 1/3 and 3/8 rules	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4
12	Trapezoidal method, Simpson 1/3 and 3/8 rules: Evaluate the numerical method integral using these methods	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4

13	Classify topics, Solve some problems, Propose solutions to some new problems	Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	2	CO3, CO4
14	Revision and Solution		2	

Mapping between CLOs and PLOs:

CLOs	Program Outcomes											
	PO-	P	P	PO-	P	P	PO	PO	PO	PO	PO	PO
CLO1	✓	✓										
CLO2				✓								
CLO3				✓			✓					
CLO4				✓								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	✓			
Mid Semester2		✓	✓	
Quiz/Performance	✓	✓	✓	✓
Class Attendance	✓	✓	✓	✓
Final	✓	✓	✓	✓

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		

Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO3
5	Question from CLO4 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Richard L. Burden, J. Douglas Faires, “Numerical Analysis”, Brooks Cole, 9th Edition, August 9, 2019.
2. Terrence J. Akai, “Applied Numerical Methods for Engineers”, Wiley, 1st Edition, November 3, 2019.

Reference Books:

3. J. Stoer, R. Bulirsch, R. Bartels and W. Gautschi, “Introduction to Numerical Analysis”, Springer, 3rd Edition, December 1, 2018.
4. S.S. Sastry, “Introductory Methods of Numerical Analysis”, Prentice-Hall of India Pvt. Ltd, 5th Edition, March 30, 2017.

CSE-2208: Design and Analysis of Algorithm

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-2208	Design and Analysis of Algorithm	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The emphasis is on developing proficiency in designing efficient algorithms, analyzing their performance, and applying these skills to real-world problem-solving scenarios.

Justification of this course in the program entity

This course is indispensable in the field of Computer Science discipline as it lays the groundwork for proficient programming and problem-solving. With a focus on algorithmic design, the course equips students with essential skills to dissect complex problems and construct efficient solutions. By delving into algorithmic complexity and optimization techniques, students gain a deep understanding of how to assess and enhance the performance of algorithms. This knowledge is vital for developing code that not only functions effectively but also optimally. Beyond theoretical concepts, the practical applications of algorithmic principles are pervasive, impacting various domains within computer science, including artificial intelligence, data science, and software engineering. Thus, the course serves as a cornerstone, empowering students with the expertise needed to navigate the complexities of modern computing.

Course Objectives

The major objectives of the course are:

- i. Develop a comprehensive understanding of fundamental algorithmic concepts, including dynamic programming, greedy algorithms, divide and conquer, and graph algorithms.
- ii. Enable students to analyze and evaluate the efficiency and correctness of algorithms, emphasizing algorithmic complexity as a key metric.
- iii. Cultivate practical problem-solving skills by applying algorithmic principles to diverse real-world scenarios, fostering the ability to address complex computational challenges.
- iv. Empower students to make informed decisions in algorithm selection and design, emphasizing the importance of optimizing solutions for enhanced computational efficiency.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CO1: To design an algorithmic approach for solving a problem;

CO2: To analyze different types of algorithms to solve a problem efficiently/optimally.

CO3: To translate an algorithm into a computer program.

CO4: To apply algorithms for automation in everyday life (e.g., social media networks, GPS applications, and so on)

CO5: To demonstrate the ability to assess and compare the performance of algorithms through empirical analysis, utilizing tools and methodologies to measure and understand their efficiency and scalability in various computational contexts.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to the role of algorithms in computing.	Lecture, Discussion	3	CO1
2	Introduction to the complexity analysis of algorithms, and methods to solve recurrences, substitution method, recursion tree method, master method.	Lecture, Problem Solving	3	CO2
3	Graph algorithms including Breadth First search, Depth First Search, Topological sort, Strongly connected components, Minimum Spanning Tree, Euler Path, and Articulation Point.	Lecture, Practical Examples	3	CO1, CO4

4	Graph algorithms for finding shortest paths using Dijkstra's shortest path algorithm, The Bellman-Ford algorithm for single source shortest path, The Floyd-Warshall algorithm for all-pair shortest path.	Lecture, Algorithm Implementation	3	CO2, CO5
5	Divide and Conquer: basic idea, properties, Applications of Divide and Conquer: Counting Inversions, Closest pair of points, etc.	Lecture, Problem-Solving Sessions	3	CO1
6	Dynamic Programming: Basic idea, Comparison with Divide and Conquer, Memorization. Application of Dynamic programming: Coin related problems, Longest Increasing Sequence (LIS), Longest Common Subsequence (LCS), 0/1 Knapsack problem, Matrix Chain Multiplication, etc.	Lecture, Coding Practice	3	CO1, CO2, CO3
7	Greedy method: Elements of greedy method basic control structure, Comparison with dynamic programming and Divide and Conquer. Application of Greedy method: Minimum spanning tree: The algorithms of Prim & Kruskal, Job sequencing with deadline.	Lecture, Case Studies	3	CO1, CO3, CO4
8	Backtracking: Basic idea behind backtracking, control structure. Application of backtracking: Permutation & Combination Generation, Graph coloring problem, n-queens problems, Hamiltonian Cycle etc. Branch and Bound.	Lecture, Problem-Solving Workshops	3	CO1, CO3
9	Network Flow: Flow networks, The Ford-Fulkerson method, maximum bipartite matching, Maxflow-Mincut Theorem. Lower bound Theory for sorting, Exhaustive Search.	Lecture, Practical Applications	3	CO2, CO4

10	Computational Geometry related Algorithms: Line segment intersection, Inclusion in a polygon, Finding Convex Hull: Graham's scan, Jarvis's March.	Lecture, Geometric Algorithm Implementations	3	CO1, CO3
11	String Matching Algorithms: Naive string matching algorithm, String matching with finite automata, The Boyer-More algorithm for string matching, Knuth-Morris-Pratt algorithm.	Lecture, Algorithm Coding Exercises	3	CO1, CO3
12	NP-Completeness: Polynomial time, Polynomial Time verification, NP-completeness and reducibility, NP-Completeness proofs, NP Complete problems.	Lecture, Theoretical Discussions	3	CO2, CO5
13	Approximation Algorithms: Introduction, Approximation Ratio, Approximation algorithms for: Vertex-Cover Problem, TSP Problem	Lecture, Algorithmic Approximations	3	CO2, CO5
14	Revision and Q/A	Review session	3	CO1, CO2, CO3, CO4, CO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M		H						H			
CLO2	M	H	H						H			
CLO3	M		H						H			
CLO4	M								H			M
CLO5	M		H	H	H				M			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
--	------	------	------	------	------

Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will

be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Introduction to Algorithms, Latest Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.

Reference Books:

1. Fundamentals of Computer Algorithms, Latest Edition by Ellis Horowitz and Sartaj Sahni.
2. Algorithms, Latest Edition by Robert Sedgewick.
3. Introduction to Algorithm, Latest Edition by Udi Manber.
4. Introduction to the Design and Analysis of Algorithms, Latest Edition by Anany V. Levitin.
5. Algorithms + Data Structures= Programs, Latest Edition by Niklaus Wirth.
6. Data Structures & Algorithms in C++, Latest Edition by Adam Drozdek.

CSEL-2209: Design and Analysis of Algorithm Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class Attendance	
CSEL-2209	Design and Analysis of Algorithm Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The primary goal of this course is to cultivate a high level of proficiency in designing algorithms that exhibit efficiency, conducting thorough analyses of their performance, and applying acquired skills to real-world problem-solving scenarios. Through a combination of theoretical learning and hands-on practical exercises, students will develop a strong foundation in algorithmic principles and problem-solving strategies.

Justification of this course in the program entity

This course holds a pivotal role within the undergraduate program in the Department of Computer Science and Engineering. As a compulsory course, it serves as a cornerstone in shaping the academic and professional journey of students. The course is strategically designed to equip learners with essential skills in algorithm design, performance analysis, and real-world problem-solving. By delving into both theoretical concepts and hands-on applications, students gain a profound understanding of efficient algorithmic strategies. The dedicated hours for individual guidance ensure personalized support, fostering a conducive environment for skill refinement. In essence, this course stands as an essential, stimulating the program's commitment to nurturing well-rounded computer science professionals capable of tackling complex challenges through systematic algorithmic approaches.

Course Objectives

The major objectives of the course are:

- Implement algorithms hands-on to translate theoretical knowledge into practical coding skills.

- ii. Foster problem-solving proficiency by applying algorithmic approaches to diverse computational challenges.
- iii. Optimize algorithmic solutions for efficiency, considering time and space complexity.
- iv. Emphasize real-world application of algorithms to enhance practical understanding and relevance.
- v. Promote collaborative learning through group projects, developing communication and interpersonal skills in algorithm development.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CO1: Implementing Algorithms with Modern Tools: Use modern tools proficiently to translate algorithmic concepts into computer programs.

CO2: Applying Algorithms for Automation: Apply algorithms to automate processes in everyday life, exemplified through applications like social media networks and GPS systems.

CO3: Team Management and Project Accomplishment: Manage effectively as a team to successfully accomplish algorithmic projects.

CO4: Evaluation and Selection of Algorithms: Develop the ability to critically evaluate and select appropriate algorithms based on specific task requirements.

CO5: Cultivation of Algorithmic Thinking: Demonstrate proficiency in algorithmic thinking, breaking down complex problems into systematic steps for efficient algorithm design.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Fundamentals of Algorithmic Programming (C/C++/Java)	Lectures, Coding Exercises	2	CO1
2	Different sorting algorithms	Theoretical Explanation, Coding Practice	2	CO1, CO4
3	Complexity calculation	Lectures, Problem-solving Sessions	2	CO4
4	Graph algorithms: BFS, DFS, topological sort, SCC, MST	Interactive Lectures, Algorithm Implementation	2	CO2

5	Graph algorithms: shortest paths using Dijkstra, Bellman-Ford, and Floyd-Warshall	Practical Examples, Coding Sessions	2	CO2, CO4
6	Pattern matching algorithms: Naïve, Rabin-Karp, KMP, and Finite Automata	Theoretical Explanation, Coding Practice	2	CO2
7	Number theory: modular arithmetic, prime number generation, fast exponential	Lectures, Problem-solving Sessions	2	CO1, CO4
8	Dynamic programming: Coin-related problems, LIS, LCS, Knapsack problems, Matrix chain multiplication, and rod cutting problems	Coding Workshops, Practical Examples	2	CO1, CO4, CO5
9	Greedy approach: elements of greedy approach and Job Sequencing problems	Theoretical Explanation, Problem-solving Sessions	2	CO4, CO5
10	Backtracking: queens problem	Interactive Sessions, Coding Exercises	2	CO5
11	Tree algorithms: BST, Red-Black tree	Lectures, Algorithm Implementation	2	CO1, CO4
12	Solving miscellaneous problems	Practical Examples, Coding Sessions	2	CO1, CO2, CO4, CO5
13	Solving NP-hard problems	Problem-solving Sessions, Coding Practice	2	CO2, CO4
14	Revision and Q/A	Review Sessions, Open Discussion	2	CO1, CO2, CO3, CO4, CO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H		H	H						M
CLO2	M		H			H						M
CLO3		H	H			H			H			M
CLO4		H	H	H		H						
CLO5			H		H	H						

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two

Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Introduction to Algorithms, Latest Edition by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest.

Reference Books:

1. Fundamentals of Computer Algorithms, Latest Edition by Ellis Horowitz and Sartaj Sahni.
2. Algorithms, Latest Edition by Robert Sedgewick.
3. Introduction to Algorithm, Latest Edition by Udi Manber.
4. Introduction to the Design and Analysis of Algorithms, Latest Edition by Anany V. Levitin.
5. Algorithms + Data Structures= Programs, Latest Edition by Niklaus Wirth.
6. Data Structures & Algorithms in C++, Latest Edition by Adam Drozdek.

3rd Year 1st Semester

CSE-3101: Theory of Computation

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3101	Theory of Computation	Theory	3	60	20	10	10	100

Course Description: Theory of Computation is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The course is intended to teach students the theory of computation and theoretical limits of computability so that will provide students with a sound theoretical view of the most fundamental concepts of computation. Language theory; finite automata: deterministic finite automata, nondeterministic finite automata, equivalence and conversion of deterministic and nondeterministic finite automata, pushdown automata; context free grammars, Chomsky Normal form, Parsing. Turing machine. Universal Turing machine and Halting problem.

Course Objectives

The major objectives of the course are:

- i. Analyze and design finite automata, pushdown automata, Turing machines, formal languages, and grammars
- ii. Examine mathematical models of computations
- iii. Use formalisms to explore the inherent limitations of computations.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Define and describe formal models of computation, such as finite automata, DFAs and NFAs, Examples of DFAs And NFAs, Conversion of DFAs to NFAs,

CLO2: Demonstrate understanding of key notions, such as algorithm, computability, decidability, and

complexity through problem solving, Mealy and Moore Models, Minimization of Finite Automata.

CLO3: Convert between finite automata, regular grammars, and regular expression representations of regular languages.

CLO4: Apply the pumping lemma for regular languages to determine if a language is regular and convert between grammars and push-down automata for context-free languages.

CLO5: Define and describe formal models like pushdown automata, and Turing machines.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Mathematical Preliminaries: Countable and Uncountable Sets, Alphabets, Strings, and Languages	i) Lecture j) Rapport building	3	CLO1

2	The Theory of Automata: DFAs and NFAs, Examples of DFAs And NFAs, Conversion of DFAs to NFAs	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO1
3	The Theory of Automata: Mealy and Moore Models, Minimization of Finite Automata, Conversion of Finite automata	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO2
4	Formal Languages: Basic Definitions and Examples, Derivations and the Language Generated by a Grammar, Parse Trees and Ambiguity	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO3
5	Regular Sets Regular Grammars: Regular Languages, Closure Properties of Regular Languages, Finite Automata and Regular Expressions, NDF As with A-moves and Regular Expressions.	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO3
6	Regular Sets Regular Grammars: Algebraic Method Using Arden's Theorem, Equivalence of Two Finite Automata, Pumping Lemma for Regular Sets.	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO4
7	Context Free Languages: Context-free Languages and Derivation Trees, Ambiguity in Context-free Grammars, Context-Free Grammar (CFG)	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO4

	And Languages, Closure Properties for CFLs.			
8	Context Free Languages: Simplification of Context-free Grammars, Construction of Reduced Grammars, Elimination of Null Productions, Elimination of Unit Production	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO4
9	Context Free Languages: Normal Forms for Context-free Grammars, Reduction of CFLs, Normal Forms CNF and GNF	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO4
10	Pushdown Automata: Basic Definitions, Pushdown Automata and Context-free Languages, Parsing and Pushdown Automata, Top-down Parsing, Bottom-up Parsing	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO5
11	Pushdown Automata and CFL: Decision Algorithm, CYK Algorithm, Equivalence of CFG and PDA	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO5
12	Turing Machines and Linear Bounded Automata: Turing Machine Model, Variants of Turing Machines, Language Acceptability by Turing Machines,	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO5
13	Decidability and Recursively Enumerable Languages: The Halting Problem, Undecidable Problems from, Language Theory, Reducibility,	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO5

	Self-Reference and The Recursive Theorem			
14	Revisions, Q & A	d) Lecture, e) Rapport building f) Cooperative learning	3	CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

- K.L.P. Mishra, N. Chandrasekaran, "Theory of Computer Science: Automata, Languages and Computation", Prentice-Hall, 2010

Reference Books:

- Introduction to Automata Theory, languages & Computation 3rd Edition by J. Hopcroft, R. Motwani, J. Ullman

3. John Peter Linz, “Introduction to Formal Language and Automata”, Jones & Bartlett Learning, 5th edition, 2011.
4. Elaine A. Rich, “Automata, Computability and Complexity: Theory and Applications”, Pearson, 1st edition, 2007.

CSE-3102: Introduction to Data Science

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3102	Introduction to Data Science	Theory	3	60	20	10	10	100

Course Description: This course is a mandatory course for the B.Sc. programme in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is an introductory-level course designed to provide students with a foundational understanding of the principles, techniques, and applications of data science. In other words, this course introduces the fundamental process of data science and provides the necessary computational and statistical foundations for further courses in the master of data science. Design thinking methodology will be utilized to approach complex data science problems as a design problem. The data science process will be practiced through case studies in a number of data-intensive domains.

Justification of this course in the program entity

This course is an essential component of the computer science curriculum, offering students a foundational understanding of data science principles and methodologies. By focusing on Python, a versatile and widely-used programming language, students not only develop crucial programming skills but also gain exposure to a language prevalent in computer science research and applications. Moreover, the integration of fundamental machine learning concepts provides students with insights into the rapidly expanding fields of artificial intelligence and machine learning, preparing them to address future challenges and opportunities in various domains. This interdisciplinary course equips students with the skills necessary to analyze complex datasets, derive actionable insights, and contribute effectively to solving real-world problems in today's data-driven world, ensuring their relevance and competitiveness in the evolving landscape of computer science.

Course Objectives

The major objectives of the course are to:

- i. Provide a foundational understanding of data science principles, methodologies, and best practices.
- ii. Introduce fundamental machine learning concepts and techniques, including classification, regression, and clustering.
- iii. Equip students with the skills to extract insights from data and communicate findings effectively through visualization and storytelling.
- iv. Foster critical thinking and problem-solving abilities in the context of real-world data science challenges, preparing students for careers in data analysis, machine learning, and artificial intelligence.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Demonstrate proficiency in Python programming for data manipulation, analysis, and visualization.

CLO2: Apply fundamental data science concepts and methodologies to collect, clean, and preprocess datasets effectively.

CLO3: Utilize exploratory data analysis techniques to uncover patterns, trends, and relationships within datasets.

CLO4: Implement basic machine learning models and algorithms for tasks such as classification, regression, and clustering.

CLO5: Communicate insights derived from data analysis through clear and compelling visualizations, effectively conveying findings to diverse audiences.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Data Science and Overview of its Applications	Lecture, discussion, case studies	2	CLO2
2	Introduction to Linear Regression	Lecture, examples, hands-on exercises	3	CLO4
3	Simple Linear Regression: Theory and Implementation	Interactive tutorials, coding workshops	3	CLO4

4	Multiple and Polynomial Regression	Demonstrations, group projects	3	CLO4
5	Feature Engineering and Transformation Techniques	Guest lectures, real-world examples	3	Partially CLO2, CLO3
6	Model Selection and Cross-Validation	Problem-based learning, coding workshops	3	CLO4
7	Regularization Methods and Bias-Variance Tradeoff	Interactive lectures, discussions	3	Partially CLO4, CLO5
8	Classification Problems and Logistic Regression	Lecture, examples, hands-on exercises	3	CLO4
9	Multi-logistic Regression and Handling Missingness	Coding workshops, group projects	3	CLO4
10	Bootstrap, Confidence Intervals, and Hypothesis Testing	Case studies, interactive tutorials	3	Partially CLO4, CLO5
11	Model Evaluation and Hyperparameter Tuning	Problem-solving sessions, coding challenges	3	CLO4
12	Advanced Topics in Data Science: Reinforcement learning, Natural language processing, Causal inference, Quantum computing and its implications for data science	Q&A sessions, review quizzes	3	CLO3, CLO4
13	Ethics and Bias in Data Science: Ethical considerations in data collection, analysis, and interpretation, Bias and fairness in machine learning algorithms, Responsible data science practices	Summary of key concepts, course reflection	2	CLO5
14	Final Review and Q&A	Reviewing key concepts, addressing queries	2	CLO1, CLO2, CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
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CLO1	H	M	M								H	H
CLO2	H	H	M	M							H	H
CLO3	H	H	M								H	H
CLO4	H	H	M	M							H	H
CLO5	M	M			H							

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
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1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. "Foundations of Data Science" by Avrim Blum, John Hopcroft, and Ravindran Kannan, 2018.

Reference Books:

1. "Python for Data Analysis" by Wes McKinney
2. "Introduction to Statistical Learning: with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
3. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" by Foster Provost and Tom Fawcett

CSEL-3103: Data Science Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-3103	Data Science Lab	Lab	1	30	10	5	5	50

Course Description: This course is a mandatory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The lab component of the Introduction to Data Science course provides students with hands-on experience in applying theoretical concepts learned in lectures to real-world data analysis tasks. Through a series of practical exercises and projects, students will gain proficiency in using programming languages such as Python and R for data manipulation, analysis, and visualization. The labs will cover a wide range of topics, including data cleaning, exploratory data analysis, statistical modeling, machine learning algorithms, and data visualization techniques

Justification of this course in the program entity

This lab course holds paramount significance within the program entity of Computer Science. By addressing the increasing demand for data scientists in various industries, this course ensures students are equipped with essential skills to navigate the data-driven landscape. Integrating theoretical concepts with practical applications, it fosters a deeper understanding of key principles while promoting interdisciplinary learning. Through hands-on projects, students develop critical problem-solving abilities crucial for addressing real-world challenges. Furthermore, this course serves as a foundation for advanced studies and aligns with industry expectations, preparing graduates to excel in data-driven roles and contribute meaningfully to organizational success. Overall, the Introduction to Data Science course plays a pivotal role in enhancing students' educational journey, empowering them for impactful careers in the rapidly evolving field of data science.

Course Objectives

The major objectives of the course are to:

- i. Establish foundational knowledge in data science concepts.
- ii. Apply data science techniques for effective data analysis.
- iii. Explore exploratory data analysis methods.
- iv. Implement basic machine learning models.
- v. Develop effective communication of data-driven insights.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Demonstrate proficiency in Python programming for data manipulation, analysis, and visualization.

CLO2: Apply fundamental data science concepts and methodologies to collect, clean, and preprocess datasets effectively.

CLO3: Utilize exploratory data analysis techniques to uncover patterns, trends, and relationships within datasets.

CLO4: Implement basic machine learning models and algorithms for tasks such as classification, regression, and clustering.

CLO5: Communicate insights derived from data analysis through clear and compelling visualizations, effectively conveying findings to diverse audiences.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lab Hours	CLOs
1	Introduction to Data Science and Python Basics	Lecture, hands-on coding exercises	2	CLO1, CLO2
2	Data Collection and Preprocessing	Case studies, group discussions	2	CLO2
3	Exploratory Data Analysis (EDA)	Interactive tutorials, data exploration tasks	2	CLO3
4	Introduction to Machine Learning	Lecture, examples, algorithm demonstrations	2	CLO4
5	Linear Regression and Feature Engineering	Problem-solving sessions, coding workshops	2	CLO4
6	Logistic Regression and Classification	Group projects, peer-to-peer learning	2	CLO4
7	Model Evaluation and Hyperparameter Tuning	Hands-on exercises, model evaluation techniques	2	CLO4
8	Clustering Techniques and Unsupervised Learning	Case studies, interactive tutorials	2	CLO4
9	Data Visualization and Communication of Insights	Lecture, data visualization tasks	2	CLO5
10	Advanced Topics: Neural Networks and Deep Learning	Guest lectures, coding challenges	2	CLO4
11	Ethical Considerations in Data Science	Debate sessions, case studies	2	CLO5
12	Time Series Analysis	Practical sessions, real-world datasets	2	CLO4

13	Natural Language Processing	Workshops, hands-on coding tasks	2	CLO4
14	Practical Applications and Project Work	Project-based learning, individual consultations	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H											M
CLO2	H	H	M					M				
CLO3		H			H							
CLO4			M									
CLO5				H			M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5

Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. "Foundations of Data Science" by Avrim Blum, John Hopcroft, and Ravindran Kannan, 2018.

Reference Books:

2. "Python for Data Analysis" by Wes McKinney
3. "Introduction to Statistical Learning: with Applications in R" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani
4. "Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking" by Foster Provost and Tom Fawcett

CSE-3104: Operating Systems

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)		
				Semester Final Exam.	Continuous Assessment	Total

					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-310 4	Operating Systems	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course is crucial as it provides a foundational understanding of how computer systems manage resources, coordinate tasks, and facilitate communication, laying the groundwork for effective software development and system administration.

Justification of this course in the program entity

The Operating Systems course is indispensable in a Computer Science program as it equips students with essential knowledge and skills to comprehend the underlying principles governing computer systems, fostering a deep understanding of resource management, process coordination, and memory allocation, thereby enabling graduates to design, optimize, and troubleshoot software applications and systems, essential for a successful and comprehensive education in computer science. Moreover, a thorough understanding of operating systems is crucial for students as it forms the basis for advanced coursework in specialized areas such as distributed systems, networking, and cybersecurity, providing a solid foundation for tackling complex challenges in these domains. Additionally, the skills acquired in the Operating Systems course are highly transferable, preparing graduates to adapt to evolving technologies and contribute effectively to a rapidly changing landscape in the field of computer science.

Course Objectives

The major objectives of the course are to:

- i. Gain a deep understanding of operating system fundamentals, enabling efficient resource management and process coordination.
- ii. Develop skills in designing and optimizing operating systems for enhanced performance.
- iii. Learn how operating systems interact with software applications, facilitating proficient software development.
- iv. Enhance troubleshooting and debugging abilities for maintaining stable computing environments.

- v. Prepare for specialized areas like distributed systems and cybersecurity through foundational knowledge in operating systems.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be

CLO1: Understanding System Fundamentals: Students will demonstrate a comprehensive knowledge of operating system principles, including process management, memory allocation, file systems, and device management.

CLO2: System Design and Optimization Skills: Students will develop the ability to design and optimize operating systems, showcasing proficiency in enhancing system efficiency, reliability, and overall performance.

CLO3: Software Development Proficiency: Students will exhibit competence in writing efficient and effective software applications by understanding the interaction between software and operating systems.

CLO4: Troubleshooting and Debugging Abilities: Students will enhance their skills in diagnosing and resolving issues related to system performance, software compatibility, and resource utilization.

CLO5: Preparation for Specialized Areas: Students will be prepared for advanced studies in specialized fields such as distributed systems and cybersecurity by acquiring foundational knowledge in operating systems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction: Operating system overview, computer system structure, structure and components of an operating system.	Lecture and Discussion	3	CLO1
2	MIPS R3000 processor: overview and programming model, Exceptions, MIPS system call, system161.	Lecture, Hands-on Programming	3	CLO1
3	System calls: class of system calls and description.	Lecture and Examples	3	CLO1, CLO2
4	Process and threads: process and thread model, process and thread creation and termination, user and kernel level thread, scheduling, scheduling algorithms, dispatcher, context switch, real-time scheduling, OS-161 switch.	Lecture, Case Studies, and OS-161 Labs	3	CLO1, CLO3

5	Concurrency and synchronization: IPC and inter-thread communication, critical region, critical section problems and solutions.	Lecture, Examples, and Group Discussions	3	CLO4
6	Resource management: introduction to deadlock, ostrich algorithm, deadlock detection and recovery, deadlock avoidance, starvation.	Lecture, Case Studies	3	CLO3, CLO4
7	Memory management: basic memory management, fixed and dynamic partition, virtual memory, segmentation, paging and swapping, MMU. Virtual memory management: paging, page table structure, page replacement, TLB, R3000 TLB and address space, R3000 TLB handling, exception vector, demand paging and segmentation, thrashing and performance.	Lecture, Hands-on Exercises	3	CLO4
8	File management: File Naming and structure, file access and attributes, system calls, file organization: OS and user perspective view of file, memory-mapped file, file directories organization.	Lecture, Case Studies	3	CLO3, CLO4
9	Case study: UNIX file access permissions and rights. File System Implementation: implementing file, allocation strategy, method of allocation, directory implementation, UNIX i-node, block management, quota.	Lecture, Group Projects	3	CLO5
10	I/O management: I/O Devices, I/O Bus architecture and controller, interrupts, DMA, programmed I/O, Evolution of I/O functions, I/O software layer, Device drivers, Device-independent I/O software, buffering.	Lecture, Lab Exercises	3	CLO4, CLO5
11	Disk I/O management: structure, performance, low-level disk formatting, Disk arm scheduling algorithm, error handling, and stable storage. RAID: RAID 0-5, HP auto RAID. Multiprocessor system: UMA MP, NUMA, SMP- structure and programming model, synchronization, scheduling	Lecture, Case Studies, and Group Discussions	3	CLO5
12	Security: threats, data security, intruders, data loss, user authentication, password security and salt, one-way function, authentication using a physical object, software threats, Trojan Horses, spoofing, trap doors, viruses, anti-virus approach and technique, sandbox	Lecture, Examples	3	CLO5

	implementation, security policy and mechanism, protection mechanism, protection domain, Access Matrix, access control list, capabilities.			
13	UNIX file management: Berkeley fast file system (FFS) Ext2fs, Ext3fs, superblocks, partition, Ext2fs and Ext3fs Directories, supporting multiple file systems, OS-161 VFS, UNIX buffer cache, file system consistency.	Lecture, Hands-on Labs	3	CLO5
14	Revision and Q/A	Review Session	3	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	H										
CLO2			H	H	H	H			M			
CLO3		H	H	M								
CLO4		H				H			H			M
CLO5			H		H		M					M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		

Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Operating System Concepts, Latest Edition, Silberchatz, Galvin, Greg

Reference Books:

1. Operating System Concepts, Latest Edition, W. Stallings

2. Modern Operating system, Latest Edition, Tanenbaum
CSEL-3105: Operating Systems Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-3105	Operating Systems Lab	Lab	1	35	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course is essential as it enables students to apply theoretical concepts through hands-on exercises, fostering the development of coding proficiency, debugging skills, and practical understanding of key OS components, preparing them for real-world challenges in system development and administration.

Justification of this course in the program entity

This lab course is mandatory, emphasizing its significance in providing students with practical experiences related to the design and implementation of fundamental operating system concepts. By focusing on system calls, CPU scheduling, process management, memory management, file systems, and deadlock handling, students gain hands-on exposure to critical components of operating systems. The utilization of the C language within a Linux environment adds a practical dimension, allowing students to apply theoretical knowledge in a real-world context, enhancing their proficiency in coding and problem-solving within the specific operating system domain. This experiential learning approach not only deepens their understanding of operating systems but also equips them with valuable skills for future endeavors in software development and system administration.

Course Objectives

The major objectives of the course are:

- Apply theoretical knowledge through practical exercises.
- Develop proficiency in the C language within a Linux environment.
- Enhance problem-solving skills in system-related scenarios.

- iv. Deepen understanding of fundamental operating system concepts.
- v. Prepare students for real-world challenges in software development and system administration.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Demonstrate the ability to implement and execute diverse system calls within the operating system kernel.

CLO2: Evaluate and compare various operating systems through practical testing methodologies using virtualization techniques.

CLO3: Simulate and analyze distinct functionalities inherent to operating systems, gaining a comprehensive understanding of their operations.

CLO4: Successfully compile the kernel of an operating system, showcasing proficiency in the compilation process and its associated components.

CLO5: Develop practical skills in diagnosing, troubleshooting, and resolving issues related to the kernel and core functionalities of operating systems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Virtual machine creation for testing different operating systems	Lecture, Hands-on Lab	2	CLO2
2	Linux basic commands	Lecture, Practical Exercises	2	CLO2
3	Linux advanced commands	Lecture, Practical Applications	2	CLO2
4	Shell Scripting: beginning and intermediate level	Lecture, Scripting Exercises	2	CLO2
5	Make system calls	Lecture, System Call Implementation	2	CLO1
6	Simulating CPU scheduling algorithm: FCFS, SJF-nonpreemptive	Lecture, Simulation Exercises	2	CLO3

7	Simulating CPU scheduling algorithm: SJF-preemptive, round robin	Lecture, Simulation Tasks	2	CLO3
8	Simulating CPU scheduling algorithm: priority	Lecture, Simulation Activities	2	CLO3
9	Simulating Deadlock problems	Lecture, Deadlock Simulation	2	CLO3
10	Designing simulator for classic problems of synchronization	Lecture, Design Exercises	2	CLO3
11	Simulation of memory management	Lecture, Memory Simulation	2	CLO3
12	Writing C program for implementing different Page replacement algorithms	Lecture, Programming Tasks	2	CLO1, CLO3
13	File management, protection and security	Lecture, Case Studies	2	CLO5
14	Revision and Q/A	Review Session	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1					H	H						
CLO2						H						
CLO3		H			H	H			M			
CLO4		H			H	H			H			
CLO5		M	H	H		H			H			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

2. Operating System Concepts, Latest Edition, Silberchatz, Galvin, Greg

Reference Books:

3. Operating System Concepts, Latest Edition, W. Stallings
4. Modern Operating system, Latest Edition, Tanenbaum

CSE-3106: Microprocessor and Assembly Language**PART A: INTRODUCTION**

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3106	Microprocessor and Assembly Language	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is a foundational course providing an introduction to the architecture, operation, and programming of microprocessors, focusing on Intel x86 architecture. The course covers fundamental concepts of digital logic, instruction set architecture, memory interfacing, input/output operations, and assembly language programming. Students gain hands-on experience in writing, debugging, and optimizing assembly language programs for real-world applications. Topics also include interrupts, I/O interfacing, and an overview of advanced microprocessor concepts. This course equips students with essential skills in low-level programming and understanding the intricacies of microprocessor operations.

Justification of this course in the program entity

This course holds significant relevance in computer science and engineering education as it provides fundamental insights into the underlying architecture and operation of computing systems. Understanding microprocessors and assembly language programming is crucial for computer scientists as it enables them to comprehend the low-level interactions between hardware and software. Mastery of these concepts equips students with the ability to optimize code efficiency, debug system-level issues, and develop embedded systems and device drivers. Moreover, knowledge of microprocessors and assembly language enhances students' critical thinking and problem-solving skills, fostering a deeper understanding of computer architecture principles essential for various fields such as software development, system design, and cybersecurity.

Course Objectives

The major objectives of the course are to:

- i. Understand the fundamental concepts of microprocessor architecture, including CPU operation, instruction execution, and memory management.
- ii. Develop proficiency in assembly language programming for Intel x86 architecture, enabling students to write, debug, and optimize low-level code.
- iii. Gain practical experience in interfacing with peripherals and external devices through input/output operations.
- iv. Explore advanced topics such as interrupts, memory interfacing, and I/O interfacing to broaden understanding of microprocessor functionality.
- v. Apply theoretical knowledge to real-world scenarios, fostering problem-solving skills and preparing students for careers in software development, system design, and embedded systems engineering.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be

CLO1: Understand core concepts of Microprocessor:

CLO2: Understand the theoretical knowledge of different types of Microprocessor

CLO3: Design, implement, and analyze Pentium Microprocessor:

CLO4: Implement basic Assembly Language Programming

CLO5: Demonstrate the ability to effectively debug and troubleshoot assembly language programs, identifying and rectifying errors in code logic and syntax.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction and Basics of a Microprocessor: Evolution of Microprocessor, overview of microcomputer structure and operation, introduction to RISC and CISC processors.	• Lecture	3	CLO1
2	8086 Microprocessor: Introduction to 8086 microprocessor, 8086 architecture, registers and other components of 8086 system	• Lecture • Group work	3	CLO1, CLO2
3	8086 Microprocessor: 8086 instruction sets, constructing machine codes for 8086 instructions. 8086 system connections,	• Lecture • Rapport building •	3	CLO2

	timing and troubleshooting,			
4	8086 Microprocessor: 8086 system connections, timing and troubleshooting, DMA	• Lecture • Exercise	3	CLO2
5	8086 Microprocessor: 8086 interrupts and interrupt applications, Architecture of 8259A	• Lecture • Group work • Exercise	3	CLO2
6	8086 Microprocessor: interrupt vector table priority interrupt controller, higher versions of 8086.	• Lecture • Exercise	3	CLO2
7	Pentium Microprocessor: Introduction to Pentium Microprocessor,	• Lecture	3	CLO3
8	Pentium Microprocessor: Pentium processor architecture, Register sets, cache, floating point operations, addressing modes, addressing,	• Lecture • Exercise	3	CLO2
9	Pentium Microprocessor: paging, Pentium process instruction set, opcode, interrupt, programming in Pentium machine	• Lecture • Demonstration • Group work	3	CLO3, CLO4
10	Pentium Microprocessor: Hardware details of Pentium, Protected mode operations, branch prediction.	• Lecture • Rapport building • Demonstration • Group work	3	CLO3
11	Assembly Language Programming: Writing programs for use with an assembler, assembly language program development tools	• Lecture • Rapport building • Demonstration • Group work	3	CLO3, CLO4, CLO5
12	Assembly Language Programming implementing standard program structures in x86 (8086, i386 and Pentium) assembly language	• Lecture • Demonstration • Group work	3	CLO4, CLO5
13	Assembly Language Programming testing and debugging an assembly language program, processing string, macros and procedures, assembler directives.	• Lecture • Rapport building • Group work	3	CLO4, CLO5

14	Revision and Solution	Interactive class Q/A	2	All CLOs
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Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H							M		
CLO2		H		H							H	
CLO3			H				H					
CLO4			H									
CLO5		H		H								M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10

Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. D.V. Hall, Microprocessors and Interfacing, latest edition.
2. M. Rafiquzzaman, Microprocessors and Microprocessor based Systems Design, latest edition.
3. Barry B. Brey, The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro-Processor /architecture, latest edition.
4. Lewis C. Eggebrech, Interfacing to the IBM Personal Computer, latest edition.

CSEL-3107: Microprocessor and Assembly Language Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)			
				Semester Final Exam.	Continuous Assessment		
					Lab Assessment/Lab Report	Class Performance	Class attendance

CSEL-3 107	Microprocessor and Assembly Language Lab	Lab	1	30	10	5	5	50
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Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course is a practical, hands-on component accompanying the theoretical study of microprocessors and assembly language programming. In this lab-based course, students apply the knowledge gained in lectures to real-world scenarios, working directly with microprocessor hardware and assembly language software. Through a series of structured exercises and projects, students learn to design, implement, and debug assembly language programs for Intel x86 architecture. They gain proficiency in interfacing with peripherals, manipulating memory, and executing input/output operations.

Justification of this course in the program entity

It is a mandatory course. Through hands-on experience with microprocessor hardware and assembly language programming, students bridge the gap between abstract concepts and real-world applications. This lab cultivates essential skills such as debugging, optimization, and system-level analysis, which are invaluable in software development, system design, and embedded systems engineering. Moreover, it fosters a deeper comprehension of computer architecture principles, enhancing students' ability to design efficient algorithms and troubleshoot complex system-level issues. By immersing students in a practical environment where they can experiment and innovate, this lab course prepares them for the demands of modern computing, ensuring they graduate with the practical expertise needed to excel in diverse fields within the realm of computer science.

Course Objectives

The major objectives of the course are:

- i. Develop a solid understanding of microprocessor architecture and its components, including the CPU, memory, and input/output systems.
- ii. Gain proficiency in assembly language programming, enabling students to write, debug, and optimize code for Intel x86 architecture.
- iii. Learn to interface with peripherals and external devices, mastering techniques for input/output operations and memory manipulation.
- iv. Explore advanced topics such as interrupts, memory management, and I/O interfacing to expand knowledge of microprocessor functionality.

- v. Apply theoretical knowledge to practical scenarios, fostering problem-solving skills and preparing students for real-world applications in software development, system design, and embedded systems engineering.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Discuss different concepts of Microprocessors

CLO2: Analyze the major component of a Computer system and steps involved in assembling, linking, and executing a program to run the system.

CLO3: Create program for microcontroller using assembly language and Programming knowledge

CLO4: Create program for Assembly Language Programming

CLO5: Demonstrate proficiency in debugging and optimizing assembly language programs, identifying and rectifying errors in code logic and syntax to ensure efficient program execution.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Assembly Language and EMU 8086	• Lab. Manual, Experiment No. 1	2	CLO1
2	Program structure and familiarization with different instructions.	• Lab. Manual, Experiment No. 1	2	CLO1
3	Arithmetic operations	• Lab. Manual, Experiment No. 2	2	CLO1, CLO2
4	Understanding Advanced 8086 I/O Instructions	• Lab. Manual, Experiment No. 3	2	CLO2, CLO3
5	Use of Loop	• Lab. Manual, Experiment No. 4	2	CLO2, CLO3
6	Use of Array	• Lab. Manual, Experiment No. 5	2	CLO2, CLO3
7	Lab Performance Test-1	• Lab. Manual, Experiment No. 1-5	2	CLO1, CLO2, CLO3
8	Understanding the use of Procedure	• Lab. Manual, Experiment No. 6	2	CLO2, CLO3

9	Familiarizing with MDA 8086 trainer Kit Commands in Serial Monitor Mode	• Lab. Manual, Experiment No. 7	2	CLO2, CLO3
10	Problem solving	• Lab. Manual, Experiment No. 8	2	CLO3, CLO5
11	Assembly Language Programming: Writing programs for use of String	• Lab. Manual, Experiment No. 9	2	CLO4, CLO5
12	Lab Performance Test-2	• Lab. Manual, Experiment No. 6-9	2	CLO2, CLO3, CLO4
13	Assembly Language Programming: Writing programs for GCD/LCM, miscellaneous problem	• Lab. Manual, Experiment No. 10	2	CLO4
14	Final Examination (Quiz, Lab test)		2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H							H		
CLO2		H		H							M	
CLO3			H				H					
CLO4			H									
CLO5		H	H	H								M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		

Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. D.V. Hall, Microprocessors and Interfacing, latest edition.
2. M. Rafiquzzaman, Microprocessors and Microprocessor based Systems Design, latest edition.
3. Barry B. Brey, The Intel Microprocessors 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro-Processor /architecture, latest edition.
4. Lewis C. Eggebrech, Interfacing to the IBM Personal Computer, latest edition.

CSE-3108: Computer Networks

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3108	Computer Networks	Theory	3	60	20	10	10	100

Course Description: Computer Networks is a compulsory course for the B.Sc. (Honours) program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course will build up student's ability to understand fundamental concepts of computer networking, diverse networking techniques that are used in the field of communication. This course will familiarize them with different network layers, their working principles and identify various issues in computer networking. This course will motivate students to learn networking skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- understand fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries, etc.
- make themselves able to write a computer program to solve specified problems.
- use the Java SDK environment to create, debug and run simple Java programs.
- build themselves as a professional developer in Java.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1- To understand the principles and concepts of computer networks.

CLO2- To Understand the use of transmission mechanism and medium, Understand the network configuration and routing mechanisms and access control methods for designing computer networks.

CLO3- To explain and learn about the basics of wireless communication

CLO4- To explain and identify security and ethical issues in computer networking.

CLO5- To understand the basic concepts how data travels from one end to other end.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	Corresponding CLOs
1	Internet Architecture, Circuit and Packet Switching, Access Systems. Delay Calculation. Protocol Layers,	• Demonstration • Discussion	3	CLO1, CLO2
2	Internet Architecture, Circuit and Packet Switching, Access Systems. Delay Calculation. Protocol Layers,	• Demonstration • Discussion • Group work	3	CLO1, CLO2
3	Application Layer: Principles of Application Layers, HTTP2, HTTP3, FTP,	• Demonstration • Discussion • Group work	3	CLO1, CLO2
4	Application Layer: E-mail, DNS, SNMP, Socket Programming, P2P Networks.	• Demonstration • Discussion • Group work	3	CLO1, CLO2
5	Transport Layer: Introduction to services, Multiplexing, UDP,	• Demonstration • Discussion • Group work	3	CLO1, CLO2
6	Transport Layer: Reliable data delivery, TCP, Congestion Control. TCP Tahoe, TCP Reno, TCP Cubic	• Demonstration • Discussion • Group work	3	CLO1, CLO2
7	Network Layer: Virtual Circuits, IP, Addressing, SDN, Data plane, Control plane	• Demonstration • Discussion • Group work	3	CLO1, CLO2
8	Network Layer: Router Internals, Routing Algorithms. SDN routing	• Demonstration • Discussion • Group work	3	CLO1, CLO2
9	Data Link Layer: Services, Error detection, Multiple Access Protocols,	• Demonstration • Discussion • Group work	3	CLO1, CLO2
10	Data Link Layer: Multiple Access Protocols, Link layer addressing, Ethernet, Switches, PPP.	• Demonstration • Discussion • Group work	3	CLO1, CLO2

11	Wireless and Mobile Networks: Wireless characteristics, CDMA,	• Demonstration • Discussion • Group work	3	CLO3
12	Wireless and Mobile Networks: Cellular Networks, Mobility.	• Demonstration • Discussion • Group work	3	CLO3
13	Network Security: Symmetric and Public Key Cryptography, Authentication, Digital Signatures, Key Distribution and certification, Firewalls.	• Demonstration • Discussion • Group work	3	CLO4
14	Revision and Solution	• Demonstration • Discussion • Group work	3	CLO1

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H		M		M							
CLO2	H		M		M							
CLO3	H		M									
CLO4	H		H									
CLO5	H	H	H									

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of

these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

1. J.F. Kurose, K.W. Ross - Computer Networking (8th Edition)
2. A.S. Tanenbaum, Computer Networks.
3. D Comer, Computer Networks.
4. L.L. Peterson, Computer Networks.

CSE-3109: Computer Networks Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessme nt/Lab Report	Class Performance	Class attendance	
CSE-3109	Computer Networks Lab	Practical	1	30	10	5	5	50

Course Description: Computer Networks lab is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 1 credits and two hours lab class duration weekly. Among the 30 hours, 24 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course concentrates on the practical part of Computer Networks by using Cisco Packet Tracer simulator, Network simulator, wireshark packet analyzer. This course allows students to design and perform simulations based on the concepts they will learn through the course by solving different problem solutions.

Course Objectives:

The objectives of the course are as follows:

1. To develop basic concepts and principles about practical tools used in computer network lab.
2. To create, implement, debug, and evaluate algorithms and techniques for solving substantial problems.

3. To understand how the theoretical algorithms works in real environments
4. To deliver and evaluate basic technical documents, presentations, and group interactions using appropriate tools.

Course Learning Outcomes (CLO)

Upon completion of the course, the students will be able to:

- a. CLO1- To develop basic concepts and principles about networking lab
- b. CLO2- To design and create different networks using different simulation tools and will understand the core concepts about the working principle
- c. CLO3- To provide solutions about network related problems
- d. CLO4- To understand the working principles of different protocol layers
- e. CLO5- To gain knowledge about how to apply them in real life problems

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Corresponding CLOs
1	(Introduction) Introduction to Packet Tracer, WireShark tools	• Demonstration • Discussion	CLO1, CLO2
2	Implementing Classful and classless IP addressing	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4, CLO5
3	LAN and VLAN setup	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4, CLO5
4	TCP/IP Socket programming	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4
5	TCP/IP Socket programming	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4, CLO5
6	TCP/IP Socket programming	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4, CLO5
7	Packet analysis using wireshark	• Demonstration • Discussion • Group work	CLO1,CLO2, CLO3,CLO4, CLO5
8	Packet analysis using wireshark	• Demonstration • Discussion • Group work	CLO1, CLO2, CLO3, CLO4
9	Server configuration (FTP)	• Demonstration • Discussion	CLO1, CLO2, CLO3, CLO5

		Group work	
10	Server configuration (HTTP)	- Demonstration - Discussion - Group work	CLO1, CLO2, CLO3, CLO4
11	Introduction to Network Simulator 3 (NS-3)	- Demonstration - Discussion - Group work	CLO1, CLO2, CLO3, CLO4
12	Implementing p2p and LAN network using NS3	- Demonstration - Discussion - Group work	CLO1, CLO2, CLO3, CLO4
13	Implementing wireless network using NS3	- Demonstration - Discussion - Group work	CLO1, CLO2, CLO3, CLO4
14	Implementing routing network using NS3	- Demonstration - Discussion - Group work	CLO1, CLO2, CLO3, CLO5

Mapping between CLOs and PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PL O-1	PL O-2	PL O-3	PL O-4	PL O-5	PL O-6	PL O-7	PL O-8	PL O-9	PL O-10	PL O-11	PL O-12
CLO-1	√				√							
CLO-2			√			√				√		
CLO-3							√				√	
CLO-4											√	√
CLO-5		√		√			√			√		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		

Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 5 marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. J.F. Kurose, K.W. Ross - Computer Networking.
2. A.S. Tanenbaum, Computer Networks.
3. D Comer, Computer Networks.
4. L.L. Peterson, Computer Networks.

CSEP-3110: Internet and Web Programming (Project)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Progress Evaluation	Class Participation	Class attendance	
CSEP-3110	Internet and Web Programming (Project)	Practical Course	1.0	30	10	5	5	50

Course Description: Internet and Web Programming (Project) is a compulsory practical course for the B.Sc. (Honours) program in the Department of Computer Science and Engineering, Jagannath University. The course carries 1.0 credits and 30 lecture hours. This is an implementation based course and hence the complete lecture hours are allocated for discussion, individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

Incorporating an Internet and Web Programming course into the curriculum is imperative to equip students with the essential skills and knowledge needed to navigate and thrive in the modern digital landscape. In this interconnected world, the internet serves as a cornerstone for communication, commerce, and collaboration, making proficiency in web programming indispensable. This course will provide students with a comprehensive understanding of web technologies, including HTML, CSS, JavaScript, and server-side scripting languages such as PHP or Python. By delving into topics such as web development frameworks, database integration, and responsive design, students will learn to create dynamic and interactive web applications that are scalable, secure, and user-friendly. Moreover, as the demand for skilled web developers continues to soar across industries, this course will not only enhance students' employability but also empower them to contribute meaningfully to the ever-evolving field of web development.

Course Objectives

The major objectives of the course are:

- Provide students with a comprehensive understanding of fundamental web technologies, including HTML, CSS, and JavaScript, as well as server-side scripting languages such as PHP, Python, or Node.js.
- Familiarize students with database management systems (DBMS) and teach them how to integrate databases into web applications using technologies such as MySQL, MongoDB, or SQLite for storing and retrieving data dynamically.

- iii. Enable students to implement client-server communication mechanisms such as AJAX (Asynchronous JavaScript and XML) or Fetch API to facilitate seamless data exchange between the client-side and server-side components of web applications.
- iv. Teach students principles of user interface (UI) and user experience (UX) design to create intuitive and user-friendly web interfaces, including concepts such as responsive design, accessibility, and usability testing.
- v. Provide students with hands-on experience through project-based learning, where they will have the opportunity to apply their knowledge and skills to develop real-world web applications, reinforcing theoretical concepts and honing practical programming skills.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: learn principles of user interface (UI) and user experience (UX) design and be able to apply these principles to create intuitive, accessible, and user-friendly web interfaces that enhance the overall user experience.

CLO2: demonstrate proficiency in front-end web development technologies, including HTML, CSS, and JavaScript, and be able to create visually appealing and responsive user interfaces for web applications.

CLO3: gain proficiency in server-side scripting languages such as PHP, Python, or Node.js, and be able to develop server-side components and dynamic web applications that interact with databases and other server resources.

CLO4: develop critical thinking and problem-solving skills essential for troubleshooting and debugging web applications, enabling them to identify and resolve issues efficiently and effectively.

CLO5: apply their knowledge and skills to develop real-world web applications, reinforcing theoretical concepts and gaining practical experience in web development.

PART B: CONTENT OF THE COURSE

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H									
CLO2		H	H									
CLO3					H							
CLO4			H		H				H			
CLO5			H			H				H		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Project Report	√	√	√		√
Viva/Quiz	√	√	√	√	√
Group Discussion	√	√	√	√	√
Final Presentation	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Project Report (10)	Viva/Quiz (5)	Group Discussion (5)
Remember			
Understand		25%	25%
Apply	25%	40%	40%
Analyze	40%	25%	25%
Evaluate	25%	10%	10%
Create	10%		

Semester Final Examination (30 Marks):

Bloom's Category	Marks
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains weekly group discussion with the course teacher and presenting the weekly work progress. In the final examination, students have to present their work as an oral presentation as well as demonstrate their implemented project in front of the examiners with necessary question answering session. The presentation and demonstration carries 20 marks whereas the viva carries 15 marks.

PART D: LEARNING RESOURCES

Text Book:

1. HTML and CSS: Design and Build Websites" by Jon Duckett
2. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5" by Robin Nixon

Reference Books:

3. Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics" by Jennifer Niederst Robbins

3rd Year 2nd Semester

CSE-3201: Compiler Design and Construction

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class Attendance	
CSE-3201	Compiler Design and Construction	Theory	3	60	20	10	10	100

Course Description: The course is intended to teach the basics related to the practice of Compiler design and construction. This course is designed to reflect both theoretical learning as well as tools and techniques necessary to perform the translation of a high-level programming language into an executable program. In addition, the course will also discuss various aspects of the run-time environment into which the high-level code is translated.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the Java programming language. It aims to develop the students logical, critical, and problem-solving skills on programming basics and also on the modern context of object-oriented programming through exercises and hands-on activities. Java is popular among professional programmers because it can be used to build visually interesting graphical user interface (GUI) and Web-based applications. This course familiarizes students with basic as well as advanced features of Java Programming Emphasis will be given to GUI and event-driven programming, Database Connectivity, Socket Programming, Servlets and JSP Technology, and Distributed Programming. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- i. to understand core concepts of compiler.
- ii. to understand the compilation process.
- iii. to design, implement, and analyze compilers.
- iv. to implement basic compiler tools

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding core concepts of compiler

CLO2: Understanding the compilation process

CLO3: Designing, implementing, and analyzing compilers

CLO4: Implementing basic compiler tools

CLO5: Analyzing compilers for improvement and research

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction and Basics of a Compiler: Introduction to the compiler and basics of compiler	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
2	Lexical Analysis: Phases of a compiler, front and back end of a compiler, Regular expressions and regular languages, Finite Automata based pattern matching, Input buffering	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture	3	CLO 2

	techniques	materials, question-answer sessions, etc. b) Problem-based class		
3	Syntax Analysis: Context free grammars, Introduction to Top-down parsing, Top-down parsing: LL(1) parsing, Recursive Descent parsing; Bottom-up parsing: LR(0) parsing, LR(1) parsing, LALR(1) parsing, SLR(1) parsing, Syntactic error recovery.	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
4	Syntax directed Definitions and Translation: Syntax directed Definitions and Translation: Attributes evaluation	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
5	Symbol Tables, Type expressions and type checking, Runtime structures: Activation Records, Static and Dynamic Scoping	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
6	Intermediate Representation: Abstract syntax trees, 3-address code, Generation of 3-address codes – Syntax directed translation for Declarations	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3

7	Intermediate Representation: Generation of 3-address codes – Assignment statements, Boolean expression, switch/case statements, Flow of Control statements, Use of Backpatching	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
8	Target Code generation: Target Code generation techniques	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
9	Optimization: Control flow graphs	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
10	Data flow Analysis: Reaching definitions and Live-variable analysis and Def-use & use-def chains, Available Expression analysis and Global common subexpression elimination, Dominators, Loops in control flow graphs, Loop invariants and code motion, Elimination of Induction variables, Partial redundancy elimination, constant folding and constant propagation, copy	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4

	propagation, Dealing with Aliasing, Interprocedural Dataflow Analysis			
11	SSA: Introduction to Static Single-Assignment (SSA) form	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
12	Global Register allocation by graph coloring, Instruction Scheduling: list scheduling	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
13	Memory Optimization: Optimization for memory hierarchies	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PL O1	PL O2	PL O3	PL O4	PL O5	PL O6	PL O7	PL O8	PL O9	PL O10	PL O11	PL O12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						

CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10

Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

- Aho, Alfred, V, Compiles: Principles, Techniques

Reference Books:

- John Wiley, Modern Compiler Design
- H. Ball, Modern Compiler Design

CSEL-3202: Compiler Design and Construction Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Attendance	Class Performance	
CSEL-3202	Compiler Design and Construction Lab	Practical	1	30	10	5	5	50

Course Description: The course is intended to make the students experiment on the basic techniques of compiler construction and tools that can be used to perform syntax-directed translation of a high-level programming language into an executable code.

Course Objectives:

The objectives of the course are as follows:

- to understand the working of lex and yacc compiler for debugging C programs
- to use lexical analyzer, Context-free grammar, and parse tree construction methods
- to create, implement, debug, and evaluate algorithms for solving substantial problems.
- to learn the new tools and technologies used for designing a compiler

Course Learning Outcomes (CLOs):

After completing the course, successful students will be able to:

CLO1: Understanding the working of lex and yacc compiler for debugging C programs

CLO2: Using lexical analyzer, Context-free grammar, and parse tree construction methods

CLO3: Learning the new tools and technologies used for designing a compiler

CLO4: Implement different algorithms and techniques related to compilation.

CLO5: Analyzing compilers for better development and research.

PART B: CONTENTS OF THE COURSE

Course description and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Compiler Design and Construction Lab: Installation and	Lectures, Coding Exercises	2	CO1

	overview of the Required SW (Lex, YACC, BISON)			
2	Lexical Analyzer: Identify the function of Lexical Analyzer, Understand the implementation of Lexical Analyzers	Theoretical Explanation, Coding Practice	2	CO1, CO4
3	Lexical Analyzer: Design a lexical analyzer for a given language and the lexical analyzer should ignore redundant spaces, tabs and newlines	Lectures, Problem-solving Sessions	2	CO4
4	Regular Expressions: Checking and implementing regular expression for various conditions	Interactive Lectures, Algorithm Implementation	2	CO2
5	Syntax Analyzer: Implement different techniques for making a parser (First, Follow, Left Recursion and Left Factoring Removal)	Practical Examples, Coding Sessions	2	CO2, CO4
6	Syntax Analyzer: Apply different parsing algorithms on grammars in C to create parsers (LR (0), LL(1))	Theoretical Explanation, Coding Practice	2	CO2
7	Syntax Analyzer: Apply different parsing algorithms on grammars in C to create parsers SLR, CLR	Lectures, Problem-solving Sessions	2	CO1, CO4
8	Syntax Analyzer: Construction of parse tree using Cation, and rod cutting problems	Coding Workshops, Practical Examples	2	CO1, CO4, CO5
9	Intermediate code generation: Implement algorithms for Intermediate code generation	Theoretical Explanation, Problem-solving Sessions	2	CO4, CO5
10	Code optimization: Implement algorithms for code optimization	Interactive Sessions, Coding Exercises	2	CO5
11	Problem Solving	Lectures, Algorithm Implementation	2	CO1, CO4
12	Problem Solving	Practical Examples, Coding Sessions	2	CO1, CO2,

				CO4, CO5
13	Problem Solving	Problem-solving Sessions, Coding Practice	2	CO2, CO4
14	Revision and Q/A	Review Sessions, Open Discussion	2	CO1, CO2, CO3, CO4, CO5

Mapping between CLOs and PLOs:

	PL O1	PL O2	PL O3	PL O4	PL O5	PL O6	PL O7	PL O8	PL O9	PL O10	PL O11	PL O12
CLO1	M		H						H			
CLO2	M	H	H						H			
CLO3	M		H						H			
CLO4	M								H			M
CLO5	M		H	H	H				M			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Aho, Alfred, V, Compiles: Principles, Techniques

Reference Books:

1. John Wiley, Modern Compiler Design
2. H. Ball, Modern Compiler Design

CSE-3203: Computer Graphics and Animation

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3203	Computer Graphics and Animation	Theory	3	60	20	10	10	100

Course Description: The course is intended to teach students about the basic principles and concepts, hardware software requirements and algorithms of computer graphics and animation; so that they can apply the techniques to develop real life applications in this field.

Justification of this course in the program entity

This course provides the beginning programmer with a guide to developing applications using the Java programming language. It aims to develop the students logical, critical, and problem-solving skills on programming basics and also on the modern context of object-oriented programming through exercises and hands-on activities. Java is popular among professional programmers because it can be used to build visually interesting graphical user interface (GUI) and Web-based applications. This course familiarizes students with basic as well as advanced features of Java Programming Emphasis will be given to GUI and event-driven programming,

Database Connectivity, Socket Programming, Servlets and JSP Technology, and Distributed Programming. This course will motivate students to learn programming skills while building a cornerstone for a Computer Science curriculum.

Course Objectives

The major objectives of the course are:

- i. understand the basics of computer graphics as well as get an understanding of applications of computer graphics.
- ii. realizing computational development of graphics with mathematics and the basics of multimedia.
- iii. identify and Apply suitable drawing algorithms to solve particular problems.
- iv. experience development of multimedia software by utilizing existing libraries and descriptions of algorithms.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understanding the basics of computer graphics as well as get an understanding of applications of computer graphics.

CLO2: Realizing the computational development of graphics with mathematics and the basics of multimedia.

CLO3: Identifying and applying suitable drawing algorithms to solve particular problems.

CLO4: Experiencing development of multimedia software by utilizing existing libraries and descriptions of algorithms.

CLO5: Analyzing graphics contents for betterment and research.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hour	CLO s
1	Introduction and Overview on Computer Graphics: Standard Graphics Primitives, Graphical User Interface; Graphics Hardware Display devices, Raster refresh graphics display	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 1
2	Introduction and Overview on Computer Graphics: Use of frame	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based	3	CLO 2

	buffer and look up table Coordinate convention Device coordinate and world coordinate system, Vector graphics and raster graphics system	lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
3	Scan Conversion Algorithms: Mid-point Line, Circle and ellipse Creation Algorithms. Slope independent line drawing using midpoint line algorithm	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
4	Polygons: Difference type of polygons, polygon filling, triangulation, polygon filling algorithm	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 2
5	Windowing and Clipping: Window Viewpoint, Zooming, panning, line, text and polygon, clipping algorithms	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
6	Transformation: Homogeneous coordination, Transformation in 3D, Transformation matrices, translation, rotation, scaling	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
7	Projection: Parallel and perspective, standard	a) Enhanced lectures containing the whiteboard-marker and/or	3	CLO 3

	projection matrices	PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class		
8	Hidden Surface Removal: Painter's algorithm, Z-Buffering, Visible surface ray-tracing algorithm	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 3
9	Illumination and Shading: Light Models, Ambient light, diffuse and specular reflection, light attenuations, Goraud and Phong shading, Recursive Ray Tracing	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
10	Monochrome and Colored Light: Monochrome light, Additive and subtractive light, Colored light-RGB, CMY, YIQ, HSV and HLS color model	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 4
11	Image File Format: PPM file, BMP file and other file formats	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5

12	Representing Curves and Surfaces: Polygonal surfaces, Parametric Cubic Curves- Hermite, Bezier and B-spline curves	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
13	Parametric Bi-cubic Surfaces: Bicubic Splines, Introduction to Graphics Programming, The nature of computer animation	a) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. b) Problem-based class	3	CLO 5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PL O1	PL O2	PL O3	PL O4	PL O5	PL O6	PL O7	PL O8	PL O9	PL O10	PL O11	PL O12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	

Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1

2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Foley, Computer Graphics: Pearson.

Reference Books:

1. Steven , Roger, Advance Graphics Theory
2. VasantHonovar, Artificial Intelligence & Neural Networks
3. A.K. Peters, Fundamentals of Computer Graphics
4. PrabhatAndleigh and KiranThakrar, Multimedia Systems Design.
5. Ralf Steinmetz, Multimedia Systems.
6. S. V. Raghavan, Satish K. Tripathi, Networked Multimedia Systems.

CSEL-3204: Computer Graphics and Animation Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)			
				Semester Final Exam.	Continuous Assessment		Total
					Two Mid-Semester Exams	Class Attendance	
CSEL-3204	Computer Graphics	Practical	1	35	10	5	50

	and Animatio n Lab						
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Course Description: The course is intended to teach students the basic principles, hardware software requirements and algorithms of computer graphics, so that they can apply the techniques to develop real life applications in this field.

Course Objectives:

The objectives of the course are as follows:

- i. to learn the principles and commonly used paradigms and techniques of computer graphics
- ii. to apply graphics programming techniques to design, and create computer graphics scenes
- iii. to create, implement, debug, and evaluate algorithms for solving substantial problems.
- iv. to develop multimedia software by utilizing existing libraries and descriptions of algorithms

Course Learning Outcomes (CLOs):

After completing the course, successful students will be able to:

CLO1: Learning the principles and commonly used paradigms and techniques of computer graphics

CLO2: Applying graphics programming techniques to design, and create computer graphics scenes

CLO3: Developing multimedia software by utilizing existing libraries and descriptions of algorithms

CLO4: Developing creativity and sense of beauty among students.

CLO5: Analyzing graphic tools to develop new one and research.

PART B: CONTENTS OF THE COURSE

Course description and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs

1	Introduction to Computer Graphics Lab: Installation and overview of the Required SW	Lectures, Coding Exercises	2	CO1
2	Digital Differential Analyzer Algorithm (DDA): Test and implement DDA Line drawing algorithm	Theoretical Explanation, Coding Practice	2	CO1, CO4
3	Digital Differential Analyzer Algorithm (DDA): Apply method to generate image using line	Lectures, Problem-solving Sessions	2	CO4
4	Bresenham's line drawing algorithm: Test and implement Bresenham's Line drawing algorithm, Apply method to generate image using line	Interactive Lectures, Algorithm Implementation	2	CO2
5	Circle Generation Algorithms: Test and implement Circle drawing algorithms	Practical Examples, Coding Sessions	2	CO2, CO4
6	Circle Generation Algorithms: Apply method to generate image using circle, Compare polynomial and trigonometric methods	Theoretical Explanation, Coding Practice	2	CO2
7	Algorithms: Test and implement ellipse drawing algorithms	Lectures, Problem-solving Sessions	2	CO1, CO4
8	Algorithms: Apply method to generate image using ellipse, Compare polynomial and trigonometric methods	Coding Workshops, Practical Examples	2	CO1, CO4, CO5
9	Creating various types of texts and fonts: Write program for output Interfacing, Draw relevant diagram, Test output Interfacing API and Device	Theoretical Explanation,	2	CO4, CO5

		Problem-solving Sessions		
10	Creating two dimensional objects: Understand the mathematics to solve problem, State scaling and translation operations, Apply the methods to generate animated image or movie	Interactive Sessions, Coding Exercises	2	CO5
11	Creating two dimensional objects: Discuss different color fill techniques of any object, Demonstrate the basic primaries and secondary colors of each system and accurately print them for reference in a sketchbook	Lectures, Algorithm Implementation	2	CO1, CO4
12	Three Dimensional Transformations: State rotation and mirror reflection operations, Understand the mathematics to solve problem, Apply the methods to generate animated images or movies.	Practical Examples, Coding Sessions	2	CO1, CO2, CO4, CO5
13	Problem Solving	Problem-solving Sessions, Coding Practice	2	CO2, CO4
14	Revision and Q/A	Review Sessions, Open Discussion	2	CO1, CO2, CO3, CO4, CO5

Mapping between CLOs and PLOs:

	PL O1	PL O2	PL O3	PL O4	PL O5	PL O6	PL O7	PL O8	PL O9	PL O10	PL O11	PL O12
CLO1	M		H						H			
CLO2	M	H	H						H			
CLO3	M		H						H			
CLO4	M								H			M

CLO5	M		H	H	H				M			
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PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10

Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Foley, Computer Graphics: Pearson.

Reference Books:

1. Steven , Roger, Advance Graphics Theory
2. VasantHonovar, Artificial Intelligence & Neural Networks
3. A.K. Peters, Fundamentals of Computer Graphics
4. PrabhatAndleigh and KiranThakrar, Multimedia Systems Design.
5. Ralf Steinmetz, Multimedia Systems.
6. S. V. Raghavan, Satish K. Tripathi, Networked Multimedia Systems.

CSE-3205: Software Engineering

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3205	Software Engineering	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is a course designed to introduce students to the principles, methodologies, and best practices involved in developing high-quality software systems. Topics typically include software development lifecycle models, requirements analysis, software design, implementation techniques, testing strategies, software maintenance, and project management. The course often emphasizes both theoretical concepts and practical applications, equipping students with the skills necessary to design, build, and manage software projects efficiently and effectively.

Justification of this course in the program entity

This course is indispensable in the field of Computer Science discipline. This course in a computer science program bridges theory and practice, providing hands-on experience in designing, developing, and maintaining software systems. It equips students with industry-standard tools, project management skills, and collaboration techniques essential for success in software development careers. By staying abreast of evolving industry trends, the course ensures graduates are adaptable and well-prepared for the dynamic field of software engineering.

Course Objectives

The major objectives of the course are to:

- Understand fundamental software engineering principles, methodologies, and best practices.
- Gain proficiency in software design techniques, including modularization and abstraction.
- Develop skills in coding, testing, debugging, and maintaining high-quality software systems.
- Learn effective project management strategies, including Agile methodologies and version control.

- v. Cultivate teamwork and communication skills essential for collaborating on software development projects.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Demonstrate a comprehensive understanding of software engineering principles, methodologies, and best practices.

CLO2: Apply software design techniques effectively to develop modular and scalable software systems.

CLO3: Implement coding, testing, debugging, and maintenance processes to ensure the reliability and quality of software products.

CLO4: Utilize project management methodologies such as Agile and version control tools to efficiently manage software development projects.

CLO5: Collaborate effectively within a team environment, demonstrating strong communication and teamwork skills essential for successful software development.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Wee k	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Definition of software and software engineering, essential attributes of good software, software product, importance of software engineering, characteristics of software, software engineering ethics, applications of software.	Lecture, Demonstration, Group work	2	CLO1
2	Legacy Software, software process layers, fundamental software engineering activities, framework activities, umbrella activities, Polya's suggestions, Hooker's principle, software myths.	Lecture, Rapport building, Demonstration, Group work	3	CLO1
3	Software Processes: Concepts of Software Process and Software process descriptions, Plan-Driven and Agile Processes, Software, Software process	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO4

	model, Waterfall Model, V-Model, Incremental Process Model, and Reuse-oriented Software Engineering. Software process activities, Coping with change.			
4	Software Processes: Software prototyping, Spiral model, Rational Unified Process, Agile Software Processes: Agility, Agile method, Plan-driven and agile development, Agility Principle, Agile Modeling, Agile Unified Process, Extreme programming (XP), refactoring, pair programming.	Lecture, Rapport building, Group work, Cooperative learning, Demonstration	3	CLO1, CLO4
5	Requirements engineering: Basic concepts of requirement engineering, Types of requirements, Functional and non-functional requirements, Software requirements documentation, Specification, Processes, Software requirements elicitation and analysis, Requirement validation and management	Lecture, Rapport building, Demonstration	3	CLO1, CLO5
6	Scenario Based Modeling: Use Case Diagram, Activity Diagram, Swim Lane Diagram Scenario Based Modeling: Use Case Diagram, Activity Diagram, Swim lane Diagram Class Based Modeling: Class Diagram, CRC, Analysis Packages.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO5
7	Behavioral Models: State diagram, Sequence Diagram Architectural Design: Software architecture, Architectural Design decisions, Software architecture and system characteristics.	Lecture, Rapport building, Demonstration, Group work	3	CLO2, CLO5
8	Architectural Design: Architectural views, Architectural pattern. Architectural Design: Model	Lecture, Rapport building,	3	CLO2, CLO5

	View-Controller (MVC) Patterns, Client-server architecture.	Demonstration, Group work		
9	Design and implementation: Design patterns, Implementation issues – Reuse, Configuration management, Host-target development, Open source development, Open source licensing Functional Independence: Cohesion and coupling, modular programming.	Lecture, Rapport building, Project work, Demonstration	3	CLO3, CLO5
10	Software quality assurance (SQA): Elements of software quality assurance, SQA tasks, SQA goals and metrics, Statistical software quality assurance, Six sigma for software engineering. Software testing strategies: Definition of testing, verification and validation (V&V), software testing strategy, Unit testing Software testing strategies: Integration testing, alpha and beta testing, White-box and black-box testing.	Lecture, Rapport building, Project work, Demonstration	3	CLO4
11	Software testing strategies: Cyclomatic Complexity, Path Testing. Software testing strategies: BPT Graph analysis, Equivalence Partitioning, Loop Testing (simple loop, nested loop)	Lecture, Rapport building, Project work, Demonstration	3	CLO4
12	Project management concepts: The management spectrum, the people, the product. Project management concepts: The process, the project etc, 90-90 rule, W5HH principle. Project scheduling: Basic concepts, Project scheduling, basic principles, Time line chart, tracking the schedule. Risk management: Reactive vs. proactive risk strategies, Software risks, Risk identification, Risk projection, Risk mitigation, monitoring and management, RMMM plan.	Lecture, Group work, Demonstration, Group work	3	CLO4

13	Process and product metrics: Metrics in the process and project domains, Project metrics. Estimation for software projects: Project planning process, Resources, Software measurement, Software project estimation	Lecture, Group work, Demonstration, Group work	3	CLO4
14	Revisions, Q&A	Discussions	2	All C LOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	H		H					H		M	
CLO2	M											
CLO3			H		H			H			H	H
CLO4								H	H		H	
CLO5						M		H	H		M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class Attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES**Reference Books:**

1. Software Engineering – Ian Sommerville (Latest Edition)
2. Software Engineering: A Practitioner's Approach – Roger S. Pressman (Latest Edition)

CSEL-3206: Software Engineering Lab**PART A: INTRODUCTION**

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-3206	Software Engineering Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. In this lab course, students will learn the practical aspects of requirement gathering, requirement elicitation, software planning, software designing, software implementation and software testing. Moreover, students will solve a real-world software engineering problem from requirement engineering to software testing as a team.

Justification of this course in the program entity

This course holds a pivotal role within the undergraduate program in the Department of Computer Science and Engineering. As a compulsory course, it serves as a cornerstone in shaping the academic and professional journey of students. It equips students with industry-standard tools, project management skills, and collaboration techniques essential for success in software development careers. By staying abreast of evolving industry trends, the course ensures graduates are adaptable and well-prepared for the dynamic field of software engineering.

Course Objectives

The major objectives of the course are:

- Practical Application:** Apply software engineering principles learned in lectures to real-world scenarios through hands-on projects and exercises.
- Tool Proficiency:** Gain proficiency in industry-standard software engineering tools and technologies used for software development, testing, and project management.
- Problem-Solving Skills:** Develop problem-solving skills by tackling challenges commonly encountered in software development projects, fostering critical thinking and innovation.
- Team Collaboration:** Cultivate teamwork and collaboration skills through group projects, encouraging effective communication, cooperation, and division of tasks.
- Quality Assurance:** Learn software testing techniques and quality assurance practices to ensure the reliability, functionality, and usability of software products developed in the lab.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Demonstrate a comprehensive understanding of software engineering principles, methodologies, and best practices.

CLO2: Apply software design techniques effectively to develop modular and scalable software systems.

CLO3: Implement coding, testing, debugging, and maintenance processes to ensure the reliability and quality of software products.

CLO4: Utilize project management methodologies such as Agile and version control tools to efficiently manage software development projects.

CLO5: Collaborate effectively within a team environment, demonstrating strong communication and teamwork skills essential for successful software development.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Software Development Tools: UML Modeling, Script Language, Framework Management	Lecture, Demonstration	2	CLO1
2	Project Selection: Project Idea Submission, Software Development Approach Selection	Discussion, Group Work	2	CLO4, CLO5
3	Provide students basic project idea about web-based project, formation of group and invite them to submit their idea through a Project Proposal	Lecture, Group Work	2	CLO5
4	Project Proposal Submission and Presentation: Presentation on Project Proposal, Lab performance evaluation	Presentation, Evaluation	2	CLO5
5	SRS report submission with UML diagram: Drawing Necessary UML Diagrams	Submission, Demonstration	2	CLO1, CLO2
6	UI design – Front-End Design (Session-I): Designing Login and Registration Page	Demonstration, Hands-on Practice	2	CLO2
7	UI design – Front-End Design (Session II): Designing Home Page and Other Sub-pages	Demonstration, Hands-on Practice	2	CLO2

8	Database Design – Back-End Evaluation (Session-I): Connect to PHP to Database Server	Demonstration, Hands-on Practice	2	CLO2, CLO3
9	Verifying Database Functions – Back-End Evaluation (Session-II)	Demonstration, Hands-on Practice	2	CLO3
10	Verifying Database Functions – Back-End Evaluation (Session-III)	Demonstration, Hands-on Practice	2	CLO3
11	Final Report Writing and Presentation Guidelines: Report Documentation, Representing Final Project Presentation Slide	Discussion, Presentation	2	CLO4, CLO5
12	Software Testing: Front-End Testing (UI design, user data entry), Back-End Testing (Database design, modeling, retrieving data from database, showing output to display)	Demonstration, Hands-on Practice	2	CLO3
13	Final presentation: Presentation of the Project by a Full Showcase of the Project and a Formal Report Submission	Presentation, Submission	2	CLO1, CLO4, CLO5
14	Final presentation: Presentation of the Project by a Full Showcase of the Project and a Formal Report Submission	Presentation, Submission	2	CLO1, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	H		H					H		M	
CLO2	M											
CLO3			H		H			H			H	H
CLO4								H	H		H	
CLO5						M		H	H		M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES**Reference Books:**

1. Software Engineering – Ian Sommerville (Latest Edition)
2. Software Engineering: A Practitioner’s Approach – Roger S. Pressman (latest Edition)

CSE-3207: IoT and Embedded System

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-3207	IoT and Embedded System	Theory	3	60	20	10	10	100

Course Description: Computer Networks lab is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 1 credits and two hours lab class duration weekly. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

To make an effective communication between the gadgets it is critical to play out every single essential apparatus and their fundamental design precisely. Therefore, to comprehend the system structure, segments and other applicable devices and procedure is an absolute necessity. This course enables the student to understand the operation of the components in a embedded system and IoT based network and the functional relationships of these components. The student will gain the ability to design and provide solutions using IoT and embedded systems to various real life problems.

Course Objectives

The major objectives of the course are:

- I. to teach students about technical aspects of the peripherals and Interfacing process
- II. to apply this knowledge in professional life
- III. to interpret the components, tools and techniques of peripheral systems.
- IV. to enable the student to understand the operation of the components in a peripheral system and the functional relationships of these components.
- V. to gain the ability to analyze, design, and improve data communication networks.

Course Learning Outcomes (CLO)

Upon completion of the course, the students will be able to:

CLO1-To **understand** the necessity of IoT devices and embedded system and to explain IoT architecture, functional blocks and different protocols used in Internet of Things

CLO2-To **analyze** different types of interfacing techniques for communicating with this devices through computer.

CLO3-To **Apply** the knowledge of IoT network architecture, IoT protocols and security in different layers of Internet of Things for **designing** embedded system for solving specific problem.

CLO-4 To analyze the knowledge in human-machine computer interactions.

CLO5 -To **design** an embedded system for solving a specific problem in real life.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	Corresponding CLOs
1	Introduction to IoT: Definition and Characteristic of IoT, M2M, M2M Block Diagram, IoT vs M2M, Evolution of IoT, Challenges in IoT.	• Demonstration • Discussion	3	CLO1, CLO2
2	IoT Network Architecture and Design: The oneM2M IoT architecture, The IoT World Forum (IoTWF) Standardized Architecture	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO4
3	IoT Network Architecture and Design: Layer 1: Physical Devices and Controllers Layer	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO4
4	IoT Network Architecture and Design: Layer 2: Connectivity Layer, Layer 3, edge Computing Layer	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO5
5	IoT Network Architecture and Design: Upper Layers: Layers 4–7, IT and OT Responsibilities in the IoT Reference Model, Additional IoT Reference	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO4

	Models, A Simplified IoT Architecture.			
6	The Core IoT Functional Stack: Layer 1, Layer2, Layer 3 Things in IoT: Sensor and Transducer Things in IoT: Actuator, Micro-Electro-Mechanical Systems (MEMS), Smart Objects Definition, Trend in Smart Objects	• Demonstration • Discussion • Group work	3	CLO1, CLO3, CLO4, CLO2
7	Sensing Network: WSN, WMSN, Behave, challenges, protocol, and application I, Sensing Network: RFID, Behave, challenges, protocol, and application II	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO4
8	Connecting Smart Objects: Communications Criteria IoT Access: IEEE 802.15.4 (WPAN), ZigBee, ZigBee IP IoT Access: IEEE 802.15.4g and 802.15.4e (LR-WPAN), IEEE 1901.2a, IEEE 802.11ah (WiFi-HaLow)	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO4
9	IoT Access: LoRaWAN, NB-IoT and Other LTE Variations, LTE Cat 0, IP as the IoT Network Layer:	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO4, CLO5
10	The Business Case for IP: The Key Advantages of Internet Protocol, Adoption or Adaptation of the Internet Protocol. The Need for Optimization: Constrained Nodes,	• Demonstration • Discussion • Group work	3	CLO1, CLO2, CLO3, CLO4

	Constrained Networks, IP Versions. Optimizing IP for IoT: 6LoWPAN and Routing in 6LoWPAN			
11	Application Protocols for IoT: CoAP, IoT Data Management and Compute Stack,	● Demonstration ● Discussion ● Group work	3	CLO1, CLO2, CLO3, CLO4
12	Application Protocols for IoT: Cloud Computing IoT Data Management and Compute Stack, Fog Computing IoT Data Management and Compute Stack: Edge Computing, The Hierarchy of Edge, Fog, and Cloud	● Demonstration ● Discussion ● Group work	3	CLO1, CLO2, CLO3, CLO4
13	Industrial IoT (IIoT) 24 Security in IoT DH: CH Data and Analytics for IoT DH: CH, Smart Grid, Home Automation	● Demonstration ● Discussion ● Group work	3	CLO1, CLO2, CLO3, CLO4
14	Industrial IoT (IIoT) 24 Security in IoT Smart Grid, Home Automation, Industrial IoT(IIoT).	● Demonstration ● Discussion ● Group work	3	CLO1, CLO2, CLO3, CLO4

Mapping between CLOs and PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO-1	PL O-2	PL O-3	P L O -4	PL O-5	P L O -6	PL O-7	P L O -8	P L O -9	P L O -10	P L O -11	PLO -12
CLO-1	H				M							
CLO-2			H			M						
CLO-3							M				H	
CLO-4							M					H

CLO-5	H		H		M		M				H	
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PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Class attendance (10)	Class performance (10)	Other forms
Remember		25%			
Understand	50%	50%			
Apply	25%	25%			
Analyze	25%				
Evaluate					
Create					

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows:

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two

Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

Text Books:

1. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things – David Hanes

Reference Book:

2. Internet of Things (A Hands-on Approach) - Vijay Madisetti and Arshdeep Bahga

CSEL-3208: IoT and Embedded System Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSEL-3208	IoT and Embedded System Lab	Lab	1	30	10	5	5	50

Course Description: The key rationale of this course is to build up student's ability to familiarize them with various IoT devices, microcontrollers and how to interface them. They will learn the basic concepts and techniques for interfacing a microprocessor and micro-controller to external devices for data collection and process control and developing the related software required.

Course Objectives

On completion of the course, students will be able to:

- i. Develop fundamental concepts and principles of IoT and embedded systems.
- ii. Design and implement IoT-based systems using microcontrollers and sensors.
- iii. Interface microcontrollers with external devices for data acquisition and control.
- iv. Analyze and solve practical problems related to IoT projects.
- v. Apply IoT and embedded system knowledge to real-life applications.

Course Learning Outcomes (CLOs)

Upon successful completion of the course, students will be able to:

CLO1: Demonstrate understanding of basic concepts and principles of IoT and embedded systems.

CLO2: Design and develop IoT-based projects using microcontrollers and interfacing techniques.

CLO3: Analyze and provide solutions to problems related to IoT system design and implementation.

CLO4: Apply IoT and embedded system knowledge to solve real-world problems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Lecture Hours	Corresponding CLOs
1	Introduction to Arduino related project using Proteus & Tinkercad	2	CLO1, CLO2, CLO3, CLO4, CLO5
2	Demonstrate how to make communication pathway between microprocessor and peripherals (Interfacing peripheral devices with microprocessor using 8255 programmable I/O controller)	2	CLO1, CLO2, CLO3, CLO4, CLO5
3	To verify and investigate how different interfacing methods work i.e (Design and write a program to toggle all the led to port and with some time delay using AVR microcontroller.)	2	CLO1, CLO2, CLO3, CLO4, CLO5
4	Make program for interfacing microcontroller with LED	2	CLO1, CLO2, CLO3, CLO4, CLO5
5	Interface 7-segment using microcontroller to display various digits	2	CLO1, CLO2, CLO3, CLO4, CLO5

6	Display characters or digits in LCD monitor	2	CLO1, CLO3, CLO5	CLO2, CLO4,
7	Design and develop a program for interfacing keyboard with microcontroller	2	CLO1, CLO3, CLO5	CLO2, CLO4,
8	Design an embedded system for temperature monitoring	2	CLO1, CLO3, CLO5	CLO2, CLO4,
9	Design an embedded system for temperature monitoring	2	CLO1, CLO3, CLO5	CLO2, CLO4,
10	Ensure security using fingerprint system	2	CLO1, CLO3, CLO5	CLO2, CLO4,
11	Interfacing DC motor, Servo Motor and ultrasonic sensor with Arduino to implement an automatic door on-off system.	2	CLO1, CLO3, CLO5	CLO2, CLO4,
12	Interfacing two or more sensors with Arduino, store their value in Thingspeak and provide notification to mobile phone, and interfacing Raspberry Pi with computer and Cloud/Fog/Edge.	2	CLO1, CLO3, CLO5	CLO2, CLO4,
13	Make a smart door locking system and Thingspeak and provide notification to mobile phone, and interfacing Raspberry Pi with computer and Cloud/Fog/Edge.	2	CLO1, CLO3, CLO5	CLO2, CLO4,
14	Solve a daily life problem	2	CLO1, CLO3, CLO4	CLO2,

Mapping between CLOs and PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PL O-1	PL O-2	P L O-3	P L O-4	PL O-5	P L O-6	P L O-7	P L O-8	P L O-9	P L O-10	P L O-11	P L O-12

CLO-1	√				√							
CLO-2			√			√						
CLO-3							√				√	
CLO-4							√					√

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Lab Report/Lab Assessment	√	√	√	
Quiz/Performance	√	√	√	√
Class attendance	√	√	√	√
Final	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3

4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things – David Hanes
2. Internet of Things (A Hands-on Approach) - Vijay Madisetti and Arshdeep Bahga

CSEP-3209: Application Design and Development (Project)

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Progress Evaluation	Class Participation	Class attendance	
CSEP-3209	Application Design and Development (Project)	Practical Course	1.0	30	10	5	5	50

Course Description: Application Design and Development (Project) is a compulsory practical course for the B.Sc. (Honours) program in the Department of Computer Science and Engineering, Jagannath University. The course carries 1.0 credits and 30 lecture hours. This is an implementation based course and hence the complete lecture hours are allocated for discussion, individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The application design and development course focuses on developing mobile applications for smartphones, tablets, and personal digital assistants, typically for the Android and iOS operating systems. The rationale of the course is to encourage students to develop knowledge and skills for creating and implementing mobile applications using android and other similar technologies.

Course Objectives

The major objectives of the course are:

- i. To understand the components and structure of mobile application development frameworks for Android and windows OS based mobiles.
- ii. To understand how to work with various mobile application development frameworks.
- iii. To learn the basic and important design concepts and issues of development of mobile applications.
- iv. To understand the capabilities and limitations of mobile devices.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Explain the basic concepts and processes of mobile application development

CLO2: Learn the fundamental and critical design concepts in UI/UX design and issues involved in developing mobile applications.

CLO3: Apply RSS Feed, Internal/External Storage, SMS, Multithreading, GPS and Analyze and discover own mobile app for simple needs using GUI and layouts.

CLO4: Applying the software engineering tools and techniques in analyzing the performance of an mobile application

CLO5: Design and implement necessary applications to build real life program.

PART B: CONTENT OF THE COURSE

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H									
CLO2		H	H									
CLO3					H							
CLO4			H		H				H			
CLO5			H			H				H		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Project Report	√	√	√	√	√
Viva/Quiz	√	√	√	√	√
Group Discussion	√	√	√	√	√
Final Presentation	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Project Report (5)	Viva/Quiz (5)	Class attendance (5)	Group Discussion (5)
Remember				
Understand		25%		25%
Apply	25%	40%		40%
Analyze	40%	25%		25%
Evaluate	25%	10%		10%
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains weekly group discussion with the course teacher and presenting the weekly work progress. In the final examination, students have to present their work as an oral presentation as well as demonstrate their implemented project in front of the examiners with necessary question answering session. The presentation and demonstration carries 20 marks whereas the viva carries 15 marks.

PART D: LEARNING RESOURCES

Text Book:

1. Beginning App Development with Flutter: Create Cross-Platform Mobile Apps 1st Edition by Rap Payne Publisher : Apress

2. React Native for Mobile Development 2nd Edition by Akshat Paul, Abhishek Nalwaya
Publisher : Apress

Reference Books:

3. Beginning Flutter: A Hands On Guide to App Development 1st Edition by Marco L. Napoli
Publisher : Wrox

CSEP-3210: Viva-Voce

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)			
				Semester Final Exam.	Continuous Assessment		Total
					Progress Evaluation	Class Participation	
CSEV-3210	Viva-Vice		1.0				50

4th Year 1st Semester

CSE-4101: Artificial Intelligence

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4101	Artificial Intelligence	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It introduces students to the theory and practice of creating intelligent systems that can mimic human behavior, reasoning, and decision-making. The course covers various topics including machine learning, natural language processing, computer vision, robotics, and expert systems. Students learn algorithms and techniques used to enable machines to perceive, understand, and respond to complex tasks and situations. Emphasis is placed on both theoretical foundations and practical applications, preparing students to develop AI-powered solutions for a wide range of domains such as healthcare, finance, gaming, and autonomous vehicles.

Justification of this course in the program entity

The inclusion of an "Artificial Intelligence" course within the Computer Science and Engineering curriculum is highly justified due to its crucial role in shaping the future of technology and computing. Artificial Intelligence (AI) has become a cornerstone of modern computer science, revolutionizing various fields such as healthcare, finance, robotics, and entertainment. Understanding AI principles and techniques is essential for computer science students as it equips them with the knowledge and skills to design, develop, and deploy intelligent systems capable of reasoning, learning, and decision-making. Moreover, AI algorithms and methodologies are integral components of many cutting-edge technologies, including machine learning, natural language processing, computer vision, and robotics. By studying AI, students gain insights into advanced problem-solving techniques, data analysis, and optimization strategies, which are invaluable assets in the rapidly evolving tech industry. Furthermore, AI intersects with various other disciplines within computer science, such as algorithms, data structures, and software engineering, making it essential for students to have a solid foundation in this area. Overall, an AI course provides students with the necessary expertise to tackle complex challenges and innovate in a wide range of domains, ensuring they are well-prepared for the demands of the ever-evolving technological landscape.

Course Objectives

The major objectives of the course are to:

- i. Introduce fundamental concepts and principles of Artificial Intelligence (AI), including machine learning, natural language processing, and computer vision.
- ii. Explore algorithms and techniques used in AI to enable machines to perceive, reason, and learn from data.
- iii. Provide hands-on experience with AI tools and frameworks for developing intelligent systems and applications.
- iv. Examine ethical considerations and societal impacts of AI technologies, fostering responsible AI development and deployment.
- v. Prepare students to apply AI methodologies to solve real-world problems across various domains, from healthcare to finance and beyond.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be

CLO1: Able to demonstrate a comprehensive understanding of fundamental concepts and principles of Artificial Intelligence (AI), including machine learning, natural language processing, and computer vision.

CLO2: Capable of applying various AI algorithms and techniques to solve complex problems, enabling machines to perceive, reason, and learn from data effectively.

CLO3: Proficient in utilizing AI tools and frameworks to develop intelligent systems and applications, demonstrating hands-on experience with AI technologies.

CLO4: Knowledgeable about ethical considerations and societal impacts associated with AI technologies, and able to incorporate responsible AI practices into development and deployment processes.

CLO5: Equipped with the skills to analyze real-world problems and apply AI methodologies to develop innovative solutions across diverse domains, contributing to advancements in technology and society.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Artificial Intelligence: Definitions, Overview of AI, Goals of AI, AI Approaches, AI Techniques, Branches of AI, Applications of AI, Intelligent Agent.	Lecture, Discussion	2	CLO1
2	General concepts of Knowledge, LISP, CLIPS, Prolog, and other AI Programming Languages.	Lecture, Demonstration	3	CLO1
3	Knowledge in AI: Definitions, Knowledge representation techniques, Predicate logic, Semantic net, Frame, Fuzzy logic, Inference and rules, IF-THEN rules, Techniques for inference, Reasoning, Symbolic reasoning, Statistical reasoning.	Lecture, Group Work	3	CLO1, CLO2
4	Expert System: Knowledge acquisition, Knowledge base, Production system model, rule based expert system, working memory, Inference engine, Expert system shells, Explanation module, Inferring procedure, forward chaining and backward chaining, Application of expert systems.	Lecture, Demonstration	3	CLO1, CLO2, CLO4
5	Learning from example: Induction, Explanation Based Learning (EBL), Discovery, Clustering, Analogy, Neural net and genetic learning, Reinforcement learning.	Lecture, Group Work	3	CLO1, CLO2, CLO5
6	Approaches and Techniques in AI: General problem solving, Search and control strategies, Exhaustive searches, informed and uninformed search, Heuristic search techniques, Problem Reduction, Constraint satisfaction problems (CSPs), Adversarial searches.	Lecture, Group Work	3	CLO1, CLO2, CLO5

7	AI in Gaming: Overview, Mini-Max search procedure, Game playing with Mini-Max, Alpha-Beta pruning, Learning, Rote learning.	Lecture, Group Work	3	CLO1, CLO2, CLO5
8	Natural Language Processing (NLP): Why NLP, NLP terminology, NLP difficulties, Language Models, Syntactic Analysis (Parsing), Semantic Interpretation, Discourse and pragmatic Processing.	Lecture, Demonstration	3	CLO1, CLO3
9	Text Classification, Discourse and pragmatic Processing, Implementation aspects of Syntactic Analysis (Parsing), Natural language analysis, Parsing, RTN.	Lecture, Demonstration	3	CLO1, CLO3
10	Introduction to selected topics in AI: pattern recognition, computer vision, expert system, artificial neural networks, robotics.	Lecture, Group Work	3	CLO1, CLO2, CLO5
11	Machine Learning: Introduction, linear classification, perceptron update rule, Perceptron convergence, generalization, Maximum margin classification, Classification errors, regularization, logistic regression, Linear regression, estimator bias and variance, active learning, Active learning, Classification.	Lecture, Group Work	3	CLO1, CLO2, CLO3, CLO5
12	Model and Features: non-linear predictions, kernel regression, Support vector machine (SVM) and kernels, kernel optimization, Model selection, Classification, Model selection criteria, Description length, feature selection, Combining classifiers, boosting, Boosting, margin, and complexity, Margin and generalization, mixture models, Text Classification.	Lecture, Demonstration	3	CLO1, CLO2, CLO3, CLO5
13	Clustering: Mixtures and the expectation maximization (EM) algorithm, EM, regularization, Clustering algorithms, Markov models, Hidden Markov models (HMMs), Bayesian networks, Learning Bayesian networks, Probabilistic inference.	Lecture, Demonstration	3	CLO1, CLO2, CLO3, CLO5
14	Revision, Q&A	Discussion, Q&A	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H					M				M		
CLO2		H	H	H								M
CLO3		H	H	H	H			M				
CLO4						H	H					
CLO5		H		H								

PART C: ASSESSMENT AND EVALUATION**Mapping/Alignment of CLOs vs Assessment Strategy:**

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class Attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions, one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Peterson, Artificial Intelligence & Expert System, Latest edition
2. Artificial Intelligence: A Modern Approach, Stuart Russell, Latest edition
3. Vasant Honavar, Artificial Intelligence & Neural Networks, Latest edition

CSEL-4102: Artificial Intelligence Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4102	Artificial Intelligence Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of

the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The Artificial Intelligence Lab is a dedicated space where students engage in hands-on learning and experimentation with various AI algorithms, techniques, and tools. In this lab, students have the opportunity to apply theoretical knowledge gained in lectures to practical scenarios. They work on projects that involve machine learning, natural language processing, computer vision, and other AI domains, gaining valuable experience in developing intelligent systems and applications. The lab provides access to state-of-the-art AI frameworks, datasets, and computing resources, fostering an environment conducive to exploration, innovation, and collaborative learning.

Justification of this course in the program entity

The Artificial Intelligence Lab course provides students with invaluable hands-on experience in applying theoretical AI concepts to real-world scenarios. Through practical exercises and projects, students develop proficiency in utilizing AI algorithms, frameworks, and tools. This lab environment fosters collaborative learning and experimentation, allowing students to explore the capabilities and limitations of AI technologies firsthand. By working on tasks such as machine learning model development, natural language processing applications, and computer vision projects, students gain practical skills that are highly sought after in the industry. This lab course not only reinforces theoretical knowledge but also cultivates problem-solving abilities and critical thinking skills essential for tackling complex AI challenges.

Course Objectives

The major objectives of the course are:

- i. Understanding AI Fundamentals: Gain a comprehensive understanding of core AI concepts, including machine learning, natural language processing, and computer vision.
- ii. Practical Application: Apply AI algorithms and techniques to solve real-world problems across various domains, fostering hands-on experience in AI development.
- iii. Tool Proficiency: Develop proficiency in utilizing AI tools and frameworks to build intelligent systems and applications, preparing students for industry demands.
- iv. Ethical Considerations: Explore ethical considerations and societal impacts associated with AI technologies, promoting responsible AI development and deployment practices.
- v. Innovation and Collaboration: Cultivate a mindset of innovation and collaboration through project-based learning, enabling students to contribute effectively to advancements in AI technology.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Understand and explain fundamental concepts and principles of artificial intelligence, including but not limited to machine learning, natural language processing, and computer vision.

CLO2: Apply various AI algorithms and methodologies to analyze, interpret, and solve complex problems across different domains, demonstrating the ability to reason and learn from data effectively.

CLO3: Utilize AI tools, frameworks, and programming languages to design, develop, and implement intelligent systems and applications capable of perceiving, reasoning, and learning from their environment.

CLO4: Evaluate ethical considerations and societal impacts associated with the development and deployment of AI technologies, integrating responsible AI practices into decision-making processes.

CLO5: Analyze real-world problems and innovate AI-based solutions, demonstrating adaptability to technological changes and contributing to advancements in AI research and application.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Prolog and LISP Programming. Lab practice on Taking input, displaying output and basic arithmetic operations in Prolog and LISP, Conditional statements (if-else), list, cut operator and debugging procedure, Loops and recursion.	Hands-on Lab Practice, Lecture	2	CLO1, CLO3
2	Implement a simple expert system in Prolog with Facts, relation and Queries. Implement an expert system for medical diagnosis in Prolog. Implement backward chaining in Prolog. Develop a simple intelligent system with LISP and perform the operations of LISP.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
3	Write a Prolog program to solve N-Queen, map coloring, monkey-banana, water-jug and travelling salesman problems, Hangman game.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
4	Learn facts, rules, and inheritance in CLIPS. Implement forward chaining, Frame based expert system in CLIPS.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
5	Lab practice on (how robots/machines take decisions using instruction) using Blockly Games. Implement the Uniformed/blind Search algorithms (BFS, Uniform-Cost Search, DFS, Depth-Limited Search)	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3

6	Implement Informed Search algorithms (Heuristics, Greedy Search, A* Tree Search, A* Graph Search). Implement Minimax algorithm with alpha beta pruning.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
7	Implement different fuzzy membership functions, Fuzzy Inference System (FIS).	Hands-on Lab Practice, Lecture	1	CLO1, CLO2, CLO3
8	Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples. Read the training data from a .CSV file For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm to output a description of the set of all hypotheses consistent with the training examples.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3, CLO4
9	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
10	Write a program to find a given function maximum or minimum, when you cannot derive using a genetic algorithm implemented with Python API.	Hands-on Lab Practice, Lecture	1	CLO1, CLO2, CLO3
11	Write a Python program to implement the following: stop words for a given passage from a text file, stemming for a given sentence, POS (Parts of Speech) tagging for the given sentence, implementing Lemmatization, classifying text for the give sentence using NLTK.	Hands-on Lab Practice, Lecture	1	CLO1, CLO2, CLO3
12	Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set. Write a program	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3, CLO4

	to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using the standard Heart Disease Data Set. You can use Java/Python ML library classes/API.			
13	Apply EM algorithm to cluster a set of data stored in a .CSV file. Use the same data set for clustering using the k-Means algorithm. Compare the results of these two algorithms and comment on the quality of clustering. You can add Java/Python ML library classes/API in the program. Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.	Hands-on Lab Practice, Lecture	2	CLO1, CLO2, CLO3
14	Project, Assignment and Continuous Assessment Evaluation.	Project Work, Assessment	-	CLO1, CLO2, CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H		H									M
CLO2	H							H	H			M
CLO3			H		H						H	
CLO4							H				M	
CLO5		H	H	H						M		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessments carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems, one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Peterson, Artificial Intelligence & Expert System, Latest edition

2. Artificial Intelligence: A Modern Approach, Stuart Russell, Latest edition
3. Vasant Honavar, Artificial Intelligence & Neural Networks, Latest edition

CSE-4103: Software Quality Assurance and Testing

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4103	Software Quality Assurance and Testing	Theory	2	30	10	5	5	50

Course Description: Software Quality Assurance is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 2 credits and 30 lecture hours. Among the 30 hours, 25 are for class teaching and 5 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The rationale for Software Quality Assurance (SQA) and Testing courses is to equip individuals with the knowledge and skills to ensure software products meet quality standards and user expectations. SQA involves overseeing the entire software development life cycle to ensure adherence to defined standards and processes, while testing focuses on identifying defects and issues within the software. These courses are crucial for maintaining software quality, minimizing defects, and improving overall user experience.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Research the state-of-the-art, and apply their findings to software testing and quality assurance.

CLO-2: Analyze different approaches to software testing and quality assurance, and select optimal solutions for different situations and projects.

CLO-3: Conduct independent research in software testing and quality assurance and apply that knowledge in their future research and practices.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Assessment Strategy	Corresponding CLOs
1	Definition of quality, Modern view of Software Quality, Aspects of Software Quality, Tools for Improving Software Quality	• Lecture • Rapport building	Test, Quiz, Assignment	CO1
2	The objectives of SQA activities in software development, Software Quality Assurance and Software Engineering	• Lecture • Rapport building • Demonstration •	Test, Quiz, Presentation	CO1, CO2
3	Critical Path & critical activities, Calculation of critical path , milestone and deliverable, CPM and PERT	• Lecture • Rapport building • Demonstration •	Test, Quiz, Assignment	CO3
4	Definition of Risk, Risk category, The risk management	• Lecture • Rapport building • Demonstration •	Test, Quiz, Presentation	CO3
5	Bug, errors & defects, defect amplification & removal, review metrics and their use ,analyzing metrics,	• Lecture • Rapport building • Demonstration •	Test, Assignment , Quiz,	CO2
6	Elements of SQA, SQA task, goals and metrics, Software reliability and availability	• Lecture • Rapport building • Demonstration •	Test, Quiz	CO2
7	Statistical SQA, Six-sigma methodology, error density, SQA related problems	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz, Assignment	CO3

			, Presentation	
8	Software Testing, Validation testing , Defect testing , V & V, Characteristics of Testable Software, Test Characteristics, The software testing process, Testing objectives, Unit testing, integration testing, regression testing, alpha & beta testing, White-box Testing,	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz	CO3
9	Path Testing, cyclomatic complexity, Graph matrices, black box testing, Testing objective oriented Applications,	• Lecture • Rapport building • Demonstration •	Test, Quiz, Presentation	CO3, CO1
10	Halstead Complexity, KLOC,LOC Metrics, Size-oriented metrics, Software Measurement, Function –based metrics, metrics for specification quality, Calculation of FP,	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz,	CO3
11	Class-oriented metrics, OO metric proposed by Lorenz and Kidd, Design metrics for WebApps,	• Lecture • Rapport building • Group work	Test, Quiz	CO3,C02
12	Activities of a Test Engineer, Testing Levels Based on Software Activity, Beizer's Testing Levels Based on Test Process Maturity, Graph Coverage Test path, Graph	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz, Assignment	CO3,CO!

	Coverage Criteria, Touring, Sidetrips, and Detours, Data Flow Criteria, Graph Coverage For Source Code,			
13	Data Flow Graph Coverage for Source Code, Overview: Logic Predicates and Clauses, Restricted Active Clause Coverage (RACC), Input Domain Modeling	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz, Presentation	CO3, CO2
14	Revision and Solution			

Mapping between CLOs and PLOs:

CLOs	Program Outcomes											
	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9	PLO -10	PLO -11	PLO -12
CLO1	√											
CLO2		√	√									
CLO3									√			√

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	√			
Mid Semester2		√	√	
Quiz/Performance	√	√	√	√
Class attendance	√	√	√	√
Final	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		

Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- c. Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- d. Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO3
5	Question from CLO4 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 60% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Roger S Pressman, "Software Engineering: A Practitioner's Approach".
2. Paul Ammann and Jeff Offutt, "Introduction to Software Testing".

Reference Books:

3. Jeff Tian, “Software Quality Engineering”.
4. Pearson, “Software Testing”.

CSEL-4104: Software Quality Assurance and Testing Lab**PART A: INTRODUCTION**

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSEL-4104	Software Quality Assurance and Testing Lab	Lab	1.0	30	10	5	5	50

Course Description: Software Quality Assurance is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 2 credits and 30 lecture hours. Among the 30 hours, 25 are for class teaching and 5 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The rationale for Software Quality Assurance (SQA) and Testing courses is to equip individuals with the knowledge and skills to ensure software products meet quality standards and user expectations. SQA involves overseeing the entire software development life cycle to ensure adherence to defined standards and processes, while testing focuses on identifying defects and issues within the software. These courses are crucial for maintaining software quality, minimizing defects, and improving overall user experience

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Research the state-of-the-art, and apply their findings to software testing and quality assurance.

CLO-2: Analyze different approaches to software testing and quality assurance, and select optimal solutions for different situations and projects.

CLO-3: Conduct independent research in software testing and quality assurance and apply that knowledge in their future research and practices.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Assessment Strategy	Corresponding CLOs
1	Definition of quality, Modern view of Software Quality, Aspects of Software Quality, Tools for Improving Software Quality	● Lecture ● Rapport building	Test, Quiz, Assignment	CO1
2	The objectives of SQA activities in software development, Software Quality Assurance and Software Engineering	● Lecture ● Rapport building ● Demonstration ●	Test, Quiz, Presentation	CO1, CO2
3	Critical Path & critical activities, Calculation of critical path , milestone and deliverable, CPM and PERT	● Lecture ● Rapport building ● Demonstration ●	Test, Quiz, Assignment	CO3
4	Definition of Risk, Risk category, The risk management	● Lecture ● Rapport building ● Demonstration ●	Test, Quiz, Presentation	CO3
5	Bug, errors & defects, defect amplification & removal, review metrics and their use ,analyzing metrics,	● Lecture ● Rapport building ● Demonstration ●	Test, Assignment , Quiz,	CO2
6	Elements of SQA, SQA task, goals and metrics, Software reliability and availability	● Lecture ● Rapport building ● Demonstration ●	Test, Quiz	CO2

7	Statistical SQA, Six-sigma methodology, error density, SQA related problems	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz, Assignment, Presentation	CO3
8	Software Testing, Validation testing , Defect testing , V & V, Characteristics of Testable Software, Test Characteristics, The software testing process, Testing objectives, Unit testing, integration testing, regression testing, alpha & beta testing, White-box Testing,	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz	CO3
9	Path Testing, cyclomatic complexity, Graph matrices, black box testing, Testing objective oriented Applications,	• Lecture • Rapport building • Demonstration •	Test, Quiz, Presentation	CO3, CO1
10	Halstead Complexity, KLOC,LOC Metrics, Size-oriented metrics, Software Measurement, Function –based metrics, metrics for specification quality, Calculation of FP,	• Lecture • Rapport building • Demonstration • Group work	Test, Quiz,	CO3
11	Class-oriented metrics, OO metric proposed by Lorenz and Kidd, Design metrics for WebApps,	• Lecture • Rapport building • Group work	Test, Quiz	CO3,C02
12	Activities of a Test Engineer, Testing Levels Based on	• Lecture • Rapport building • Demonstration	Test, Quiz, Assignment	CO3,CO!

	Software Activity, Beizer's Testing Levels Based on Test Process Maturity, Graph Coverage Test path, Graph Coverage Criteria, Touring, Sidetrips, and Detours, Data Flow Criteria, Graph Coverage For Source Code,	Group work		
13	Data Flow Graph Coverage for Source Code, Overview: Logic Predicates and Clauses, Restricted Active Clause Coverage (RACC), Input Domain Modeling	- Lecture - Rapport building - Demonstration - Group work	Test, Quiz, Presentation	CO3, CO2
14	Revision and Solution			

Mapping between CLOs and PLOs:

CLOs	Program Outcomes											
	PLO -1	PLO -2	PLO -3	PLO -4	PLO -5	PLO -6	PLO -7	PLO -8	PLO -9	PLO -10	PLO -11	PLO -12
CLO1	√											
CLO2		√	√									
CLO3									√			√

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4
Mid Semester 1	√			
Mid Semester2		√	√	
Quiz/Performance	√	√	√	√
Class attendance	√	√	√	√
Final	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- e. Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- f. Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO2 or Question from CLO2
4	Question from CLO3 or Question from CLO3
5	Question from CLO4 or Question from CLO4

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions, one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Roger S Pressman, “Software Engineering: A Practitioner’s Approach”.
2. Paul Ammann and Jeff Offutt, “Introduction to Software Testing”.

Reference Books:

3. Jeff Tian, “Software Quality Engineering”.
4. Pearson, “Software Testing”.

CSE-4105: Digital Image Processing

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4105	Digital Image Processing	Theory	3	60	20	10	10	100

Course Description: The Digital Image Processing course offers a deep dive into the fundamental concepts and techniques involved in the manipulation and analysis of digital images. This course covers a wide range of topics, from basic image processing operations to advanced techniques in image analysis, enhancement, restoration, and compression. Students will learn about image representation, spatial and frequency domain processing, filtering, edge detection, segmentation, and morphological operations. Practical assignments and projects will provide hands-on experience with image processing tools and libraries, emphasizing applications in various domains such as computer vision, medical imaging, and multimedia.

Justification of this course in the program entity

Foundation of Image Analysis: Understanding digital image processing is essential for students aiming to excel in fields requiring image analysis and manipulation, including computer vision, artificial intelligence, and medical imaging. This course lays the foundational skills necessary for analyzing and enhancing digital images.

Skills in High Demand: The techniques taught are widely used in various industries, such as healthcare, security, and entertainment, making the skills acquired highly relevant and in demand.

Critical Thinking and Problem Solving: Through the exploration of image processing algorithms and their applications, students develop critical thinking and problem-solving skills, enabling them to approach complex visual data challenges effectively.

Preparation for Advanced Topics: Mastery in digital image processing prepares students for more advanced topics in computer science and engineering, such as machine learning, computer vision, and pattern recognition, providing a solid foundation for academic growth and research.

Innovation and Creativity: The course encourages innovation and creativity, as students learn to apply image processing techniques to solve real-world problems, potentially leading to new discoveries and advancements in technology.

Course Objectives

The major objectives of the course are:

- i. Understand the basic concepts and principles of digital image processing.
- ii. Implement and apply image processing techniques using programming tools.
- iii. Analyze and enhance images in both spatial and frequency domains.
- iv. Develop algorithms for image segmentation, restoration, and compression.
- v. Apply image processing techniques to solve real-world problems.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Explain the core concepts of digital image processing and apply various image enhancement techniques.

CLO-2: Utilize spatial and frequency domain methods for image analysis and enhancement.

CLO-3: Design and implement algorithms for image restoration, segmentation, and compression.

CLO-4: Evaluate the performance of image processing techniques and algorithms.

CLO-5: Apply digital image processing concepts and techniques to practical applications and projects.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction to Digital Image Processing, Basics of Images	a) Lecture, Demonstrations, Group Discussions	3	CLO1
2	Digital Image Fundamentals, Image Enhancement in the Spatial Domain	a) Lecture, Lab Exercises	3	CLO1, CLO2

3	Image Enhancement in the Frequency Domain	a) Lecture, Hands-on Practice	3	CLO2
4	Image Restoration	a) Collaborative Projects, Case Studies	3	CLO3
5	Color Image Processing	a) Lecture, Group Discussions	3	CLO1, CLO2
6	Wavelets and Multiresolution Processing	a) Interactive Sessions, Lab Exercises	3	CLO2, CLO3
7	Image Compression and Watermarking	a) Lecture, Demonstrations	3	CLO3
8	Morphological Image Processing	a) Group Projects, Hands-on Practice	3	CLO1, CLO3
9	Image Segmentation	a) Case Studies, Lab Exercises	3	CLO3
10	Object Recognition	a) Projects, Demonstrations	3	CLO3, CLO5
11	Advanced Topics and Current Trends in Image Processing	a) Lecture, Guest Lectures	3	CLO4, CLO5
12-14	Project Work and Revision	a) Project Presentation, Review Sessions	9	CLO1, CLO2, CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H	M	H	M	H	M	M	M	M	M	M
CLO2	H	H	H	H	M	H	M	M	M	M	M	M
CLO3	H	H	H	H	H	H	M	M	M	M	M	M
CLO4	M	H	H	H	H	H	M	M	M	M	M	M
CLO5	H	H	M	H	M	H	H	M	H	H	M	H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3

4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing," 4th Edition, 2018.

Reference Books:

1. Anil K. Jain, "Fundamentals of Digital Image Processing," 1st Edition, 1989.
2. William K. Pratt, "Digital Image Processing: PIKS Inside," 3rd Edition, 2007.
3. John C. Russ, "The Image Processing Handbook," 7th Edition, 2016.

CSE-4106: Digital Image Processing Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4106	Digital Image Processing Lab	Lab	1.0	30	10	5	5	50

Course Description: The Digital Image Processing Lab complements the theoretical knowledge gained in CSE-4105 Digital Image Processing. This laboratory course focuses on the practical application and implementation of image processing techniques. Students will work with image processing software and tools to apply the concepts learned in lectures, including image manipulation, enhancement, restoration, segmentation, and compression. Through a series of lab assignments and projects, students will gain hands-on experience in handling real-world image processing tasks.

Justification of this course in the program entity

Practical Skills Development: This lab course is crucial for developing practical skills in digital image processing, enabling students to apply theoretical concepts to solve real-world problems.

Technical Proficiency: Hands-on experience with image processing tools and software prepares students for technical challenges in their future careers.

Innovative Problem-Solving: The lab encourages creative thinking and problem-solving by allowing students to work on projects that may lead to innovative solutions.

Career Preparedness: Gaining proficiency in digital image processing techniques through practical experience enhances students' employability in various fields such as computer vision, medical imaging, and multimedia applications.

Course Objectives

The major objectives of the course are:

- i. To apply digital image processing techniques using practical tools and software.
- ii. To enhance understanding of image processing concepts through hands-on experiments.
- iii. To develop algorithms for processing and analyzing images in practical applications.
- iv. To foster problem-solving skills by working on image processing projects.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Apply image processing techniques using software tools for image enhancement and manipulation.

CLO-2: Implement algorithms for image restoration, segmentation, and compression in lab environments.

CLO-3: Analyze and solve real-world problems using digital image processing techniques.

CLO-4: Demonstrate the ability to work effectively in teams on collaborative image processing projects.

CLO-5: Present and report on lab projects and assignments, showing proficiency in communicating technical information.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1-2	Introduction to Image Processing Software and Tools	Hands-on Training, Demonstrations	6	CLO-1
3-4	Image Enhancement Techniques	Hands-on Exercises, Group Work	6	CLO-1, CLO-2

5-6	Image Restoration and Compression	Project-Based Learning	6	CLO-2
7-8	Image Segmentation Techniques	Collaborative Projects, Peer Review	6	CLO-2, CLO-3
9-10	Object Recognition and Analysis	Case Studies, Hands-on Implementation	6	CLO-3
11-12	Advanced Image Processing Projects	Independent Projects, Mentorship	6	CLO-3, CLO-4
13-14	Project Presentations and Review	Presentations, Feedback Sessions	6	CLO-4, CLO-5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H	M	H	M	H	M	M	M	H	M	M
CLO2	H	H	H	H	M	H	M	M	M	H	M	M
CLO3	H	H	H	H	H	H	M	M	M	H	M	M
CLO4	M	H	M	H	M	M	M	M	H	H	M	M
CLO5	M	H	M	M	M	M	M	M	H	H	M	M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10%)	Mid Semester 2 (10%)	Quiz/Performance (10%)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		

Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing," 4th Edition, 2018.

Reference Books:

2. Anil K. Jain, "Fundamentals of Digital Image Processing," 1st Edition, 1989.
3. William K. Pratt, "Digital Image Processing: PIKS Inside," 3rd Edition, 2007.
4. John C. Russ, "The Image Processing Handbook," 7th Edition, 2016.

CSE-4107: Data Mining and Machine Learning

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4107	Data Mining and Machine Learning	Theory	3	60	20	10	10	100

Course Description: Data Mining and Machine Learning is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The course is intended to teach students about the basic concept of data mining, issues and techniques. Data warehouse and OLTP technologies for data mining, Classification of data mining techniques and models, data pre-processing, data mining primitives, query languages and system architecture, characterization and comparison. Mining association rules in a large database. Cluster analysis, multidimensional analysis and descriptive mining of complex data object. Data mining in distributed heterogeneous database systems. Data mining applications and future research issues.

Course Objectives

The major objectives of the course are:

- i. Understand about data mining and its importance as well as application of data mining in real life situations.
- ii. Understand types of data mining techniques.
- iii. How to apply the techniques of clustering, classification, association finding, feature selection and visualization on real world data.

- iv. How to set up a data mining process for an application, including data preparation, modelling and evaluation.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: To demonstrate advanced knowledge of data mining concepts and techniques and understand, use, and determine appropriate machine learning methods/algorithms suitable for different types of learning problems, i.e. know about their most important weaknesses and advantages.

CLO2: Apply pattern mining algorithms for data mining tasks after processing raw input data, data preprocessing, Association rules mining, Regression Analysis.

CLO3: Understand and use a range of machine learning algorithms, supervised algorithms like classification with their strengths and weaknesses.

CLO4- To apply the techniques of clustering, feature selection and visualization on real world data and data mining software and toolkits in a range of applications.

CLO5: Examine and analyze appropriate software tools to implement algorithms in a range of real-world applications; sequential pattern mining and write report to represent the analysis results.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction: Introduction: Definition of data mining, its application and challenges; Area and types of data mining.	b) Lecture c) Rapport building	3	CLO1
2	Machine Learning: Machine learning, Applications of machine Learning.	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO1
3	Data Warehouse and OLAP: Define Data Warehouse and DBMS, discuss multidimensional data model, derive OLAP operations, Example: loan data set.	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO1

4	Data preprocessing: Types of data and data set; Data cleaning, selection, Data transformation, Data reduction, Discretization and linearization.	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO2
5	Exploratory analysis: Descriptive statistics; Visualization using Plots and diagrams and OLAP, Attribute relevance, Class comparison, Statistical measures	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO2
6	Data mining algorithms Association rules: Explain the basic idea: item sets, Explain generating item sets and rules efficiently, State Correlation analysis.	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO2
7	Association rules: Apriori Algorithm, FP growth, AprioriTID Algorithm	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO2
8	Regression Analysis: Linear Regression, Perception Algorithm, Logistic Regression	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO2
9	Classification: Types of classification, Hunts algorithms for Decision trees classification. Validating a model.	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO3, CLO4
10	Classification: Naive Bayes, Support Vector Machine (SVM)	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO3, CLO4

11	Clustering: Clustering, K-means Hierarchical and Density based clustering, Advanced discussion on clustering and EM	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO3, CLO4
12	Feature Selection and Reduction: Gaussian Mixture Models, Feature Scaling, Dimensionality Reduction	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO3, CLO4
13	Sequential Pattern Mining: Sequential Pattern Mining, GSP, SPADE, Text Mining, NLP	g) Lecture, h) Rapport building i) Cooperative learning	3	CLO5
14	Model Evaluation: Training and Testing models, Evaluation Metrics, Evaluation and Validation, Choose the best model using cross-validation and grid search	g) Lecture, h) Rapport building i) Cooperative learning	3	CO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	

Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (30 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Other forms
Remember	10%	10%		
Understand	15%	15%	25%	
Apply	20%	20%	40%	
Analyze	15%	15%	25%	
Evaluate	20%	20%	10%	
Create	20%	20%		

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination,

students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Jiawei Han, Micheline Kamber “Data Mining. Concepts and Techniques”, 3rd Edition.

Reference Books:

2. Morgan Kaufmann “Data Mining: Practical Machine Learning Tools and Techniques”, 2nd Edition, 2005.
3. Aurelien Geron “Hands-On Machine Learning with Scikit-Learn and TensorFlow”, O'Reilly Media, Inc., 2nd edition, 2017.
4. Andreas C. Müller, Sarah Guido, “Introduction to Machine Learning with Python: A Guide for Data Scientists”, 1st Edition, 2016

CSEL-4108: Data Mining and Machine Learning Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4108	Data Mining and Machine Learning Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The objective of this course is to enhance students' capacity to extract information from data and recognize patterns and trends through the creation of a data warehouse and the utilization of data mining techniques for classification, clustering, and association analysis.

Justification of this course in the program entity

The primary goal of this course is to study the basic concepts of data warehousing and the required skills to develop and use them. It emphasizes on employing data warehousing to support the decision-making process. It also covers the architectures of data warehousing and the infrastructural settings to develop them. It explains various ways of extracting, analyzing data to support the decision-making process. This course is intended to develop the student's ability to extract information from data and identify patterns and trends by designing a data warehouse and by applying data mining methods for classification, clustering, and association analysis. It aims to give students a solid understanding of the main abstractions and reasoning techniques used of Machine Learning.

Course Objectives

The major objectives of the course are:

- i. Understanding how to collect, clean, and preprocess data from various sources, including databases, APIs, and raw files. This involves tasks such as data cleaning, feature selection, and transformation.
- ii. Learning techniques to explore and visualize datasets to gain insights into underlying patterns, trends, and relationships.
- iii. Developing proficiency in building machine learning models using algorithms such as regression, classification, clustering, and association rule mining. Understanding model selection, hyperparameter tuning, and evaluation metrics is crucial.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Exhibit proficiency in data preprocessing techniques, including data cleaning, missing value imputation, feature scaling, and encoding categorical variables, to prepare datasets for machine learning model training.

CLO2: Demonstrate a comprehensive understanding of fundamental machine learning concepts, including supervised and unsupervised learning, feature engineering, model evaluation, and optimization techniques.

CLO3: Understand, use, and determine appropriate machine learning methods/algorithms suitable for different types of learning problems, i.e. know about their most important weaknesses and advantages.

CLO4: Able to select appropriate machine learning models based on the characteristics of the dataset and evaluate their performance using relevant metrics such as accuracy, precision, recall, F1-score, and ROC curves.

CLLO5: Develop problem-solving skills by identifying suitable machine learning approaches for different types of problems, troubleshooting model performance issues, and iteratively refining models based on feedback..

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Machine learning programming and environment such as Python, R, Google Collaboratory.	Lectures, Coding Exercises	2	CLO1
2	Study of Python Basic Libraries such as Statistics, Math, Numpy and Scipy	Theoretical Explanation, Coding Practice	2	CLO1
3	Study of Python Libraries for ML application such as Pandas and Matplotlib	Lectures, Problem-solving Sessions	2	CLO1, CLO2
4	Data preprocessing: Data cleaning, Missing Value Handling, Encoding.	Interactive Lectures, Algorithm Implementation	2	CLO2, CLO3
5	Data preprocessing: Data selection, Data transformation, Data reduction, Normalization	Practical Examples, Coding Sessions	2	CLO2, CLO3
6	Exploratory analysis: Central Tendency Measures: Mean, Median, Mode Measure of Dispersion: Variance, Standard Deviation	Theoretical Explanation, Coding Practice	2	CLO2, CLO3
7	Association rules: Apriori Algorithm, FP growth	Lectures, Problem-solving Sessions	2	CLO2, CLO3
8	Regression Analysis: Linear Regression, Perception Algorithm, Logistic Regression	Coding Workshops, Practical Examples	2	CLO2, CLO3
9	Classification: Decision tree using sklearn and its parameter tuning, KNN using sklearn.	Theoretical Explanation, Problem-solving Sessions	2	CLO2, CLO3, CLO4, CLO5
10	Model Evaluation: Training and Testing	Interactive Sessions, Coding Exercises	2	CLO3, CLO4, CLO5

	models, Evaluation Metrics, Evaluation and Validation, Choose the best model using cross-validation and grid search			
11	Classification: Naive Bayes, Support Vector Machine (SVM)	Lectures, Algorithm Implementation	2	CLO3, CLO4, CLO5
12	Clustering: Implementation of K-Means Clustering, Elbow method other method	Practical Examples, Coding Sessions	2	CLO3, CLO4, CLO5
13	Feature Selection and Reduction: Feature Scaling, Dimensionality Reduction, PCA	Problem-solving Sessions, Coding Practice	2	CLO3, CLO4, CLO5
14	Implementation of Ensemble method	Review Sessions, Open Discussion	2	CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H		H	H						M
CLO2	M		H			H						M
CLO3		H	H			H			H			M
CLO4		H	H	H		H						
CLO5			H		H	H						

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
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Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

1. Aurélien Géron “Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow”, 2nd Edition.

Reference Books:

2. Morgan Kaufmann “Data Mining: Practical Machine Learning Tools and Techniques”, 2nd Edition, 2005.
3. Aurelien Geron “Hands-On Machine Learning with Scikit-Learn and TensorFlow”, O'Reilly Media, Inc., 2nd edition, 2017.
4. Andreas C. Müller, Sarah Guido, “Introduction to Machine Learning with Python: A Guide for Data Scientists”, 1st Edition, 2016

CSE-4109: Cryptography and Network Security

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4109	Cryptography and Network Security	Theory	3	60	20	10	10	100

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course explores the fundamental principles and applications of cryptography in the context of network security. Students will delve into the theoretical foundations of cryptographic techniques, including symmetric and asymmetric key algorithms, digital signatures, and hash functions. The course also covers practical aspects of network security, focusing on secure communication protocols, threat analysis, access controls, and security measures for both wired and wireless networks. Through hands-on exercises and case studies, students will gain practical skills in implementing secure communication solutions and understanding the challenges associated with safeguarding information in modern networked environments. Emphasis will be placed on current industry standards and best practices in the field of cryptography and network security.

Justification of this course in the program entity

This course is mandatory and plays a pivotal role in the computer science program by equipping students with indispensable knowledge and skills to navigate the complex landscape of cybersecurity. In today's digital age, where the integrity and confidentiality of information are paramount, the course provides a comprehensive understanding of cryptographic techniques and network security principles. This knowledge is not only crucial for protecting sensitive data but also forms the bedrock for designing secure and resilient software systems. The industry relevance of the course is undeniable, as cybersecurity concerns continue to escalate, making graduates with expertise in cryptography and network security highly sought after. Ultimately, the course serves as a cornerstone in preparing computer science students to address the

ever-evolving challenges of ensuring the security and reliability of information in networked environments.

Course Objectives

The major objectives of the course are to:

- i. Understand the basic principles of cryptography, including encryption and decryption techniques, key management, and cryptographic protocols.
- ii. Differentiate between symmetric and asymmetric key algorithms and comprehend their respective applications in securing communication.
- iii. Explore the role of digital signatures and hash functions in ensuring data integrity, non-repudiation, and authentication.
- iv. Examine common network security protocols and their implementation in securing data transmission over various network architectures.
- v. Learn about access controls, authentication mechanisms, and security measures to protect against unauthorized access and data breaches.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Cryptographic Proficiency: Demonstrate proficiency in applying cryptographic principles, including encryption, decryption, and key management, to secure information in various computing scenarios.

CLO2: Encryption Expertise: Apply knowledge of symmetric and asymmetric key algorithms to implement robust encryption techniques, ensuring secure communication in diverse computing environments.

CLO3: Data Integrity Assurance: Utilize digital signatures and hash functions effectively to ensure data integrity, non-repudiation, and authentication in information exchange processes.

CLO4: Network Security Competence: Evaluate, implement, and optimize network security protocols to secure data transmission across different network architectures, demonstrating a comprehensive understanding of their functionalities.

CLO5: Security Implementation Skills: Demonstrate practical skills in implementing security measures, access controls, and authentication mechanisms to safeguard against unauthorized access and potential data breaches in computing systems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
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1	Introduction to information security in computer systems	Lecture, Discussion	3	CLO1
2	Introduction to Symmetric Ciphers, Block Ciphers, and the Data Encryption Standard	Lecture, Practical Examples	3	CLO2
3	Introduction to Number Theory in computer security	Lecture, Problem Solving	3	CLO3
4	Mathematical Tools for Cryptography: Substitutions and Permutations, Modular Arithmetic, Euclid's Algorithm, Finite Fields, Polynomial Arithmetic, Discrete Logarithms	Lecture, Mathematical Exercises	3	CLO3
5	Classical Encryption Techniques and Conventional Symmetric Encryption Algorithms	Lecture, Case Studies	3	CLO2, CLO4
6	Modern Symmetric Encryption Algorithms: Data Encryption Standard (DES)	Lecture, Hands-on Implementation	3	CLO2, CLO4
7	Modern Symmetric Encryption Algorithms: Advanced Encryption Standard (AES)	Lecture, Practical Applications	3	CLO2, CLO4
8	Contemporary Symmetric Ciphers, Confidentiality Using Symmetric Encryption Public, Key-Encryption	Lecture, Group Discussions	3	CLO2, CLO4
9	Public-Key Cryptography and RSA, Key Management; Other Public-Key Cryptosystems	Lecture, Problem-Solving Sessions	3	CLO2, CLO4
10	Message Authentication and Hash Functions	Lecture, Hands-on Exercises	3	CLO3, CLO5
11	Digital Signatures, and Certificates	Lecture, Case Studies	3	CLO3, CLO5
12	User authentication and key management; Security Handshake Protocols, Network	Lecture, Practical Scenarios	3	CLO1, CLO2
13	Security Practice, Application Electronic Mail Security, IP Security, Web Security, Electronic Commerce Security, System Security, Intruders, Malicious Software, Firewalls	Lecture, Real-world Applications	3	CLO1, CLO2, CLO4
14	Review and discussion	Review Session	3	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H				M		H	M				M

CLO2		H	H		M							
CLO3		H			M		H					
CLO4		H			H							
CLO5		H	H					M				

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
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1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Cryptography and network security, W. Stallings, Latest edition.
2. Cryptography & Network Security, Behrouz A. Forouzan, Latest edition.

Reference Books:

3. MC Nab, Network Security Assessment, Latest edition.
4. Chuvakin, Peikari, Security Warrior.
5. Lockhart, Network Security Hacks.
6. Charis Brenton. Mastering Network Security.
7. John E. Canavan, The Fundamentals of Network Security.

CSEL-4110: Cryptography and Network Security Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4110	Cryptography and Network Security Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of

the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is a hands-on, practical extension of the theoretical concepts covered in the Cryptography and Network Security course. This lab is designed to provide students with real-world experiences in implementing security measures, cryptographic protocols, and network security solutions. Students will gain proficiency in utilizing various tools and techniques to secure communication, authenticate users, and protect data integrity. The lab exercises will cover a range of topics, including cryptographic algorithm implementation, network protocol analysis, and the deployment of security mechanisms in networked environments. Through practical scenarios and case studies, students will develop practical skills essential for addressing cybersecurity challenges in modern computing.

Justification of this course in the program entity

This lab course is mandatory and stands as an indispensable component within the computer science discipline, strategically designed to fortify theoretical knowledge with hands-on experiences. By providing practical applications of cryptographic principles and network security measures, the lab ensures that students are adept at translating conceptual understanding into tangible skills. This practical approach is essential in preparing graduates for the dynamic landscape of cybersecurity, where real-world scenarios demand not only theoretical comprehension but also the ability to implement security measures, analyze network protocols, and respond to potential threats. The lab's emphasis on user authentication, intrusion detection, and security mechanism deployment aligns seamlessly with industry demands, cultivating a skill set that is highly relevant in today's cybersecurity landscape. Ultimately, the Cryptography and Network Security Lab enhances the computer science program by fostering a comprehensive education that equips students with the tools and expertise necessary to navigate the complex challenges of securing digital systems and networks effectively.

Course Objectives

The major objectives of the course are to:

- i. Gain hands-on experience in implementing cryptographic algorithms, including symmetric and asymmetric key encryption, hash functions, and digital signatures.
- ii. Analyze network protocols and identify vulnerabilities, providing insights into the practical application of secure communication principles.
- iii. Deploy security mechanisms, access controls, and user authentication protocols in networked environments, emphasizing practical implementation skills.
- iv. Develop proficiency in detecting and responding to network intrusions, understanding the practical aspects of maintaining network security.
- v. Familiarize with and utilize security tools and technologies for assessing, monitoring, and enhancing network security.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Solve Problems Using Number Theory in Terms of Information Security: Students will demonstrate the ability to apply number theory concepts to address problems related to information security, emphasizing the practical application of mathematical principles.

CLO2: Implement Security Mechanisms, Including Encryption, Decryption, Digital Signature, and Hash Function: Students will implement various security mechanisms, including encryption, decryption, digital signatures, and hash functions, showcasing practical skills in securing information and communications.

CLO3: Analyze Different Cryptographic Algorithms: Students will critically analyze and evaluate different cryptographic algorithms, gaining a deep understanding of their strengths, weaknesses, and suitability for specific security requirements.

CLO4: Design and Implement Secure Network Protocols: Students will design and implement secure network protocols, demonstrating the ability to create communication channels with a focus on confidentiality, integrity, and authenticity.

CLO5: Troubleshoot and Respond to Security Incidents: Students will develop the skills to troubleshoot and respond effectively to security incidents, showcasing the ability to identify vulnerabilities, mitigate risks, and implement corrective measures in a timely manner.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Designing mathematical model for cryptographic algorithms	Lecture, Problem Solving	2	CLO1
2	Implementation of traditional cryptography: Caesar cipher, Affine cipher, Vigenere cipher	Practical Implementation, Case Studies	2	CLO2
3	Implementation of traditional cryptography: Hill cipher, Playfair cipher	Hands-on Exercises, Group Discussions	2	CLO2
4	Using brute force and statistical attack to break security algorithms	Practical Demonstrations, Group Discussions	2	CLO3

5	Encryption and decryption using DES algorithm	Hands-on Implementation, Case Studies	2	CLO2, CLO4
6	Encryption and decryption using AES algorithm	Practical Exercises, Problem Solving	2	CLO2, CLO4
7	Encryption and decryption using RSA algorithm	Hands-on Implementation, Group Discussions	2	CLO2, CLO4
8	Encryption and decryption using ElGamal algorithm	Practical Demonstrations, Case Studies	2	CLO2, CLO4
9	Encryption and decryption using Rabin-crypto system	Hands-on Exercises, Problem Solving	2	CLO2, CLO4
10	Encryption and decryption using Elliptic curve system	Practical Implementation, Group Discussions	2	CLO2, CLO4
11	Implementation of SHA-512 hash function	Hands-on Exercises, Case Studies	2	CLO2, CLO4
12	Implementation of MD-5 hash function	Practical Demonstrations, Problem Solving	2	CLO2, CLO4
13	Implementation of digital signature scheme: RSA, ElGamal	Hands-on Implementation, Group Discussions	2	CLO2, CLO4
14	Implementation of digital signature scheme: DSS	Practical Exercises, Case Studies	2	CLO2, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M		H	H								
CLO2		H	H	H								
CLO3		H			H							
CLO4			H	H	H				M			
CLO5		H			H	H			M			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two

Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Cryptography and network security, W. Stallings, Latest edition.
2. Cryptography & Network Security, Behrouz A. Forouzan, Latest edition.

Reference Books:

3. MC Nab, Network Security Assessment, Latest edition.
4. Chuvakin, Peikari, Security Warrior.
5. Lockhart, Network Security Hacks.
6. Charis Brenton. Mastering Network Security.
7. John E. Canavan, The Fundamentals of Network Security.

CSE-4207: Network Routing and Switching

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4207	Network Routing and Switching	Theory	3	60	20	10	10	100

Course Description: Network Routing and Switching is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

This course is designed to provide students with a comprehensive understanding of internetworking and network protocols, including IP addressing, routing of IP packets, unicast and multicast routing techniques. Moreover, the course will also discuss multimedia data transmission over Internet and QoS routing technologies. In addition, the course will emphasis on application and demonstration of skills acquired in pre-requisite courses.

Course Objectives

The major objectives of the course are:

- i. Introduce students to the fundamental concepts of network routing and switching, including the OSI model, TCP/IP protocols, subnetting, and addressing.
- ii. Teach students how to configure and manage routers and switches, including setting up routing protocols such as OSPF, RIP, and BGP, as well as VLANs, trunking, and port security.
- iii. Provide students with the skills to design and implement efficient and scalable network architectures using routers, switches, and other networking devices.
- iv. Explore various routing protocols in depth, including their operation, advantages, and limitations, and teach students how to select and configure the appropriate routing protocol for different network scenarios.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Understand the core concepts of IPv4 and IPv6 address structure

CLO2: Understand the working procedure of different routing protocols

CLO3: Design effective and efficient IP networks based on combinations of the techniques like variable length subnet mask (VLSM), network address translation (NAT) and dynamic host configuration protocol (DHCP).

CLO4: Configure and implement various wide area networks (WANs) using VLSM concept and routing protocols.

CLO5: develop the skills to troubleshoot common network issues, including connectivity problems, routing errors, and performance issues, using diagnostic tools and techniques to identify and resolve network problems effectively.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction and Classful Addressing	d) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. e) Problem-based class	3	CLO 1

2	Subnetting and Supernetting in Classful Addressing	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 2
3	Classless Addressing and VLSM	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 2
4	IPv6 Addressing	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 2
5	Delivery and forwarding of IP Packets	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 3
6	Routing in Unicast Domain (RIP)	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students,	3	CLO 3

		including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class		
7	Routing in Unicast Domain (OSPF)	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 3
8	Routing in Unicast Domain (BGP)	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 3
9	Routing in Multicast Domain (IGMP)	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4
10	ICMP and ICMPv6	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 4

11	Routing for MANET / Ad-hoc Network	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 5
12	QoS Routing	c) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. d) Problem-based class	3	CLO 5
13	Real Time Traffic over Internet	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 5
14	Revision and Solution		3	

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
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Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Books:

1. TCP/IP Protocol Suite, 4th Edition by Behrouz A. Forouzan, Publisher: McGraw Hill
2. Computer Networking: A Top-Down Approach, 7th Edition by James F. Kurose, Keith W. Ross, Publisher: Pearson.

Reference Books:

1. Internetworking with TCP/IP, Principles, Protocols and Architecture, 6th Edition by Douglas E. Comer, Publisher: Pearson.
2. CCNA Routing and Switching Study Guide: Exams 100-101, 200-101, and 200-120, 1st Edition by Todd Lammle, Publisher: Sybex.

CSE-4208: Network Routing and Switching Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Continuous Evaluation	Class Participation	Class attendance	
CSE-4208	Network Routing and Switching Lab	Practical	1.0	30	10	5	5	50

Course Description: Network Routing and Switching Lab is a optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The course carries 1.0 credits and 30 lecture hours. Among the 30 hours, 24 are for computer based practical implementation and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

A Network Routing and Switching Lab is indispensable in the curriculum as it provides students with the practical skills and experience necessary to excel in the field of networking. In this lab, students have the opportunity to apply theoretical concepts learned in lectures to real-world scenarios, gaining hands-on experience in configuring, managing, and troubleshooting network routers, switches, and other networking devices. Through guided lab exercises and simulations, students can explore various routing and switching protocols, design and implement network architectures, and troubleshoot common network issues in a controlled environment.

Course Objectives

The major objectives of the course are:

- v. Provide students with hands-on experience in configuring, managing, and troubleshooting network routers, switches, and other networking devices in a laboratory setting.
- vi. Enable students to configure and manage routers and switches, including setting up routing protocols, VLANs, trunking, port security, and other networking features.
- vii. Teach students how to implement and manage routing protocols such as OSPF, EIGRP, BGP, and RIP, including configuring routing parameters and troubleshooting routing issues.
- viii. Guide students in designing and optimizing network architectures to meet specific requirements, considering factors such as scalability, redundancy, and performance, using routing and switching techniques.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: demonstrate proficiency in configuring and managing network routers, switches, and other devices, including setting up routing protocols, VLANs, trunking, and port security.

CLO2: implement and manage routing protocols such as OSPF, EIGRP, BGP, and RIP, including configuring routing parameters and troubleshooting routing issues.

CLO3: design and optimize network architectures to meet specific requirements, considering factors such as scalability, redundancy, and performance, using routing and switching techniques.

CLO4: develop strong troubleshooting skills by diagnosing and resolving common network issues using appropriate tools and techniques, enhancing their problem-solving abilities.

CLO5: collaborate with peers to solve complex networking challenges, fostering teamwork and communication skills essential for success in the field of networking.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs

1	(Introduction) Introduction to Packet Tracer, Basic Commands, and constructing a simple network topology on Packet Tracer.	f) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. g) Problem-based class	3	CLO 1
2	(VLAN) Implement the given network topology using VLAN and VLAN trunk in packet tracer.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 2
3	(Classful addressing with static routing) Implement the given network topology using classful addressing in packet tracer and enable data transfer among the hosts using static routing.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 2
4	(Subnetting in Classful addressing with static routing) Subnet the given network according to the specification using classful addressing, implement it in packet tracer and test data transmission among different networks using Static routing only.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 2
5	(Subnetting in Classless addressing with static routing) Subnet the given network according to the specification using classless addressing,	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over	3	CLO 3

	implement it in packet tracer and test data transmission among different networks using Static routing only.	lecture materials, question-answer sessions, etc. h) Problem-based class		
6	(Subnetting in Classful addressing with RIP routing) Subnet the given network according to the specification using classful addressing, implement it in packet tracer and test data transmission among different networks using RIP routing only.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 3
7	(Subnetting in Classless addressing with RIP routing) Subnet the given network according to the specification using classless addressing, implement it in packet tracer and test data transmission among different networks using RIPv2 routing only.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 3
8	(Subnetting in Classless addressing with OSPF routing) Subnet the given network according to the specification using classless addressing, implement it in packet tracer and test data transmission among different networks using OSPF routing only.	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 3

9	(Subnetting in Classless addressing with Multi-area OSPF routing) Subnet the given network according to the specification using classless addressing, implement it in packet tracer and test data transmission among different networks using Multi-area OSPF routing only.	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 4
10	(IPv6 addressing with static routing) Implement the network topology in packet tracer according to the specification using IPv6 addressing and test data transmission among different networks using Static routing only.	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 4
11	(IPv6 address auto-configuration with static routing) Implement the network topology in packet tracer according to the specification using IPv6 auto-configuration and test data transmission among different networks using Static routing only.	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. f) Problem-based class	3	CLO 5
12	(IPv6 addressing with RIPng routing) Implement the network topology in packet tracer according to the specification using IPv6 addressing and test data transmission among	e) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc.	3	CLO 5

	different networks using RIPng routing only.	f) Problem-based class		
13	(IPv6 addressing with MOSPF routing) Implement the network topology in packet tracer according to the specification using IPv6 addressing and test data transmission among different networks using MOSPF routing only.	g) Enhanced lectures containing the whiteboard-marker and/or PowerPoint presentation-based lectures that include the following elements: pausing for discussion among students, including tests/quizzes over lecture materials, question-answer sessions, etc. h) Problem-based class	3	CLO 5
14	(Classful addressing with BGP routing) Implement a given network topology using classful addressing in packet tracer and enable data transfer among different networks using BGP routing.		3	

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H				H							
CLO2						H						
CLO3			H									
CLO4							H				H	
CLO5											H	H

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report	√	√	√	√	√
Viva/Quiz	√	√	√	√	√
Lab Performance	√	√	√	√	√

Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Lab Report (5)	Quiz (5)	Class attendance (5)	Lab Performance (5)
Remember				
Understand		25%		25%
Apply	25%	40%		40%
Analyze	40%	25%		25%
Evaluate	25%	10%		10%
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains weekly problem based lab tasks carrying 20% marks, and the class performance/quiz/viva/presentation carries 10% marks Class attendance carrying 10% marks. In the final examination, students have to solve one/two problems. The complete solution of the problem with demonstration contains 45% marks whereas viva/question answering related to the problem contains 25% marks. The marks of final examination and continuous assessment are added, and converted to the letter grade.

PART D: LEARNING RESOURCES

Text Books:

- TCP/IP Protocol Suite, 4th Edition by Behrouz A. Forouzan, Publisher: McGraw Hill

4. Computer Networking: A Top-Down Approach, 7th Edition by James F. Kurose, Keith W. Ross, Publisher: Pearson.

Reference Books:

3. Internetworking with TCP/IP, Principles, Protocols and Architecture, 6th Edition by Douglas E. Comer, Publisher: Pearson.
4. CCNA Routing and Switching Study Guide: Exams 100-101, 200-101, and 200-120, 1st Edition by Todd Lammle, Publisher: Sybex.

CSE-4213: Neural Network and Fuzzy System

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4213	Neural Network and Fuzzy System	Theory	3	60	20	10	10	100

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This is a course that explores two fundamental paradigms in artificial intelligence and computational intelligence. Neural networks are computational models inspired by the structure and function of the human brain, capable of learning complex patterns and relationships from data. Fuzzy systems, on the other hand, deal with uncertainty and imprecision in data by employing fuzzy logic to model human-like reasoning. This course typically covers topics such as neural network architectures, learning algorithms, fuzzy set theory, fuzzy logic operations, and their applications in pattern recognition, control systems, and decision-making. Students gain insights into how these techniques can be combined or used independently to solve real-world problems in various domains such as robotics, finance, healthcare, and more.

Justification of this course in the program entity

This course offers a compelling justification for its study in today's technological landscape. With the increasing complexity of real-world problems across diverse domains, there is a growing need for intelligent systems capable of handling uncertainty, learning from data, and making decisions akin to human reasoning. Neural networks provide a powerful framework for learning complex patterns and relationships from data, enabling tasks such as image recognition, natural language processing, and predictive analytics. On the other hand, fuzzy systems excel in modeling and reasoning with uncertain or imprecise information, offering robust solutions for

control systems, decision-making, and pattern recognition in uncertain environments. By combining these two paradigms, students gain a comprehensive understanding of how to tackle real-world problems that traditional computational approaches struggle to address. Moreover, the interdisciplinary nature of this course prepares students to apply these techniques across a wide range of fields, including engineering, finance, healthcare, and beyond, making it a crucial component of modern artificial intelligence and computational intelligence education.

Course Objectives

The major objectives of the course are to:

- i. Understand the theoretical foundations and practical applications of neural networks and fuzzy systems.
- ii. Develop proficiency in designing and implementing neural network architectures for various tasks such as pattern recognition, classification, and regression.
- iii. Gain expertise in employing fuzzy logic techniques to handle uncertainty and imprecision in decision-making and control systems.
- iv. Explore advanced topics such as hybrid systems combining neural networks and fuzzy logic for enhanced performance.
- v. Apply learned concepts to real-world problems through hands-on projects and case studies.
- vi. Foster critical thinking and problem-solving skills by analyzing and evaluating the effectiveness of neural network and fuzzy system solutions in different scenarios.
- vii. Prepare students for careers in artificial intelligence, machine learning, robotics, and related fields by providing a solid foundation in these essential techniques.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Demonstrate a comprehensive understanding of the theoretical foundations and principles underlying neural networks and fuzzy systems, including their architectures, learning algorithms, and applications in various domains.

CLO2: Design, implement, and optimize neural network architectures for tasks such as pattern recognition, classification, regression, and time-series analysis, utilizing appropriate activation functions, loss functions, and optimization techniques.

CLO3: Apply fuzzy logic principles to model and reason with uncertain or imprecise data, designing fuzzy inference systems and membership functions to address complex decision-making and control problems effectively.

CLO4: Analyze and evaluate the performance of neural network and fuzzy system solutions using appropriate evaluation metrics, considering factors such as accuracy, precision, recall, convergence rate, and computational efficiency.

CLO5: Synthesize and integrate neural network and fuzzy system techniques to develop hybrid intelligent systems, demonstrating proficiency in combining the strengths of both paradigms to address challenging real-world problems effectively.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Artificial Neural Network: Introduction, Model of a neuron	Lecture, Rapport building, Demonstration, Group work	3	CLO1
2	Problem Domain and Application of ANN, Activation functions	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2
3	Perceptron: Introduction, Linear Classification, Learning Alg.	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2
4	Multi-layer Perceptron: Architecture, I/O Mapping, Supervised Learning	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2
5	Backpropagation Algorithm: Introduction, Mathematical Expression	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2, CLO4
6	Backpropagation: Mathematical expression, Training for NN	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2, CLO4
7	Introduction to Fuzzy Logic, Fuzzy Arithmetic, Fuzzy Relations	Lecture, Rapport building, Demonstration	2	CLO1, CLO3
8	Possibility Theory, Fuzzy Inference, Defuzzification Techniques	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO3
9	Radial Basis function Network, Self-Organizing Maps	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO2
10	Convolutional Neural Network: Introduction, Architecture	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2
11	State-of-the-art CNN, Hopfield Network, Intro to RNN	Lecture, Rapport building, Project work, Demonstration	3	CLO1, CLO2
12	Intro to LSTM, Back Propagation in LSTM	Lecture, Rapport building, Project work, Demonstration	3	CLO1, CLO2

13	Introduction to GAN, Problem Solving with GAN	Lecture, Group work, Demonstration, Group work	3	CLO1, CLO5
14	Revisions, Q&A	Lecture, Rapport building, Group work, Demonstration	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H		H								
CLO2		H	H	H				M				
CLO3	M	H	H					M				
CLO4				H				H	H		M	
CLO5		H	H	H								M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10

Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Intelligent Control Systems Using Soft Computing Methodologies; Editor: Ali Zilouchian, Mo Jamshidi
2. Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville
3. Fuzzy and Neural Approaches in Engineering by Lefteri H. Tsoukalas, Robert E.Uhrig, Lotfi A. Zadeh
4. Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurélien Géron

CSEL-4214: Neural Network and Fuzzy System Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)
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				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4 214	Neural Network and Fuzzy System Lab	Lab	1	30	10	5	5	50

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. The Neural Network and Fuzzy System Lab provides students with hands-on experience in implementing, training, and fine-tuning neural network architectures and fuzzy logic systems. In this lab, students have the opportunity to work with various software tools and programming languages commonly used in the field, such as TensorFlow, Keras, MATLAB, or Python libraries like scikit-fuzzy.

Justification of this course in the program entity

The course on Neural Networks and Fuzzy Systems Lab is highly justified in today's technological landscape due to its critical role in addressing complex real-world problems. Neural networks offer powerful tools for learning intricate patterns and relationships from data, enabling advancements in areas such as image recognition, natural language processing, and predictive analytics. Similarly, fuzzy systems excel in handling uncertainty and imprecision, providing robust solutions for decision-making and control in dynamic environments. By combining these two paradigms, students gain a comprehensive understanding of how to tackle a wide range of challenges across various domains. This course equips students with essential skills in designing, implementing, and optimizing neural network architectures and fuzzy logic systems, preparing them for careers in artificial intelligence, machine learning, robotics, and beyond. Furthermore, the interdisciplinary nature of the course ensures that students are well-equipped to apply these techniques across diverse fields, making it a crucial component of modern computational intelligence education.

Course Objectives

The major objectives of the course are:

- i. To provide a thorough understanding of the theoretical foundations and practical applications of neural networks and fuzzy systems.
- ii. To develop proficiency in designing, implementing, and optimizing neural network architectures for various tasks such as pattern recognition, classification, and regression.
- iii. To gain expertise in employing fuzzy logic techniques to handle uncertainty and imprecision in decision-making and control systems.

- iv. To explore advanced topics such as hybrid systems combining neural networks and fuzzy logic for enhanced performance.
- v. To apply learned concepts to real-world problems through hands-on projects and case studies, fostering critical thinking and problem-solving skills.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Execute practical implementations of neural network architectures and fuzzy logic systems using appropriate software tools and programming languages.

CLO2: Demonstrate proficiency in training, fine-tuning, and optimizing neural network models for various tasks, including pattern recognition, classification, and regression, through hands-on experimentation.

CLO3: Apply fuzzy logic principles to design and implement fuzzy inference systems, including defining membership functions, constructing fuzzy rules, and developing fuzzy controllers, for decision-making and control applications.

CLO4: Analyze and evaluate the performance of neural network and fuzzy system solutions using real-world datasets, considering factors such as accuracy, precision, recall, and computational efficiency, through practical experimentation.

CLO5: Integrate neural network and fuzzy system techniques to develop hybrid intelligent systems, showcasing the ability to combine the strengths of both paradigms to solve complex real-world problems effectively in a laboratory setting.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to Tensorflow with Python	Lecture, Hands-on	2	CLO1, CLO2
2	Introduction to Tensorflow with Python (continued)	Lecture, Hands-on	2	CLO1, CLO2
3	Introduction to Keras Library	Lecture, Hands-on	2	CLO1, CLO2
4	Introduction to Keras Library (continued)	Lecture, Hands-on	2	CLO1, CLO2
5	Image Classification from Scratch	Hands-on	2	CLO1, CLO2
6	Structured Data Classification from Scratch	Hands-on	2	CLO1, CLO2

7	Time Series Forecasting, Anomaly Detection	Hands-on	2	CLO1, CLO2, CLO4
8	NLP Application: Text Classification	Hands-on	2	CLO1, CLO2, CLO4
9	Audio Data: Speaker Recognition	Hands-on	2	CLO1, CLO2, CLO4
10	GAN: Develop a DCGAN for Grayscale Handwritten Digits	Hands-on	2	CLO1, CLO5
11	Moving Square Video Prediction Problem with LSTM	Hands-on	2	CLO1, CLO2, CLO4
12	Implement a Fuzzy Inference System to Design a Medical Diagnostic System	Hands-on	2	CLO1, CLO3, CLO5
13	Project Evaluation, Presentation and Viva	Presentation, Viva	2	CLO1, CLO2, CLO3, CLO4, CLO5
14	Lab Performance Evaluation	Assessment	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1		H			H							
CLO2				H	H			M				
CLO3			H	H	M							M
CLO4		H		M	M							H
CLO5		H	H									M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Intelligent Control Systems Using Soft Computing Methodologies; Editor: Ali Zilouchian, Mo Jamshidi
2. Deep Learning by Ian Goodfellow and Yoshua Bengio and Aaron Courville
3. Fuzzy and Neural Approaches in Engineering by Lefteri H. Tsoukalas, Robert E.Uhrig, Lotfi A. Zadeh
4. Hands-On Machine Learning with Scikit-Learn and TensorFlow by Aurélien Géron

CSE-4221: Natural Language Processing

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4221	Natural Language Processing	Theory	3	60	20	10	10	100

Course Description: This course is a optional course for the B.Sc. programme in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. Natural Language Processing (NLP) is a cutting-edge field at the intersection of computer science, artificial intelligence, and linguistics, aiming to enable machines to understand, interpret, and generate human language. This course provides a comprehensive exploration of NLP techniques and applications, covering fundamental concepts such as tokenization, part-of-speech tagging, syntactic and semantic analysis, and sentiment analysis. Students will delve into machine learning models for language processing, including neural networks and deep learning architectures. Practical hands-on exercises and projects will reinforce theoretical concepts, allowing students to build applications that process and comprehend human language, such as chatbots, sentiment analyzers, and machine translation systems.

Justification of this course in the program entity

This course is mandatory and plays a pivotal role in the computer science program by equipping students with indispensable knowledge and skills to navigate the complex landscape of cybersecurity. In today's digital age, where the integrity and confidentiality of information are paramount, the course provides a comprehensive understanding of cryptographic techniques and network security principles. This knowledge is not only crucial for protecting sensitive data but also forms the bedrock for designing secure and resilient software systems. The industry relevance of the course is undeniable, as cybersecurity concerns continue to escalate, making graduates with expertise in cryptography and network security highly sought after. Ultimately, the course serves as a cornerstone in preparing computer science students to address the ever-evolving challenges of ensuring the security and reliability of information in networked environments.

Course Objectives

The major objectives of the course are to:

- i. Understand NLP Fundamentals: Gain a solid understanding of the fundamental concepts and techniques in Natural Language Processing, including language representation, syntax, and semantics.
- ii. Explore NLP Applications: Explore practical applications of NLP, such as sentiment analysis, named entity recognition, and machine translation, to comprehend the real-world impact of language processing.
- iii. Implement NLP Algorithms: Develop proficiency in implementing NLP algorithms, including tokenization, part-of-speech tagging, and syntactic analysis, using both traditional and modern machine learning approaches.
- iv. Dive into Deep Learning for NLP: Explore the integration of deep learning models, including recurrent neural networks (RNNs) and transformers, in NLP tasks, emphasizing hands-on implementation and experimentation.
- v. Build NLP Applications: Apply acquired knowledge to build practical NLP applications, such as chatbots, language translators, and sentiment analyzers, through individual and collaborative projects.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Design, Implement, and Understand Appropriate Neural Network Models: Students will proficiently design, implement, and comprehend neural network models using modern programming languages, demonstrating a sound understanding of their architecture and applications.

CLO2: Identify Semantic Relationships with Dependency Parsing: Students will demonstrate the ability to identify semantic relationships between words in a sentence through dependency parsing, showcasing expertise in syntactic and semantic analysis.

CLO3: Make Large-Scale Word Predictions with Language Models and RNNs: Students will showcase proficiency in making large-scale word predictions using language models, recurrent neural networks (RNNs), and neural machine translation, emphasizing the practical application of these techniques.

CLO4: Improve NLP Models and Pre-trained Transformers: Students will enhance NLP models and pre-trained transformers to achieve more efficient natural language processing and understanding, showcasing the ability to optimize existing models for improved performance.

CLO5: Apply Named Entity Recognition Techniques: Students will demonstrate the ability to implement and apply Named Entity Recognition techniques, identifying and classifying entities such as names, locations, and organizations in a given text.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction: Regular Expressions, Text Normalization, Edit Distance	Lecture, Practical Exercises	2	CLO1, CLO5
2	N-gram Language Models	Lectures, Hands-on Implementation	3	CLO3
3	Naive Bayes and Sentiment Classification	Lectures, Case Studies	3	CLO2
4	Logistic Regression, Neural Network	Lectures, Practical Demonstrations	3	CLO1
5	Vector Semantics and Embeddings, WordNet	Lectures, Group Discussions	3	CLO4
6	Neural Networks and Neural Language Models	Lectures, Hands-on Exercises	3	CLO1, CLO3
7	Part-of-Speech Tagging	Lectures, Practical Implementation	3	CLO2
8	Sequence Processing with Recurrent Networks	Lectures, Case Studies	3	CLO3
9	Encoder-Decoder Models, Attention, and Contextual Embeddings	Lectures, Group Discussions	3	CLO3
10	Machine Translation using Constituency Grammars	Lectures, Practical Demonstrations	3	CLO4
11	Constituency Parsing, Statistical Constituency Parsing, and Dependency Parsing	Lectures, Hands-on Exercises	3	CLO2
12	Logical Representations of Sentence Meaning	Lectures, Group Discussions	3	CLO4
13	Information Extraction: Named Entity Recognition, Relation Extraction, and Event Entity Extraction	Lectures, Practical Implementation	3	CLO5
14	Summarization: Question Answering, Dialogue Systems, Chatbots, and Phonetics	Lectures, Case Studies	2	CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1		H	H	M								
CLO2		H	H		H							
CLO3			H	H	M							
CLO4		H	H		M	M						
CLO5	M			H	M							

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	15
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
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1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Natural Language Processing with Python, Steven Bird et al., O'Reilly Media (2009).
2. Speech and Language Processing, Daniel Jurafsky, James H. Martin, 2nd Edition.

CSEL-4222: Natural Language Processing Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4222	Natural Language Processing Lab	Lab	1	30	10	5	5	50

Course Description: This course is a optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is a hands-on, practical extension of the theoretical concepts covered in the Natural Language Processing (NLP). It is designed to complement theoretical concepts learned in the NLP course by providing hands-on experience with tools and technologies used in natural language processing applications. Through a series of practical

exercises and projects, students will gain proficiency in implementing and experimenting with various NLP algorithms and models. The lab will cover essential aspects such as text processing, sentiment analysis, named entity recognition, and machine translation. Additionally, students will have the opportunity to work on real-world NLP challenges, fostering critical thinking and problem-solving skills in the domain of language processing.

Justification of this course in the program entity

This lab course is mandatory and stands as an indispensable component within the computer science discipline, strategically designed to fortify theoretical knowledge with hands-on experiences. By providing practical applications of cryptographic principles and network security measures, the lab ensures that students are adept at translating conceptual understanding into tangible skills. This practical approach is essential in preparing graduates for the dynamic landscape of cybersecurity, where real-world scenarios demand not only theoretical comprehension but also the ability to implement security measures, analyze network protocols, and respond to potential threats. The lab's emphasis on user authentication, intrusion detection, and security mechanism deployment aligns seamlessly with industry demands, cultivating a skill set that is highly relevant in today's cybersecurity landscape. Ultimately, the Cryptography and Network Security Lab enhances the computer science program by fostering a comprehensive education that equips students with the tools and expertise necessary to navigate the complex challenges of securing digital systems and networks effectively.

Course Objectives

The major objectives of the course are to:

- i. Implement NLP Algorithms: Develop practical skills in implementing NLP algorithms, including but not limited to tokenization, part-of-speech tagging, and syntactic analysis.
- ii. Experiment with Sentiment Analysis: Gain hands-on experience in sentiment analysis by working on projects that involve classifying the sentiment expressed in textual data.
- iii. Apply Named Entity Recognition (NER) Techniques: Implement and apply Named Entity Recognition techniques to identify and classify entities such as names, locations, and organizations in a given text.
- iv. Explore Machine Translation: Work on projects related to machine translation, experimenting with techniques to translate text between different languages.
- v. Develop Conversational Agents: Gain practical insights into building conversational agents or chatbots using NLP techniques, enhancing skills in natural language understanding and generation.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Practical Implementation of NLP Algorithms: Students will demonstrate the ability to practically implement NLP algorithms, including tokenization, part-of-speech tagging, and syntactic analysis, showcasing proficiency in applying theoretical concepts.

CLO2: Hands-on Experience with Sentiment Analysis: Successful students will gain hands-on experience in sentiment analysis, effectively classifying the sentiment expressed in textual data, and understanding the nuances of sentiment classification.

CLO3: Application of Named Entity Recognition (NER) Techniques: Students will implement and apply Named Entity Recognition techniques, showcasing the ability to identify and classify entities such as names, locations, and organizations in a given text.

CLO4: Experimentation with Machine Translation: Successful students will work on projects related to machine translation, experimenting with techniques to translate text between different languages and gaining practical insights into the challenges of translation.

CLO5: Development of Conversational Agents: Students will develop conversational agents or chatbots using NLP techniques, demonstrating proficiency in natural language understanding and generation for practical applications.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction of Python: python library NLTK	Lecture, Hands-on Coding	2	CLO1
2	Accessing text corpora and lexical resources using Python	Lecture, Practical Exercises	2	CLO1
3	Text pre-processing: regular expression for detecting word pattern, tokenization, and text normalization	Lecture, Coding Sessions	2	CLO1, CLO3
4	Text pre-processing: part of speech tagger (POST)	Lecture, Practical Implementation	2	CLO3
5	Designing N-gram model: unigram, bigram, and n-gram model for identifying chunks of texts	Lecture, Group Discussions	2	CLO3
6	Sentiment analysis using BoW, TFIDF, and word embedding models	Lecture, Hands-on Sentiment Analysis	2	CLO2

7	Dimensionality reduction using PCA, LDA, NMF, Random Projection, Autoencoder, T-SNE	Lecture, Practical Dimensionality Reduction	2	CLO2
8	Text classification using supervised learning models	Lecture, Coding Sessions	2	CLO2
9	Named Entity Recognition from texts	Lecture, Hands-on NER Implementation	2	CLO3
10	Topic modeling using LDA and LSA	Lecture, Group Discussions	2	CLO4
11	Text sequence generation	Lecture, Practical Sequence Generation	2	CLO4
12	Designing machine translators	Lecture, Hands-on Translation Projects	2	CLO4
13	Text summarization from newspapers and articles	Lecture, Practical Summarization	2	CLO4
14	Designing a Chatbot	Lecture, Group Chatbot Development	2	CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H	H		H			M			
CLO2		H	H	H	M	H						
CLO3	M	H	H	H		H						
CLO4		H	H	H	H	H			M			
CLO5				H	H	H		M	H			

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√

Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Natural Language Processing with Python, Steven Bird et al., O'Reilly Media (2009).
2. Speech and Language Processing, Daniel Jurafsky, James H. Martin, 2nd Edition.

CSE-4223: System Design and Analysis

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4223	System Design and Analysis	Theory	3	60	20	10	10	100

Course Description: System Design and Analysis is a compulsory course for the B.Sc. (Honours) programme in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and 45 lecture hours. Among the 45 hours, 39 are for class teaching and 6 hours are allotted for individual guidance and contact of the students with the course teacher.

Justification of this course in the program entity

The main goal of this course is to provide students with a solid background in information systems analysis and design techniques through a combination of theory and practice. It introduces the vital logical and design considerations addressed during system and application software development. The course is intended to teach students about system design Systems and models; Project management; Tools for determining system requirements; data flow diagrams; decision table and decision trees; Systems analysis: systems development life cycle models. Object oriented analysis: use-case modeling, Unified Modeling Language. Feasibility analysis, Structured analysis; systems prototyping; system design and implementation: application architecture, user interface design. Front-end and backend design; database design; software management and hardware selection. Case Studies of Information Systems.

Course Objectives

The major objectives of the course are:

- i. Understand the system development life cycle and be familiar with a variety of problem-solving tools and approaches for the design and analysis of information systems.
- ii. Draw Entity Relationship diagrams to represent common business situations.
- iii. Understand database design and tools related to system analysis.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Define various systems analysis and design concepts and terminologies

CLO2: Apply system development lifecycle; Analyze, justify and construct object-oriented software.

CLO3: Prepare and analyze software requirements for developing software projects fulfilling user requirement.

CLO4: Compare, use and synthesize different conceptual modeling techniques for systems analysis (including EROs, DFDs and UML).

CLO5: Use and select appropriate project management techniques for developing software projects and apply analysis and design methodologies for real world systems development such as Clinical Information Systems.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Introduction: Data and Information, Types of Information, Need for an Information System, Qualities of Information	h) Lecture i) Rapport building	3	CLO1
2	Introduction: System concept and Definition, Characteristics, Attributes, Elements of a system, Types of System	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO1
3	System development life cycle (SDLC): Steps in SDLC, Different types of SDLC, Waterfall Model, Prototype Model.	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO2
4	Role of System Analyst: Define System Analyst, Role of a system analyst, attributes, multifunctional role of system analyst, skills required by system analyst	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO2

5	Requirements Engineering: User Requirement Definition, System Requirements Specification, Functional and non-functional requirements, Requirement Engineering process, Feasibility Study.	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO2
6	Information Gathering : Definition, Fact-Finding, Information gathering tools, Review of Literature, Procedures and Forms On Site Observation Interviews and Questionnaires Desk research, Brainstorming Group Discussion	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO3
7	Tools for structured Analysis: Structured Analysis, Tools for Structured Analysis, Diagram (DFD) and why DFD, DFD notations, DFD level's and layers, Physical and logical DFD Decision tree, Decision Table.	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO3
8	Software Project Management: Management activities, Project planning Project scheduling, Risk management	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO4
9	Software Project Management: critical path scheduling, Gantt charts, and Network diagrams	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO4

10	Risk Analysis: Risk Management, Study Various Kind of Risks, Method Apply to Manage Several Risks, Risk Management process	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO4
11	Interface Design: Input Design, Output Design, File and database design, System architecture,	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO5
12	System Models: System modelling, UML, Class diagram, Use case Diagram, Activity diagram, Database Design, ERD diagram.	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO5
13	Software Measurement: Introduction to Measurement, Examples of Various Metrics, Measures and Indicators, LOC, Function Point.	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO5
14	Software Testing and Maintenance: Software Testing, Test case Design, Systems Support and Maintenance, Types of System Maintenance	j) Lecture, k) Rapport building l) Cooperative learning	3	CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	M	M	M	M		M						
CLO2	H	M	M	M		M						
CLO3	H	H	H	M		M						
CLO4	H	H	H	M		M						
CLO5	H	H	H	M		M	M					

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- c. Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- d. Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4

5	Question from CLO5 or Question from CLO5
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Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

5. Elias M. Awad, "Systems Analysis and Design", 2nd Edition.

Reference Books:

3. Jeffrey L. Whitten and Lonnie D. Bentley, "System Analysis and Design Methods", 7th Edition or Latest One, McGraw Hill Publications.
4. Gary B. Shelly and Harry J. Rosenblatt, "System Analysis and Design", Ninth Edition
5. Dennis, A., Wixom, B, H, and Roth, R, M., "systems Analysis and Design", Latest edition, John Wiley & Sons.
6. Roger Pressman, "Software Engineering: A Practitioner's Approach", Current Edition
7. Ian Sommerville, "Software Engineering", Ninth Edition

CSEL-4224: System Design and Analysis Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-204	Database Management System Lab	Lab	1	30	10	5	5	50

Course Description: This course is a compulsory course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. System design and analysis serve

as foundational concepts in software engineering. Understanding how to break down complex systems into manageable components, design effective architectures, and analyze trade-offs is crucial for developing high-quality software systems. This course provides students with the theoretical knowledge and practical skills necessary to excel in software engineering roles.

Justification of this course in the program entity

The main goal of this course is to provide students with a solid background in information systems analysis and design techniques through a combination of theory and practice. It introduces the vital logical and design considerations addressed during system and application software development. System design and analysis involve solving complex problems by identifying requirements, designing system architectures, and evaluating design choices. By engaging in hands-on projects and case studies, students develop critical thinking and problem-solving skills essential for addressing real-world challenges in diverse domains.

Course Objectives

The major objectives of the course are:

- i. Provide hands-on experience in applying system design principles and methodologies.
- ii. Foster proficiency in using design tools and techniques such as UML and software modeling tools.
- iii. Encourage collaborative problem-solving and decision-making through group-based projects.
- iv. Reinforce understanding of system design concepts learned in theoretical courses.
- v. Develop practical skills in creating system models, designing architectures, and evaluating design alternatives.
- vi. Prepare students for careers in software engineering, systems engineering, and project management.
- vii. Emphasize effective communication, collaboration, and critical thinking skills in system design and analysis.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able

CLO1: Elicit, analyze, and prioritize system requirements from stakeholders, considering both functional and non-functional aspects, and document them effectively using appropriate techniques such as use cases, user stories, and requirement prioritization.

CLO2: To formulate systematic database design approaches covering conceptual design, logical design and an overview of physical design.

CLO3: Demonstrate proficiency in creating system models using standard modeling languages (e.g., UML), including use case diagrams, class diagrams, sequence diagrams, and activity diagrams, and translate these models into detailed system designs that meet

CLO4: Compare, use and synthesize different conceptual modeling techniques for systems analysis (including EROs, DFDs and UML).

CLO5: Use and select appropriate project management techniques for developing software projects and apply analysis and design methodologies for real world systems development such as Clinical Information.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	To find the requirement specification (both functional and nonfunctional) of a given Problem.	Lectures, Coding Exercises	2	CLO1
2	Design Dataflow diagram for small application such as order processing system, Results management System	Theoretical Explanation, Coding Practice	2	CLO1, CLO2
3	System design: Sample diagrams: Class diagram, sequence diagram.	Lectures, Problem-solving Sessions	2	CLO1, CLO2, CLO3
4	System design: Sample diagrams: Use case diagram, Activity Diagram.	Interactive Lectures,	2	CLO2, CLO3
5	Implement Basic COCOMO model (Software Project Measurement)	Practical Examples, Coding Sessions	2	CLO2, CLO3
6	Software Project Management Analysis Using Microsoft Project	Theoretical Explanation, Coding Practice	2	CLO2, CLO3, CLO4
7	Software Project Management Analysis Using Microsoft Project: Scheduling	Lectures, Problem-solving Sessions	2	CLO2, CLO3, CLO4
8	Software Project Management Analysis Using Microsoft Project: CPM	Coding Workshops, Practical Examples	2	CLO2, CLO3, CLO4

9	Software Project Measurement: FP and LOC	Theoretical Explanation, Problem-solving Sessions	2	CLO2, CLO3, CLO4, CLO5
10	Database Design: ERD design.	Interactive Sessions, Coding Exercises	2	CLO3, CLO4, CLO5
11	Database Design: ERD to Database development.	Lectures, Algorithm Implementation	2	CLO3, CLO4, CLO5
12	Feasibility Analysis for sample project.	Practical Examples, Coding Sessions	2	CLO3, CLO4, CLO5
13	SRS design for sample project	Problem-solving Sessions, Coding Practice	2	CLO3, CLO4, CLO5
14	Revise and Q/A	Review Sessions, Open Discussion	2	CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1			H		H	H						M
CLO2	M		H			H						M
CLO3		H	H			H			H			M
CLO4		H	H	H		H						
CLO5			H		H	H						

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 15. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggested Text Books:

4. Elias M. Awad, "Systems Analysis and Design", 2nd Edition.

Reference Books:

1. Jeffrey L. Whitten and Lonnie D. Bentley, "System Analysis and Design Methods", 7th Edition or Latest One, McGraw Hill Publications.
2. Gary B. Shelly and Harry J. Rosenblatt, "System Analysis and Design", Ninth Edition
3. Dennis, A., Wixom, B. H., and Roth, R. M., "systems Analysis and Design", Latest edition, John Wiley & Sons.
4. Roger Pressman, "Software Engineering: A Practitioner's Approach", Current Edition
5. Ian Sommerville, "Software Engineering", Ninth Edition
6. Ramez Elmasri and Shamkant Navathe, "Fundamentals of Database Systems", Pearson, 7th Edition, March 2017.

CSE-4225: Blockchain Technology and Application

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4225	Blockchain Technology and Application	Theory	3	60	20	10	10	100

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course provides an in-depth exploration of blockchain technology and its diverse applications across various industries. Students will gain a comprehensive understanding of the fundamental principles behind blockchain, including cryptographic techniques, consensus mechanisms, and distributed ledger technology. Through a combination of theoretical lectures and practical exercises, participants will learn how blockchain enables secure, transparent, and immutable record-keeping, revolutionizing traditional systems of trust and verification.

Justification of this course in the program entity

This course provides a thorough examination of blockchain's core principles and practical implications. Students will explore cryptographic techniques, consensus mechanisms, and distributed ledger technology, gaining insight into how blockchain ensures secure, transparent, and immutable record-keeping. Through a mix of theoretical lectures and hands-on exercises, participants will delve into smart contracts, decentralized applications, and the nuances between permissioned and permissionless blockchains. Real-world use cases across various sectors like

finance, supply chain management, and healthcare will be explored, illuminating the potential transformative power of blockchain. By the course's end, students will be equipped to navigate the intricate landscape of blockchain technology and identify innovative opportunities within their respective domains.

Course Objectives

The major objectives of the course are to:

- i. Understand the foundational principles of blockchain technology.
- ii. Explore the practical applications of blockchain across different industries.
- iii. Analyze the impact of blockchain on traditional systems of trust and verification.
- iv. Develop hands-on skills in implementing blockchain solutions, including smart contracts and decentralized applications.
- v. Identify opportunities for innovation and integration of blockchain technology within various domains.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Develop blockchain-based decentralized applications (DApps) proficiently, showcasing the ability to design, implement, and deploy decentralized solutions leveraging blockchain technology effectively.

CLO2: Build decentralized storage and internet application layer services using blockchain technologies, demonstrating a comprehensive understanding of distributed systems architecture and blockchain protocols.

CLO3: Understand the design and development principles of various cryptocurrency systems, including Bitcoin, Ethereum, Hyperledger, etc., enabling critical analysis and informed decision-making in blockchain-based projects.

CLO4: Design and deploy smart contract-based applications on Ethereum using Solidity language, demonstrating expertise in writing secure and efficient smart contracts to automate business logic on the blockchain.

CLO5: Analyze and evaluate real-world applications of blockchain technology across diverse industries and use cases, demonstrating a deep understanding of blockchain's potential, limitations, and implications for digital transformation.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
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1	Introduction to cryptocurrencies, decentralization, and Bitcoin blockchain.	Lecture, Rapport building, Demonstration, Group work	3	CLO1
2	The blockchain basics: blockchain layers, blockchain components, transactions, blocks, security, integrity, confidentiality, and privacy issues of a decentralized System.	Lecture, Rapport building, Group work, Demonstration	3	CLO2
3	The blockchain basics: blockchain layers, blockchain components, transactions, blocks, security, integrity, confidentiality, and privacy issues of a decentralized System.	Lecture, Rapport building, Group work, Demonstration	3	CLO2
4	The blockchain cryptography, public/private key, hash, Merkle tree, digital signature, the development of a basic blockchain using Python/Java/Go.	Lecture, Rapport building, Group work, Cooperative learning, Demonstration	4	CLO1, CLO4
5	The blockchain cryptography, public/private key, hash, Merkle tree, digital signature, the development of a basic blockchain using Python/Java/Go.	Lecture, Rapport building, Group work, Cooperative learning, Demonstration	4	CLO1, CLO4
6	Introduction to Ethereum blockchain, Hyperledger blockchain.	Lecture, Rapport building, Demonstration	3	CLO1
7	Introduction to smart contracts, programming in Solidity.	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4
8	Introduction to smart contracts, programming in Solidity.	Lecture, Rapport building, Demonstration, Group work	3	CLO1, CLO4
9	Programming a decentralized App.	Lecture, Rapport building, Demonstration, Group work	3	CLO1
10	Permissioned blockchain, security issue, utility tokens.	Lecture, Rapport building, Project work, Demonstration	3	CLO2
11	Decentralized storage via Blockchain.	Lecture, Rapport building, Project	3	CLO2

		work, Demonstration		
12	Consensus protocol, distributed consistency, alternatives to Proof of Work.	Lecture, Group work, Demonstration	3	CLO1
13	Blockchain and Internet of Things, blockchain and Cloud of Things, blockchain and Fog of Things, blockchain and SDN.	Lecture, Group work, Demonstration	3	CLO1
14	Revisions, Q&A, List your findings, Classify topics, Solve some problems, Modify those using alternatives, Propose solutions to some new problems	Lecture, Rapport building, Group work, Demonstration	3	CLO1, CLO2, CLO4

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H		H				M				M	
CLO2			H	H								
CLO3		H	H	M								M
CLO4			H	H							M	M
CLO5		H	H								H	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		

Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Alan T. Norman(2016). Blockchain Technology Explained: The Ultimate Beginner's Guide About Blockchain Wallet, Mining, Bitcoin, Ethereum, Litecoin, Zcash, Monero, Ripple, Dash, IOTA And Smart Contracts.

2. Vivek Acharya, Anand Eswararao Yerrapati, Nimesh Prakash(2019). Oracle Blockchain Quick Start Guide, Packt Publishing
3. Chris Dannen(2017). Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginner, ACM
4. William Mougayar, Vitalik Buterin. The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology.
5. Don Tapscott, Alex Tapscott(2016). Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World.
6. Bikramaditya SinghalGautam Dhameja Priyansu Sekhar Panda(2018), Beginning Blockchain, A Beginner's Guide to Building Blockchain Solutions, Springer.

CSEL-4226: Blockchain Technology and Application Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4226	Blockchain Technology and Application Lab	Lab	1	30	10	5	5	50

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It is an immersive course designed to provide students with hands-on experience and practical skills in blockchain technology. Through a series of interactive sessions, students will explore the fundamental principles of blockchain, including cryptographic techniques, distributed ledger technology, and smart contract development. The lab emphasizes real-world applications of blockchain across various industries, allowing students to design and implement decentralized solutions, develop smart contracts, and analyze blockchain use cases. By engaging in project-based learning and collaborative exercises, participants will gain a deep understanding of blockchain's potential impact and learn how to leverage this transformative technology to address contemporary challenges.

Justification of this course in the program entity

This lab course provides hands-on learning to bridge theoretical knowledge with practical skills in blockchain. Participants delve into cryptographic techniques, distributed ledgers, and smart contract development. Through interactive sessions and projects, students explore real-world applications across industries, fostering critical thinking. This immersive experience empowers

students to leverage blockchain's transformative potential, contributing to its evolution in the digital realm.

Course Objectives

The major objectives of the course are to:

- i. Develop proficiency in implementing blockchain solutions through practical exercises and project-based learning.
- ii. Equip students with the skills to design, deploy, and test blockchain-based decentralized applications (DApps).
- iii. Explore and analyze real-world blockchain use cases, gaining insight into their practical applications and challenges.
- iv. Foster collaboration and teamwork in the development of blockchain projects, enhancing problem-solving abilities.
- v. Prepare students for industry relevance by providing hands-on experience with blockchain technology, enabling them to contribute effectively to blockchain projects and initiatives.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Demonstrate proficiency in setting up and configuring blockchain networks, including understanding the deployment process, network architecture, and consensus mechanisms.

CLO2: Apply cryptographic techniques effectively within blockchain systems, including encryption, hashing, and digital signatures, to ensure data security and integrity.

CLO3: Design and develop decentralized applications (DApps) using blockchain platforms such as Ethereum, Hyperledger, or others, showcasing proficiency in smart contract programming and decentralized application architecture.

CLO4: Analyze and evaluate real-world blockchain use cases across diverse industries, identifying opportunities for blockchain integration, and assessing the potential benefits and challenges.

CLO5: Collaborate with peers to solve complex problems and complete practical blockchain projects, demonstrating effective teamwork, communication, and project management skills in a laboratory setting.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
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1	To study different cryptocurrencies including Bitcoin, Ethereum blockchain and open account in blockchain wallet.	• Lecture • Demonstration • Group discussion	2	CLO1
2	Implement the basic component of a blockchain using Python or any programming language.	• Hands-on coding exercises • Peer-to-peer learning	2	CLO2
3	Implement the basic component of a blockchain using Python or any programming language.	• Hands-on coding exercises • Peer-to-peer learning	2	CLO2
4	Implement public/private key, hash, Merkle tree, digital signature, the development of a basic blockchain using Python/Java/Go.	• Problem-solving sessions • Coding workshops	2	CLO2, CLO3
5	Implement cryptography, public/private key, hash, Merkle tree, digital signature, the development of a basic blockchain using Python/Java/Go.	• Problem-solving sessions • Coding workshops	2	CLO2, CLO3
6	Install Ganache, Truffle to develop app on Ethereum blockchain, Hyperledger blockchain.	• Demonstration • Interactive tutorials	2	CLO1
7	Execute smart contracts developed using programming in Solidity.	• Hands-on exercises • Model evaluation techniques	2	CLO3
8	Execute smart contracts developed using programming in Solidity.	• Hands-on exercises • Model evaluation techniques	2	CLO3
9	Design and develop a decentralized App.	• Project-based learning • Individual consultations	2	CLO1, CLO3
10	Create a permissioned blockchain and generate tokens.	• Case studies • Interactive tutorials	2	CLO1, CLO2

11	Develop a decentralized storage via Blockchain.	- Group projects - Peer-to-peer learning	2	CLO1, CLO2
12	Implement different consensus protocols including Proof of Work, proof of stake.	- Case studies - Interactive tutorials	2	CLO1
13	Implement a decentralized application in IoT.	- Workshops - Hands-on coding tasks	2	CLO1
14	Revisions, Q&A, List your findings, Classify topics, Solve some problems, Modify those using alternatives, Propose solutions to some new problems.	- Debate sessions - Case studies - Interactive tutorials	2	CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H											
CLO2		H		H								
CLO3			H									
CLO4		H				H						
CLO5								H	H	H		

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		

Create	10%			
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Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Suggest Text Books:

1. Alan T. Norman(2016). Blockchain Technology Explained: The Ultimate Beginner's Guide About Blockchain Wallet, Mining, Bitcoin, Ethereum, Litecoin, Zcash, Monero, Ripple, Dash, IOTA And Smart Contracts.
2. Vivek Acharya, Anand Eswararao Yerrapati, Nimesh Prakash(2019). Oracle Blockchain Quick Start Guide, Packt Publishing
3. Chris Dannen(2017). Introducing Ethereum and Solidity: Foundations of Cryptocurrency and Blockchain Programming for Beginner, ACM
4. William Mougayar, Vitalik Buterin. The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology.

5. Don Tapscott, Alex Tapscott(2016). Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World.
6. Bikramaditya SinghalGautam Dhameja Priyansu Sekhar Panda(2018), Beginning Blockchain, A Beginner's Guide to Building Blockchain Solutions, Springer.

CSE-4227:VLSI Design

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4227	VLSI Design	Theory	3	60	20	10	10	100

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 45 hours, including 39 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. This course covers fundamental concepts of digital logic design, semiconductor device physics, and integrated circuit fabrication processes. Students learn to use industry-standard Electronic Design Automation (EDA) tools for schematic capture, simulation, synthesis, and layout of VLSI circuits. Topics include transistor-level design, timing analysis, power optimization, and design for testability. Through hands-on projects and practical exercises, students gain the skills necessary to design complex digital systems at the chip level, preparing them for careers in integrated circuit design, semiconductor industry, and related fields.

Justification of this course in the program entity

The study of VLSI Design holds supreme importance in computer science education as it delves into the core of modern computing infrastructure. Understanding VLSI principles is essential for computer scientists as it underpins the development of high-performance and energy-efficient computing systems. Through this discipline, students grasp intricate concepts of digital logic, semiconductor physics, and integrated circuit design, enabling them to innovate and optimize hardware solutions for a wide array of applications. Moreover, VLSI Design intersects with various other domains such as artificial intelligence, internet of things, and embedded systems, making it indispensable for aspiring computer scientists to comprehend. By mastering VLSI Design, students not only expand their technical prowess but also enhance their problem-solving skills, ensuring they are well-equipped to tackle the complexities of modern computing challenges.

Course Objectives

The major objectives of the course are to:

- i. Understand the fundamental principles of VLSI design, including transistor-level circuitry, logic synthesis, and physical layout.
- ii. Gain proficiency in using Electronic Design Automation (EDA) tools for designing, simulating, and verifying VLSI circuits.
- iii. Learn techniques for optimizing VLSI designs for performance, power consumption, and area efficiency.
- iv. Explore advanced topics such as timing analysis, design for manufacturability, and testing methodologies in VLSI systems.
- v. Apply theoretical knowledge to practical projects, fostering hands-on experience in designing and implementing VLSI circuits, preparing students for careers in semiconductor industry and digital system design.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Demonstrate an understanding of semiconductor physics and the operation of basic electronic devices utilized in VLSI circuits, including transistors and diodes.

CLO2: Analyze and interpret the characteristics and behavior of MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) and CMOS (Complementary Metal-Oxide-Semiconductor) technology, fundamental to VLSI design.

CLO3: Evaluate the impact of process variations, parasitic elements, and technology scaling on the performance and reliability of VLSI circuits.

CLO4: Explain the principles of digital logic design and boolean algebra as they apply to VLSI circuitry, including combinational and sequential logic circuits.

CLO5: Explore the concepts of circuit timing, propagation delay, and clocking methodologies essential for designing synchronous digital systems in VLSI.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to MOS technology: POMS, NMOS and CMOS, transistors, CMOS Fabrication	Lecture, Discussion	3	CLO1, CLO2
2	Design Approaches: Fabrication steps, stick diagrams, design rules, layout, contact cuts	Lecture, Examples	3	CLO3

3	MOS circuits, Delay Analysis: Inverter delay analysis, delay of sequential and combinational circuits	Lecture, Problem Solving	3	CLO4
4	Sequential System: Susperbuffer, Dynamic MOS circuits, Scaling of MOS circuits	Lecture, Case Studies	3	CLO5
5	Switch logic: pass transistors and transmission gates, Gate logic design	Lecture, Examples	3	CLO1, CLO2
6	Design of parity generator, multiplexers, Registers, Counters, Memory realizations	Lecture, Problem Solving	3	CLO3
7	One transistor and three transistors dynamic RAM cell design	Lecture, Examples	3	CLO4
8	Hierarchical view of VLSI System Design: Behavioral description, High level Synthesis	Lecture, Discussion	3	CLO5
9	Logic synthesis: multilevel minimization, PLA reduction, regular structure circuits	Lecture, Problem Solving	3	CLO1, CLO2
10	Synthesis of FSM-ASM chart representation and realization	Lecture, Examples	3	CLO3
11	Layout synthesis, Placement and routing, Testing of VLSI	Lecture, Case Studies	3	CLO4
12	Testing of stuck-at fault, Testing of PLAs RAM	Lecture, Problem Solving	3	CLO5
13	Introduction to Reversible Logic: Theory of reversibility, Reversible gates	Lecture, Discussion	3	CLO1, CLO2
14	Reversible circuits, reversible logic synthesis, FPGA: Introduction and VHDL programming	Lecture, Examples, Hands-on Exercises	3	CLO3, CLO4, CLO5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	H	H	H							M	
CLO2	H			H								
CLO3	H	H	H									
CLO4	H	H		H								
CLO5				H							M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
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Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	25%	25%	30%		
Analyze	20%	20%	30%		
Evaluate	20%	20%	15%		
Create	10%	10%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will

be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Steven M. Rubin, Computer Aids for VLSI Design, latest edition.
2. Wayne Wolf, Modern VLSI Design: System-On-Chip Design, latest edition.
3. Sabib H Gerez, Algorithms for VLSI Design Automation, latest edition.

CSEL-4228: VLSI Design Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Lab Assessment/Lab Report	Class Performance	Class attendance	
CSEL-4228	VLSI Design Lab	Lab	1	30	10	5	5	50

Course Description: This course is an optional course for the B.Sc. program in the Department of Computer Science and Engineering, Jagannath University. The total duration of the course is 34 hours, including 28 hours of classroom instruction and 6 hours allocated for individual guidance and interaction with the course instructor. It offers practical hands-on experience in designing and implementing Very Large Scale Integrated (VLSI) circuits. Students apply theoretical knowledge from lectures to real-world scenarios, utilizing industry-standard Electronic Design Automation (EDA) tools for schematic capture, simulation, synthesis, and layout of VLSI circuits. Through structured lab exercises and projects, students gain proficiency in transistor-level design, timing analysis, power optimization, and design for testability. They learn to interface with peripherals, manipulate memory, and implement complex digital systems on silicon. This course enables students to deepen their understanding of VLSI design principles, fostering critical thinking and problem-solving skills essential for careers in semiconductor industry and digital system design.

Justification of this course in the program entity

The "VLSI Design Lab" course is an integral component of the CSE program, providing students with practical skills essential for success in the field. In today's technology-driven world, the demand for high-performance and energy-efficient computing systems is ever-increasing. This

lab course equips students with the expertise needed to design and implement Very Large Scale Integrated (VLSI) circuits, which form the backbone of modern digital electronics. By immersing students in hands-on projects using industry-standard Electronic Design Automation (EDA) tools, the course bridges the gap between theoretical knowledge and practical application. Through designing and optimizing VLSI circuits, students gain insights into semiconductor physics, digital logic design, and circuit timing, enhancing their problem-solving abilities and critical thinking skills.

Course Objectives

The major objectives of the course are:

- i. Develop a comprehensive understanding of the principles and methodologies involved in Very Large Scale Integrated (VLSI) circuit design, including transistor-level operation, digital logic design, and timing analysis.
- ii. Gain proficiency in utilizing industry-standard Electronic Design Automation (EDA) tools for schematic capture, simulation, synthesis, and layout of VLSI circuits.
- iii. Acquire hands-on experience in designing, implementing, and testing VLSI circuits through structured lab exercises and projects.
- iv. Explore advanced topics such as power optimization, design for testability, and technology scaling to enhance the performance and reliability of VLSI systems.
- v. Foster critical thinking and problem-solving skills by analyzing and optimizing VLSI designs, considering factors such as area efficiency, power consumption, and timing constraints.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to:

CLO1: Demonstrate proficiency in designing and implementing Very Large Scale Integrated (VLSI) circuits using industry-standard Electronic Design Automation (EDA) tools.

CLO2: Analyze and interpret the behavior and characteristics of MOSFETs and CMOS technology, fundamental to VLSI design.

CLO3: Evaluate the impact of process variations, parasitic elements, and technology scaling on the performance and reliability of VLSI circuits.

CLO4: Apply principles of digital logic design and boolean algebra to design and optimize VLSI circuits, including combinational and sequential logic circuits.

CLO5: Explore concepts of circuit timing, propagation delay, and clocking methodologies essential for designing synchronous digital systems in VLSI.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents/Experiments	Teaching Strategy	Lecture Hours	CLOs
1	Introduction to VLSI Design Tools and Environment	Lecture, Demonstration	2	CLO1
2	Transistor-Level Design: Basic Logic Gates	Lab Session	2	CLO1, CLO2, CLO4
3	Designing Combinational Logic Circuits	Lab Session	2	CLO1, CLO4
4	Sequential Logic Design: Flip-Flops and Registers	Lab Session	2	CLO1, CLO4
5	Timing Analysis and Optimization Techniques	Lecture, Lab Session	2	CLO5
6	Layout Design and Physical Verification	Lab Session	2	CLO1, CLO3
7	Power Analysis and Optimization Techniques	Lab Session	2	CLO3, CLO4, CLO5
8	Technology Scaling and Process Variations	Lecture, Lab Session	2	CLO2, CLO3, CLO4
9	Testing and Design for Testability	Lab Session	2	CLO3, CLO4
10	Project Work: Design and Implementation of VLSI Circuits	Lab Session, Project Work	2	CLO1, CLO2, CLO3, CLO4
11	Project Work: Testing and Optimization of VLSI Circuits	Lab Session, Project Work	2	CLO3, CLO4, CLO5
12	Project Presentation and Review	Presentation, Discussion	2	CLO1, CLO2, CLO3, CLO4, CLO5
13	Advanced Topics: Emerging Trends in VLSI Design	Lecture, Discussion	2	CLO1, CLO2, CLO3
14	Revision and Recapitulation	Review, Q&A Session	2	All CLOs

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12

CLO1	H			H								
CLO2	H	H	H	H								
CLO3	H	H	H	H								H
CLO4	H	H	H	H								
CLO5	H	H	H	H								

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Report/Lab Assessment	√	√	√		√
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (20 Marks):

Bloom's Category	Mid Assessment (10)	Quiz/Performance (5)	Class attendance (5)	Other forms
Remember	10%			
Understand	15%	25%		
Apply	25%	30%		
Analyze	20%	30%		
Evaluate	20%	15%		
Create	10%			

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2

3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester/single-mid assessment carrying 10 marks, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 5 marks out of 20. In the final examination, students have to solve 5 problems one from each CLO. There will be alternatives for each question. Two examiners will assess the final lab tests and the average of their given marks will be taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Reference Books:

1. Steven M. Rubin, Computer Aids for VLSI Design, latest edition.
2. Wayne Wolf, Modern VLSI Design: System-On-Chip Design, latest edition.
3. Sabib H Gerez, Algorithms for VLSI Design Automation, latest edition.

CSE-4229: Cloud Computing

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4229	Cloud Computing	Theory	3	60	20	10	10	100

Course Description: Cloud Computing offers an extensive overview of cloud computing technologies and their impact on the IT landscape. It covers distributed systems, the evolution of cloud computing, its definitions, benefits, challenges, and risks. The course dives into cloud computing architecture, service models (IaaS, PaaS, SaaS, and XaaS), deployment models (public, private, hybrid, and community clouds), and the underlying technologies enabling cloud computing, including virtualization and cloud storage options. Students will explore cloud security, management, and resource scheduling, along with hands-on exposure to leading cloud platforms such as AWS, Azure, and Google Cloud. The course will also touch upon emerging trends like serverless computing, microservices, and the role of cloud computing in mobile environments.

Justification of this course in the program entity

Cloud Computing is foundational to modern software development and deployment strategies, offering scalable, flexible, and cost-effective solutions. Understanding cloud technologies is critical for students aiming to pursue careers in software engineering, systems architecture, and data science. This course aligns with industry needs by providing students with practical knowledge and skills in cloud computing, preparing them for the rapidly evolving tech landscape.

Course Objectives

The major objectives of the course are:

- iv. To introduce the principles and architectures of cloud computing.
- v. To familiarize students with cloud service and deployment models.
- vi. To develop skills for designing and implementing cloud-based solutions.
- vii. To understand cloud security measures and compliance requirements.
- viii. To analyze the impact and potential of emerging cloud computing trends.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Describe cloud computing architectures, service models, and deployment models.

CLO-2: Implement and manage solutions on various cloud platforms.

CLO-3: Identify and mitigate security risks in cloud computing environments.

CLO-4: Evaluate the use of cloud computing technologies in different application scenarios.

CLO-5: Communicate effectively about cloud technologies and their applications in IT.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Background of Cloud Computing Distributed systems, issues in distributed systems, Basic Computing Paradigms	● Lecture ● Rapport building ● Demonstration ● Group work	3	CLO-1
2	Introduction to Cloud Computing	● Lecture ● Multimedia	3	CLO-1

	Definition and applications including benefits, challenges, and risks, historical overview, Enabling Technologies and System Models for Cloud Computing	• Projector • Hand Note		
3,4	Cloud Computing Architecture and management Architectural design of Cloud computing, cloud service models (Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) and emerging XaaS), type of clouds ((Public clouds, Private clouds, Hybrid or heterogeneous clouds, Community clouds)	• Lecture • Group work • Demonstration	6	CLO-1, CLO-2
5,6	Cloud Resource Virtualization Resource virtualization and virtual Machine, File virtualization, type of virtualization and Hardware/software support for virtualization, type of network virtualization, Performance of virtual machines, VM software and platforms, Hypervisors, Operating system level virtualization, Dockers.	• Lecture, • Rapport building • Multimedia Projector • Cooperative learning	6	CLO-2
7,8	Cloud Resource Management and Scheduling Cloud Computing Reference Architecture, Policies and mechanisms, Resource utilization and energy efficiency, Application resource management, Models for cloud-based web services, scheduling algorithms for the cloud	• Lecture • Case Study • Group work	6	CLO-2, CLO-4
9	Cloud Storage Overview of storage systems and models, Distributed File Systems, Cloud Databases, Data storage for online transaction processing systems	• Lecture • Case Study • Group work	3	CLO-2
10	Cloud Security	• Lecture • Group work • Hand Note	3	CLO-3

	Cloud-security risks, Privacy and trust on the cloud, Cloud data encryption, Security of cloud infrastructure			
11	Cloud Platforms Amazon AWS, Microsoft Azure, Google App Engine, Google MapReduce, Yahoo Hadoop, Eucalyptus, Nimbus, OpenStack etc	• Lecture • Case Study • Group work	3	CLO-2
12	Mobile Cloud Computing Architecture and applications of MCC, Code partitioning, Code offloading and VM migration techniques.	• Lecture • Group work • Hand Note	3	CLO-2, CLO-4
13,14	Emergent Trends and Practices Hybrid Multicloud, Serverless Computing, Microservices, Virtual Desktop Infrastructure, Desktop as a Service, Virtual Desktop Delivery, Future Cloud Computing Architectures, Edge computing, and Fog Computing.	• Lecture g) Case Study h) Group work i) Research Presentation	6	CLO-4, CLO-5

Mapping between CLOs and PLOs:

	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1	H	M	M	H		M			M	M	M	
CLO2	H	H		H		H		M	M	H	M	M
CLO3	M	H	H	H	M	M	M	H	M	M		
CLO4	M	H	H	H	M	M		M	M	M	M	
CLO5		M						M	H	H	M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
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Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

Question Pattern:

- Mid Semester Examinations: There will be two/three questions, all of which to be answered.
- Semester Final Examination: 5 questions to be answered from 5 different CLOs.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl, Ricardo Puttini, Zaigham Mahmood.

Reference Books:

1. "Cloud Computing: Principles, Systems & Applications" edited by Nick Antonopoulos, Lee Gillam.
2. "Amazon Web Services in Action" by Andreas Wittig, Michael Wittig.
3. "Mastering Cloud Computing" by Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi.

CSE-4230: Cloud Computing Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				
				Semester Final Exam.	Continuous Assessment			Total
					Two Mid-Semester Exams	Class Performance	Class attendance	
CSE-4230	Cloud Computing Lab	Lab	1	30	10	5	5	50

Course Description: The Cloud Computing Lab complements the theoretical concepts covered in the Cloud Computing course by providing hands-on experience with cloud computing technologies. Students will engage in lab exercises and projects that involve using cloud services and platforms like AWS, Azure, and Google Cloud to implement and manage cloud-based solutions. The lab will cover virtual machine creation, deployment of applications in cloud environments, configuration of cloud networks, and implementation of cloud security measures.

Through these practical exercises, students will gain proficiency in navigating cloud platforms and understanding the architectural and operational aspects of cloud solutions.

Justification of this course in the program entity

This lab course is crucial for bridging the gap between theoretical knowledge and practical skills in cloud computing. It offers students the opportunity to apply cloud concepts in real-world scenarios, enhancing their understanding of cloud architectures, services, and security practices. The lab prepares students for industry roles that require hands-on cloud computing skills and fosters innovation in cloud technology applications.

Course Objectives

The major objectives of the lab are:

- i. To apply principles of cloud computing in practical settings using leading cloud platforms.
- ii. To develop skills in designing, deploying, and managing cloud-based applications.
- iii. To understand and implement cloud security and compliance measures in lab projects.
- iv. To analyze and solve problems using cloud computing technologies.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO-1: Implement and configure cloud services and deployment models using cloud platforms.

CLO-2: Deploy and manage cloud-based applications effectively.

CLO-3: Apply security measures to protect cloud resources and data.

CLO-4: Utilize cloud computing for solving practical problems in various application scenarios.

CLO-5: Document lab projects and communicate the outcomes effectively.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hour	CLOs
1	Lab Environment Setup and Cloud Services Introduction	a) Demonstration, Hands-on	3	CLO-1
2	Working with Virtual Machines and Containers	a) Hands-on, Lab Assignment	3	CLO-1 , CLO-2
3-4	Cloud Storage and Databases	a) Hands-on, Lab Assignment	6	CLO-2
5-6	Cloud Networking and Security Basics	a) Hands-on, Lab Assignment	6	CLO-3

7-8	Deploying Web Applications in the Cloud	a) Hands-on, Lab Assignment	6	CLO-2 , CLO-4
9-10	Serverless Architecture and Microservices	a) Hands-on, Lab Assignment	6	CLO-2 , CLO-4
11	Cloud Monitoring and Management Tools	a) Hands-on, Lab Assignment	3	CLO-2
12	Mobile Cloud Computing Applications	a) Hands-on, Lab Assignment	3	CLO-4
13	Project Work on Emerging Cloud Trends	a) Hands-on, Project Work	3	CLO-4 , CLO-5
14	Lab Report Writing and Presentation	a) Presentation, Evaluation	3	CLO-5

Mapping between CLOs and PLOs:

	PL O1	PL O2	PL O3	PL O4	PL O5	PL O6	PL O7	PL O8	PL O9	PL O 10	PL O 11	PL O 12
CLO1	H	H	M	H		H		M	M	H	M	
CLO2	H	H		H	M	H	M		M	H	M	M
CLO3	M	H	H	H	M	M	H	H	M	M		
CLO4	H	H	M	H	H	M	M	M	M	H	H	M
CLO5		M		M				M	H	H	M	

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√

Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40% Marks):

Bloom's Category	Mid Semester 1 (10%)	Mid Semester 2 (10%)	Quiz/Performance (10%)	Class attendance (10%)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (30% Marks):

Bloom's Category	Marks (30)
Remember	5
Understand	5
Apply	5
Analyze	5
Evaluate	5
Create	5

Question Pattern:

The semester final examination will consist of five questions, one from each CLO.

Students will have options to choose questions from each CLO.

Each question will be designed to assess different levels of Bloom's taxonomy.

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Text Book:

1. Provided resources online through cloud platforms' official documentation and tutorials (AWS, Azure, Google Cloud).
2. Online resources

Reference Books:

3. "AWS Certified Solutions Architect Official Study Guide" by Joe Baron et al.
4. "Azure for Architects" by Ritesh Modi.
5. "Google Cloud Platform in Action" by JJ Geewax.

CSE-4231: Machine Learning

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)				Total
				Semester Final Exam.	Continuous Assessment			
					Two Mid-Semester Exams	Class Performance	Class attendance	

CSE-423 1	Machine Learning	Theory	3	60	20	10	10	100
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Course Description: Machine Learning is an optional course for the B.Sc. (Honours) program in the Department of Computer Science and Engineering, Jagannath University. The course carries 3 credits and consists of 45 lecture hours. It covers fundamental concepts of machine learning, including algorithms, models, and practical applications. The course aims to develop students' understanding of machine learning principles and their ability to apply them in real-world scenarios.

Justification of this course in the program entity

Machine Learning is a crucial aspect of modern computing, empowering students with the ability to create intelligent systems and make data-driven decisions. This course introduces students to foundational machine learning concepts, providing them with essential skills for various applications, such as data analysis, pattern recognition, and predictive modeling.

Course Objectives

The major objectives of the course are:

- i. Understand fundamental machine learning concepts and algorithms.
- ii. Apply machine learning techniques to solve real-world problems.
- iii. Develop proficiency in implementing machine learning models using programming languages.
- iv. Analyze and evaluate the performance of machine learning models.
- v. Gain insight into advanced machine learning topics and emerging trends.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Apply fundamental machine learning algorithms and syntax for solving problems.

CLO2: Implement and evaluate machine learning models for various types of data.

CLO3: Develop proficiency in programming languages used in machine learning.

CLO4: Analyze and interpret the performance metrics of machine learning models.

CLO5: Explore advanced topics and emerging trends in the field of machine learning.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lecture Hours	CLOs
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1	Introduction to Machine Learning: History, Types, and Applications	Enhanced lectures, Problem-based class	3	CLO1
2	Supervised Learning: Regression and Classification	Enhanced lectures, Problem-based class	3	CLO2
3	Unsupervised Learning: Clustering and Dimensionality Reduction	Enhanced lectures, Problem-based class	3	CLO2
4	Introduction to Programming in Machine Learning (e.g., Python)	Enhanced lectures, Hands-on coding sessions	3	CLO3
5	Model Evaluation and Metrics	Enhanced lectures, Practical exercises	3	CLO4
6	Feature Engineering and Selection	Enhanced lectures, Group projects	3	CLO3
7	Advanced Topics: Neural Networks and Deep Learning	Enhanced lectures, Coding projects	3	CLO5
8	Reinforcement Learning	Enhanced lectures, Case studies	3	CLO5
9	Natural Language Processing and Machine Learning	Enhanced lectures, Text analysis projects	3	CLO2
10	Ethical Considerations in Machine Learning	Enhanced lectures, Group discussions	3	CLO4
11	Machine Learning in Practice: Industry Applications	Guest lectures, Case studies	3	CLO5
12	Project Work: Applying Machine Learning to Real-World Data	Project guidance, Hands-on implementation	3	CLO1, CLO2, CLO3, CLO4, CLO5
13	Challenges and Future Trends in Machine Learning	Enhanced lectures, Research discussions	3	CLO5
14	Revision and Q/A	Review sessions, Exam preparation	3	All CLOs

Mapping between CLOs and PLOs:

CLOs	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
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CLO1	H	M	M	M	M			M				H
CLO2		H							H			M
CLO3		M	H									
CLO4				H	H			H			H	
CLO5												

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Mid Semester 1	√	√			
Mid Semester2			√	√	
Quiz/Performance	√	√	√	√	√
Class attendance	√	√	√	√	√
Final	√	√	√	√	√

Continuous Assessment (40 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Class attendance (10)	Other forms
Remember	10%	10%			
Understand	15%	15%	25%		
Apply	20%	20%	40%		
Analyze	15%	15%	25%		
Evaluate	20%	20%	10%		
Create	20%	20%			

Semester Final Examination (60 Marks):

Bloom's Category	Marks (60)
Remember	10
Understand	10
Apply	10
Analyze	10
Evaluate	10
Create	10

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
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1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the rest 40% is for continuous assessment. The continuous assessment contains two Mid-semester examinations carrying 10% marks each, Class attendance carrying 10% marks and the class performance/quiz/viva/presentation carries 10% marks. In the final examination, students have to answer 5 questions one from each CLO. There will be alternatives for each question. Two examiners will examine the final script and the average of their given marks will be taken. If the difference between their marks is more than or equal to 20% of the total marks, the script will be examined by a third examiner and in that case, the average of the nearest two of these three marks are taken. The sum of the marks of the final examination and continuous assessment is added, and the final marks are converted to the letter grade.

PART D: LEARNING RESOURCES

Textbook:

1. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, O'Reilly Media.

Reference Books:

1. "Pattern Recognition and Machine Learning" by Christopher M. Bishop, Springer.
2. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy, MIT Press.
3. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller, Sarah Guido, O'Reilly Media.
4. "Python Machine Learning" by Sebastian Raschka, Vahid Mirjalili, Packt Publishing.
5. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press.

CSEL-4232: Machine Learning Lab

PART A: INTRODUCTION

Course Code	Course Title	Course Type	Credit Point	Evaluation (Marks Distribution)
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				Semester Final Exam.	Continuous Assessment		Total
					Two Mid-Semester Exams	Class Performance	
CSEL-4 231	Machine Learning Lab	Lab	1	30	15	5	50

Course Description: Machine Learning Lab complements the theory course by providing hands-on experience with real datasets, implementation of algorithms, and model evaluation. Students will use tools like Python, NumPy, Pandas, Scikit-learn, and TensorFlow.

Justification of this course in the program entity

This lab develops practical problem-solving ability, programming expertise, and analytical thinking required for modern machine learning applications.

Course Objectives

The major objectives of the course are:

- I. Implement machine learning algorithms practically.
- II. Perform preprocessing and feature engineering.
- III. Evaluate models using metrics.
- IV. Apply ML in real-world problems.
- V. Improve reporting and presentation skills.

Course Learning Outcomes (CLOs)

After the completion of the course, successful students will be able to

CLO1: Implement ML algorithms.

CLO2: Perform preprocessing & feature engineering.

CLO3: Evaluate ML models.

CLO4: Apply ML on real datasets.

CLO5: Present and document results.

PART B: CONTENT OF THE COURSE

Course description with teaching strategy and alignment of course content with CLOs:

Week	Course Contents	Teaching Strategy	Lab Hours	CLOs
1	Python Basics, NumPy, Pandas	Hands-on	2	CLO1
2	Visualization (Matplotlib)	Hands-on	2	CLO2
3	Data Cleaning & Preprocessing	Hands-on	2	CLO2
4	Linear Regression	Problem-based	2	CLO1
5	Logistic Regression	Problem-based	2	CLO1
6	k-NN	Hands-on	2	CLO1
7	Decision Tree	Hands-on	2	CLO3
8	Random Forest	Hands-on	2	CLO3
9	K-Means Clustering	Hands-on	2	CLO3
10	PCA	Hands-on	2	CLO2
11	Model Evaluation	Exercises	2	CLO3
12	Neural Networks	Coding	2	CLO4
13	Project Implementation	Project	2	All CLOs
14	Viva & Review	Discussion	2	All CLOs

Mapping between CLOs and PLOs:

CLOs	PL O1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO1 0	PLO1 1	PLO1 2
CLO1	H	M	M		H						H	H
CLO2	M	H	M	M	H						H	H
CLO3	M	H		H	H						M	H
CLO4	H	H	H	H	H	M		M			H	H
CLO5				M			M	H	H	M		M

PART C: ASSESSMENT AND EVALUATION

Mapping/Alignment of CLOs vs Assessment Strategy:

	CLO1	CLO2	CLO3	CLO4	CLO5
Lab Performance	√	√	√		
Assignments			√	√	√
Project	√	√	√	√	√
Viva	√	√	√	√	√

Continuous Assessment (30 Marks):

Bloom's Category	Mid Semester 1 (10)	Mid Semester 2 (10)	Quiz/Performance (10)	Other forms
Remember	10%	10%		
Understand	15%	15%	25%	
Apply	20%	20%	40%	
Analyze	15%	15%	25%	
Evaluate	20%	20%	10%	
Create	20%	20%		

Semester Final Examination (30 Marks):

Bloom's Category	Marks (30)
Remember	4
Understand	4
Apply	6
Analyze	6
Evaluate	5
Create	5

The question pattern of the semester final examination will be as follows

Question	Options for summative assessment
1	Question from CLO1 or Question from CLO1
2	Question from CLO2 or Question from CLO2
3	Question from CLO3 or Question from CLO3
4	Question from CLO4 or Question from CLO4
5	Question from CLO5 or Question from CLO5

Assessment/Evaluation Procedure: 60% of the total marks is allotted for the final examination and the remaining 40% is for continuous assessment. The continuous assessment consists of class performance/quiz/viva/presentation and lab reports/assignments, carrying a total of 20 marks. In the final examination (30 marks), students are required to answer questions or perform tasks covering all CLOs, with alternatives where applicable. Two examiners will evaluate the

final scripts and the average of their awarded marks will be considered; if the difference between their marks is greater than or equal to 20% of the total marks, the script will be evaluated by a third examiner, and the average of the nearest two marks among the three will be taken. The total marks obtained from the final examination and continuous assessment will be summed and converted into the final letter grade.

PART D: LEARNING RESOURCES

Textbook:

1. "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, O'Reilly Media.

Reference Books:

1. "Pattern Recognition and Machine Learning" by Christopher M. Bishop, Springer.
2. "Machine Learning: A Probabilistic Perspective" by Kevin P. Murphy, MIT Press.
3. "Introduction to Machine Learning with Python: A Guide for Data Scientists" by Andreas C. Müller, Sarah Guido, O'Reilly Media.
4. "Python Machine Learning" by Sebastian Raschka, Vahid Mirjalili, Packt Publishing.
5. "Deep Learning" by Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press.